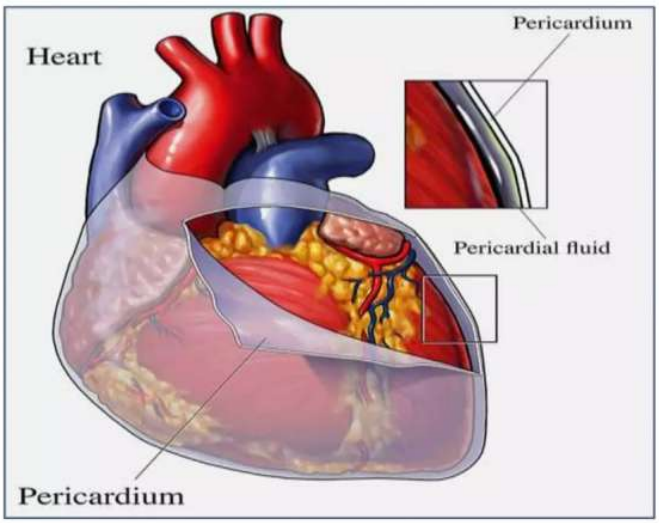
Pericardium

Dr.m.almusaibli

Introduction

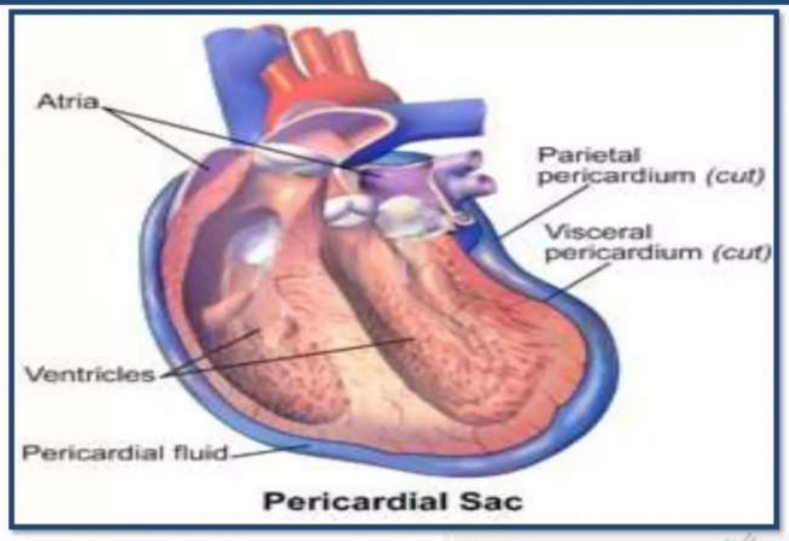
- The pericardium (Greek “around heart”) is a fibroserous fluid filled sac
- Pericardium surrounds the heart and the roots of the great vessels(aorta, pulmonary artery, Pulmonary veins ,SVC and IVC .

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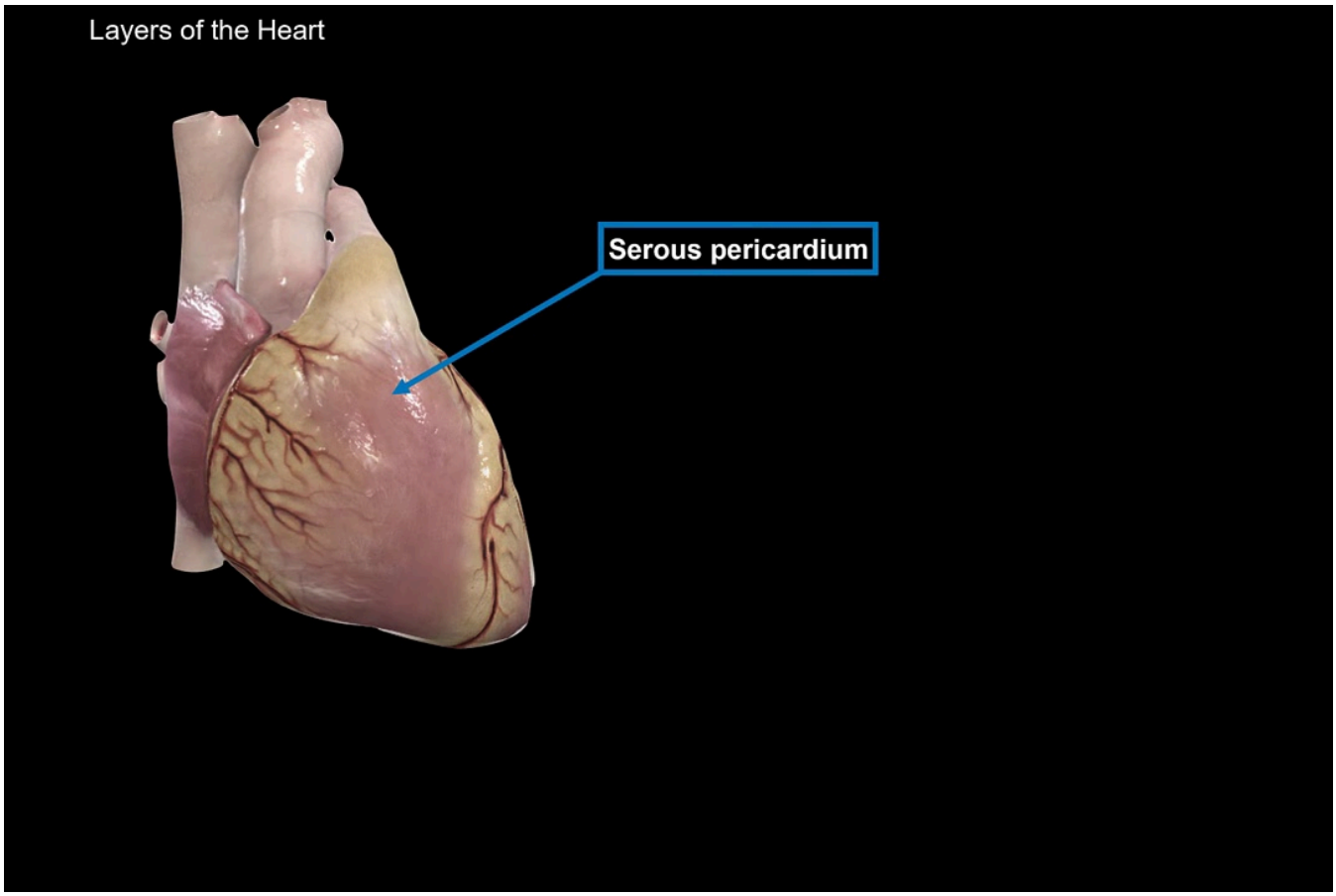


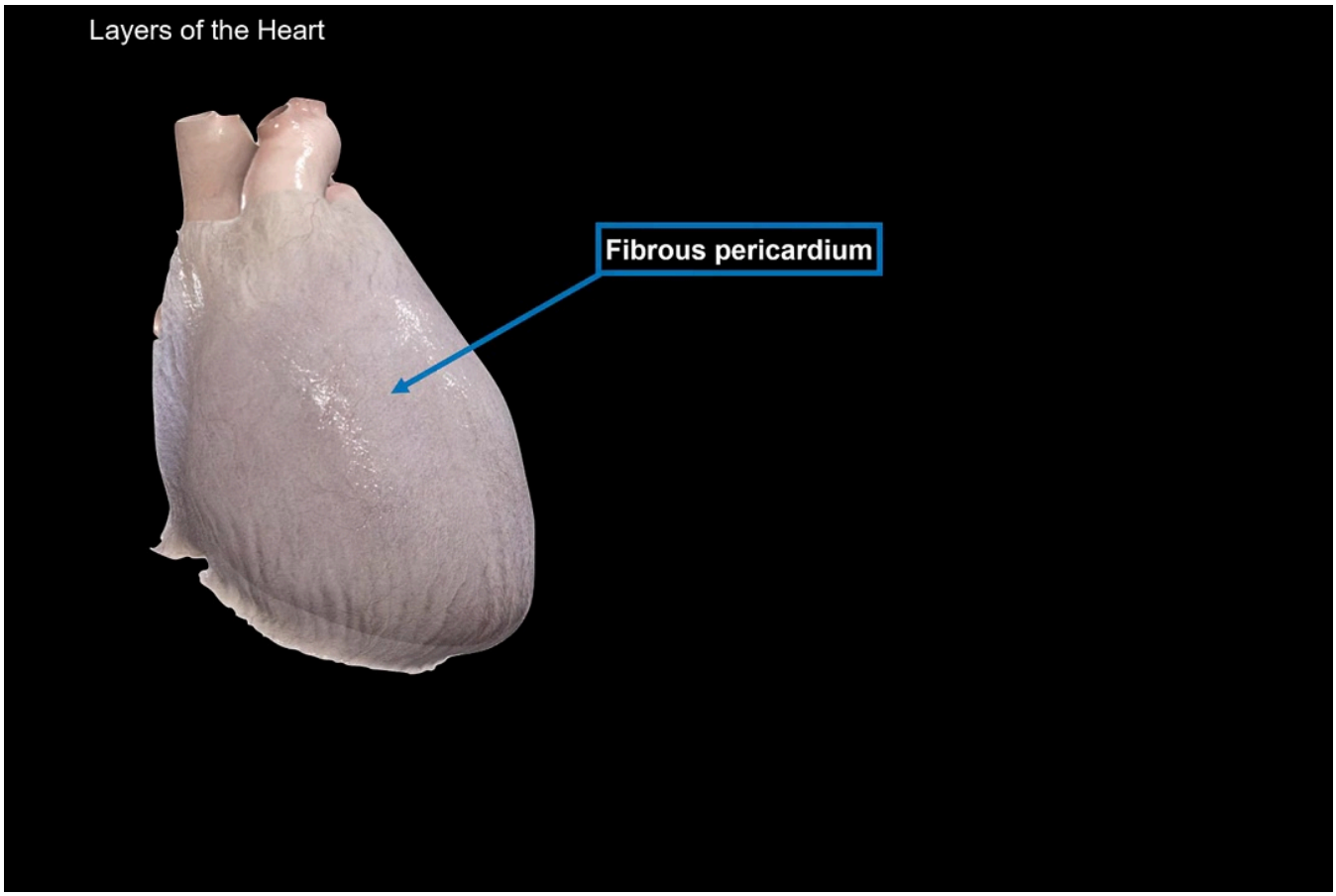
Pericardium

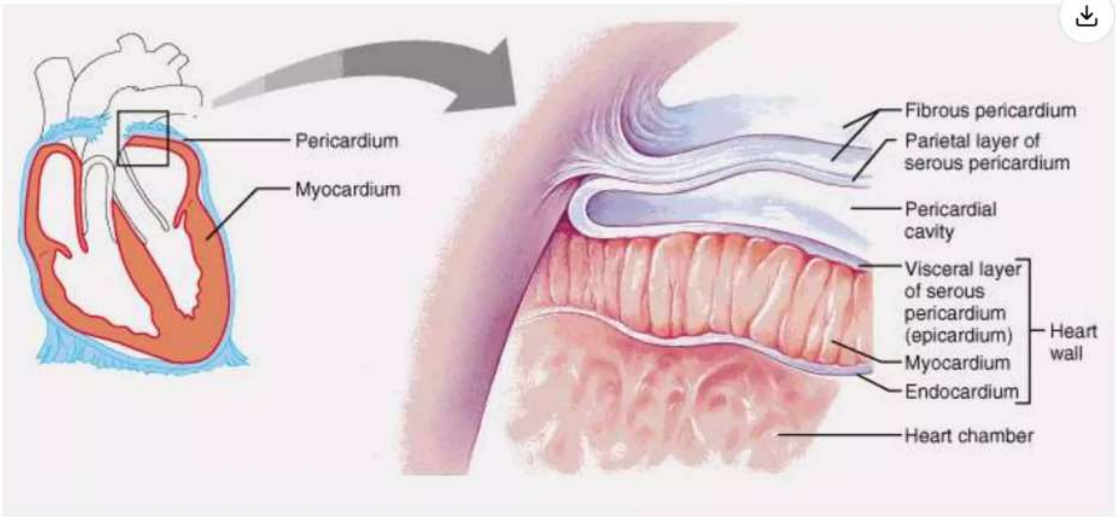
▪It is situated in the middle mediastinum.

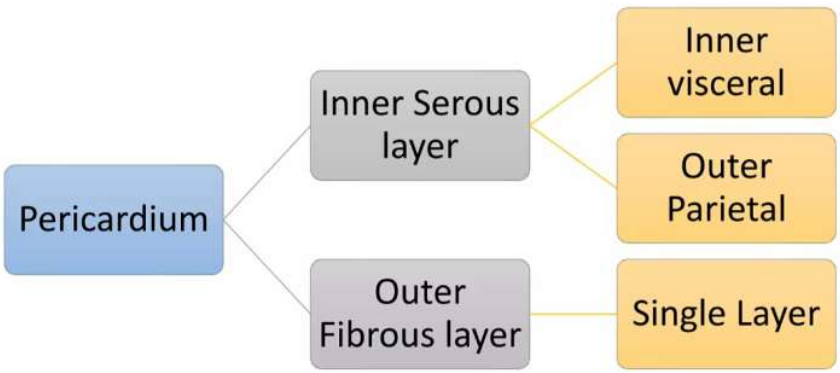


Pericardium

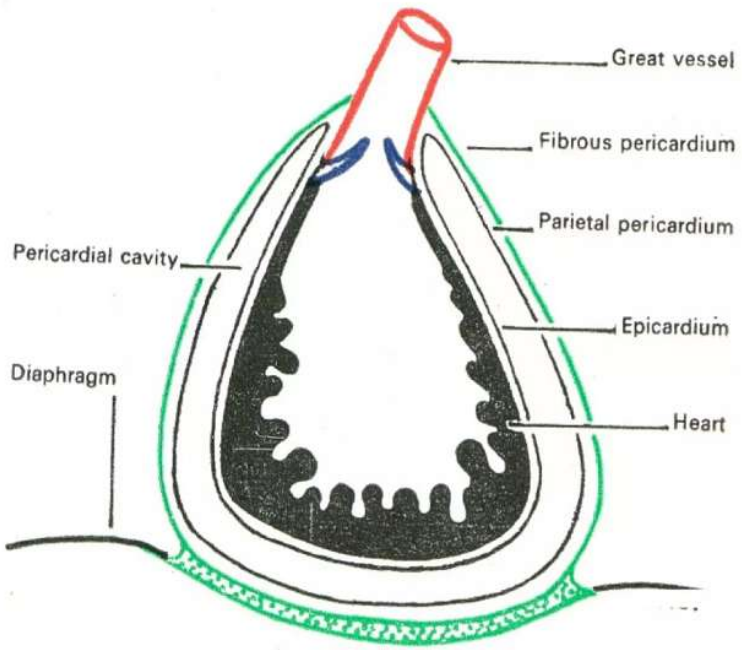

Layers of the Heart
Serous pericardium


Layers of the Heart
Fibrous pericardium


Pericardium
Myocardium
Fibrous pericardium
Parietal layer of serous pericardium
Pericardial cavity
Visceral layer of serous pericardium (epicardium)
Myocardium
Endocardium
Heart wall
Heart chamber


Pericardium
Inner Serous layer
Outer Fibrous layer
Inner visceral
Outer Parietal
Single Layer

Layers of Pericardium



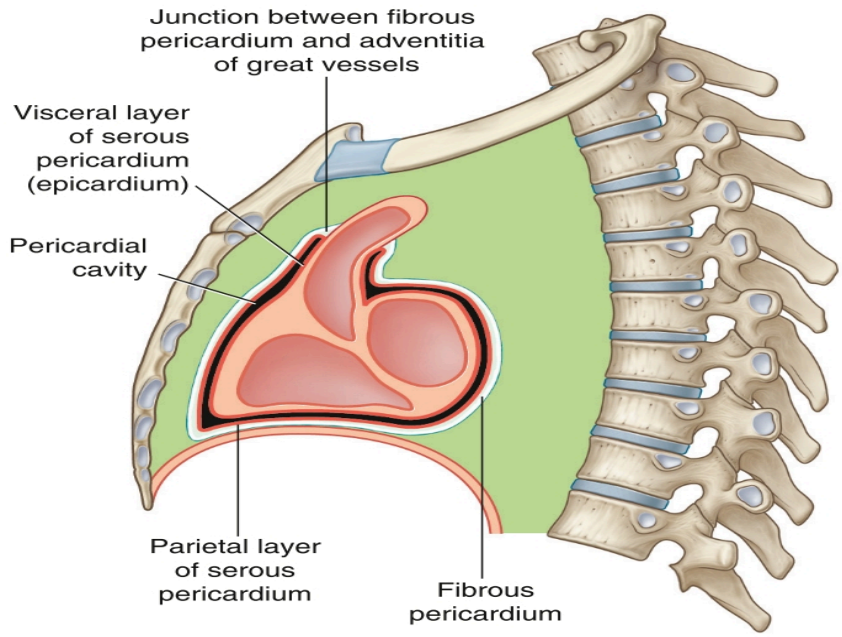


Fig. 3.61 Sagittal Section of the Pericardium.

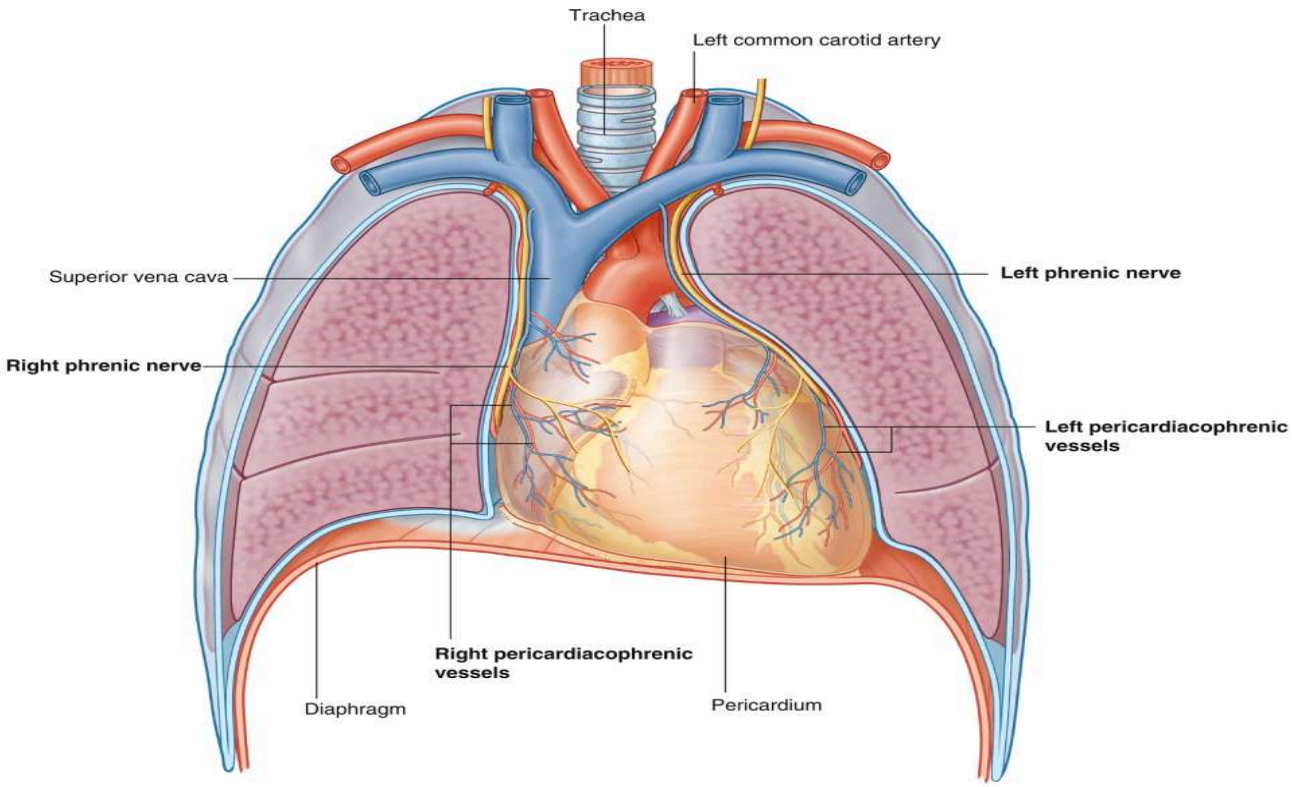


Fig. 3.62 Phrenic Nerves and Pericardiacophrenic Vessels.

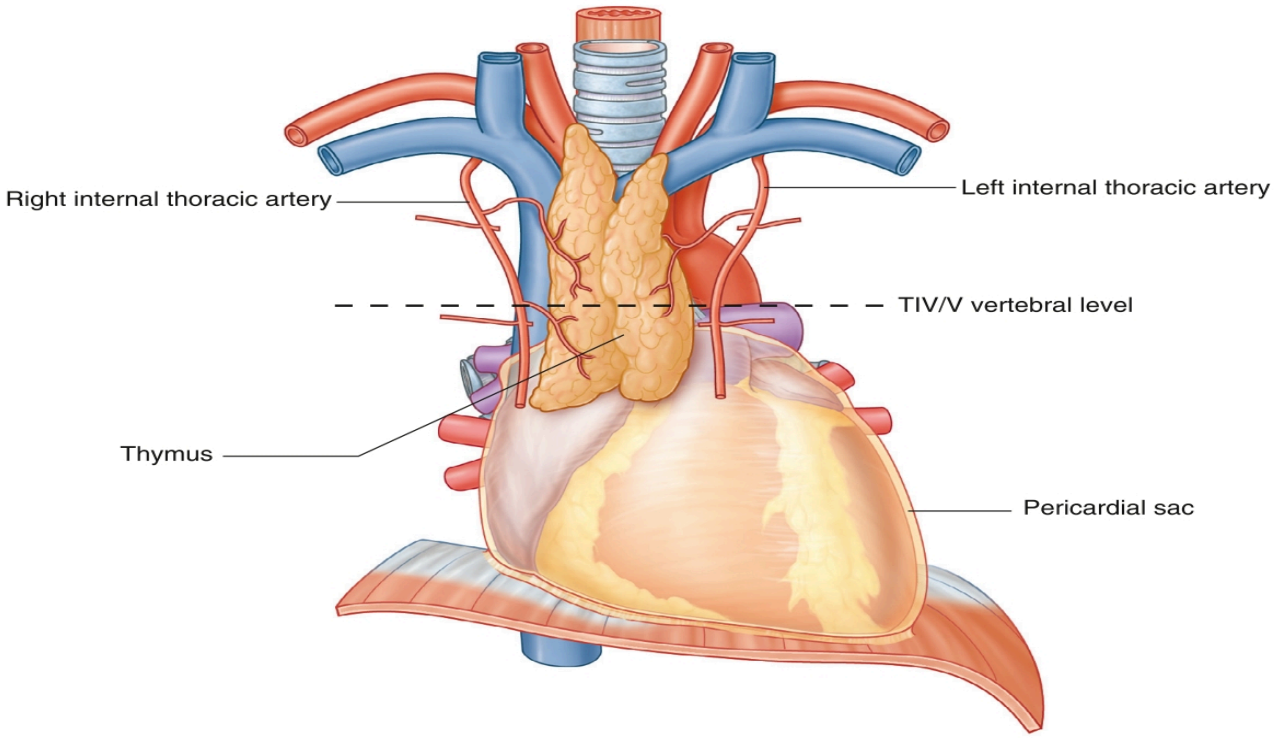


Fig. 3.60 Thymus.

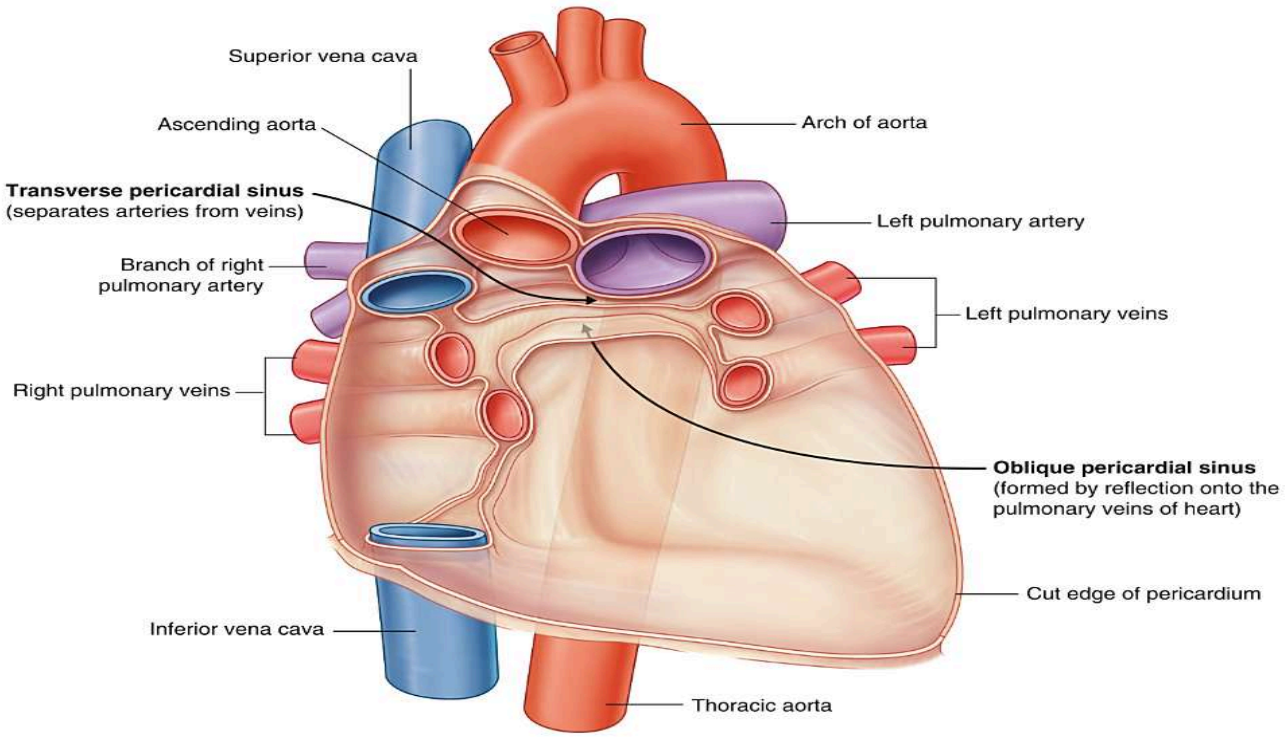


Fig. 3.63 Posterior Portion of Pericardial Sac Showing Reflections of Serous Pericardium.

Middle mediastinum (pericardium and heart)

Pericardium

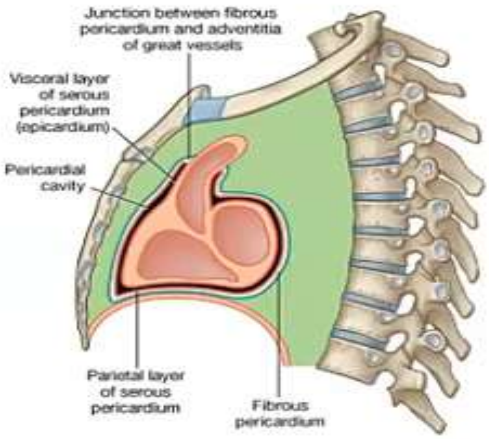
It is the fibro-serous sac surrounding the heart and the roots of great vessels.

Fibrous



Serous pericardium

Serous



- It is **closed** serous **sac** that is **invaginated** by **heart** to be **divided** into:
- A) visceral layer: **adherent** to the **heart** (**epicardium**).
- B) parietal layer: **lines** the **inner** surface of **fibrous** pericardium.
- **Both** layers are **continuous** around **roots** of great **vessels**.

Middle mediastinum (pericardium and heart)

Pericardium

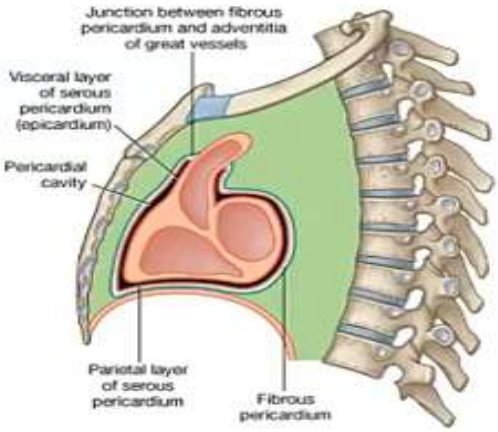
It is the fibro-serous sac surrounding the heart and the roots of great vessels.

Fibrous

- It is the tough outer layer.
- It defines the boundaries of the middle mediastinum.



Serous pericardium



Serous

- It is closed serous sac that is invaginated by heart to be divided into:
- A) visceral layer: adherent to the heart (epicardium).
- B) parietal layer: lines the inner surface of fibrous pericardium.
- Both layers are continuous around roots of great vessels.

Middle mediastinum (pericardium and heart)

Pericardium

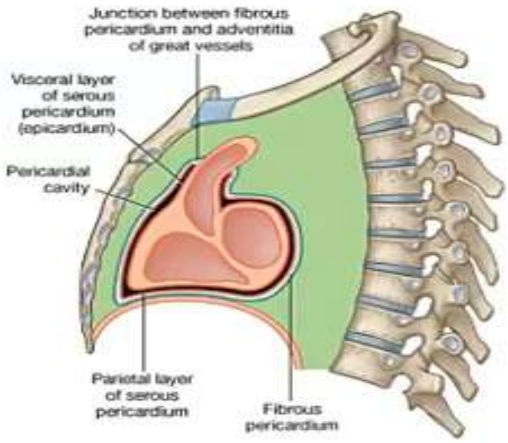
It is the fibro-serous sac surrounding the heart and the roots of great vessels.

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Serous pericardium



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- A) visceral layer: adherent to the heart (epicardium).
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- Both layers are continuous around roots of great vessels.

Middle mediastinum (pericardium and heart)

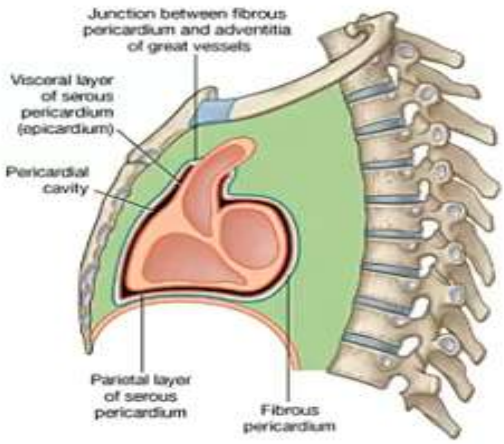
Pericardium

It is the fibro-serous sac surrounding the heart and the roots of great vessels.

Fibrous

- It is the tough outer layer.
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Serous

- It is closed serous sac that is invaginated by heart to be divided into:
- A) visceral layer: adherent to the heart (epicardium).
- B) parietal layer: lines the inner surface of fibrous pericardium.
- Both layers are continuous around roots of great vessels.

Fibrous pericardium:

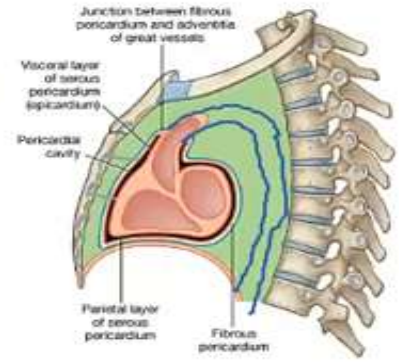
- It is cone shaped sac.

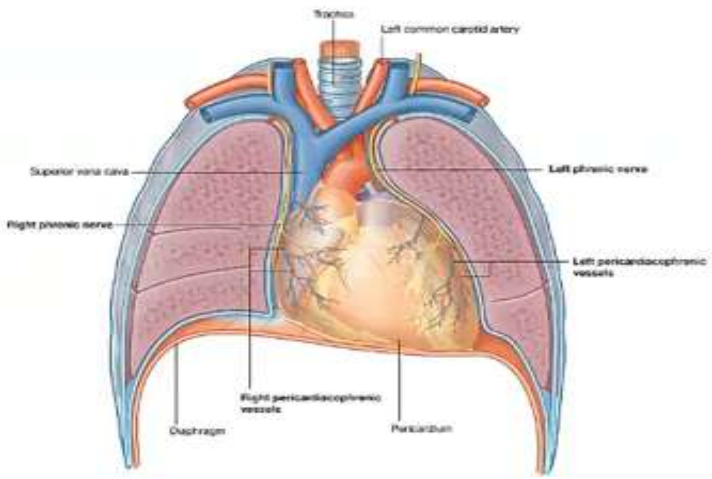
- Superiorly, its apex surrounds the great vessels and adherent to their walls.
- Inferiorly, its base fuses with central tendon of the diaphragm (pierced by IVC).

- On either sides, it is related to phrenic nerves and pericardiophrenic vessels and medial surfaces of both lungs.

- Anteriorly, it is related to body of sternum, attached to it by two sterno-pericadial ligaments (superior and inferior).
- Posteriorly, it is related to posterior mediastinal structures (esophagus and descending thoracic aorta).

Notice the deviation of left lung and pleura. What is the clinical significance?





Pericardial sinuses (transverse and oblique):

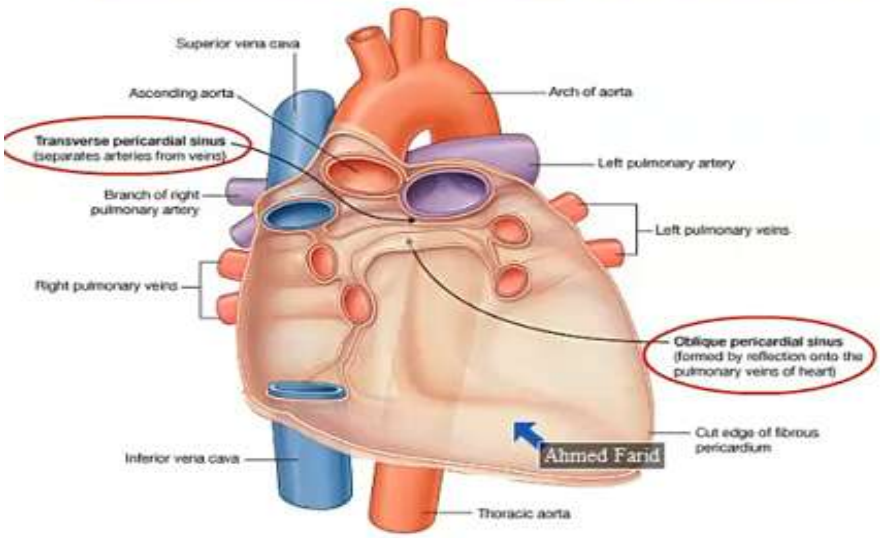
- The parietal layer of serous pericardium is continuous with the visceral layers of serous pericardium around the roots of the great vessels. These reflections of serous pericardium occur in two locations:

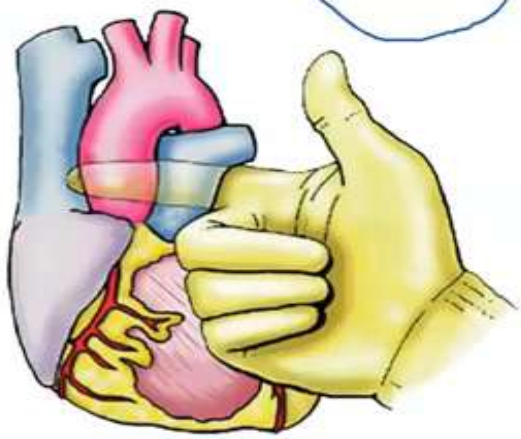
a) One superiorly: surrounding the arteries (Aorta and pulmonary trunk).

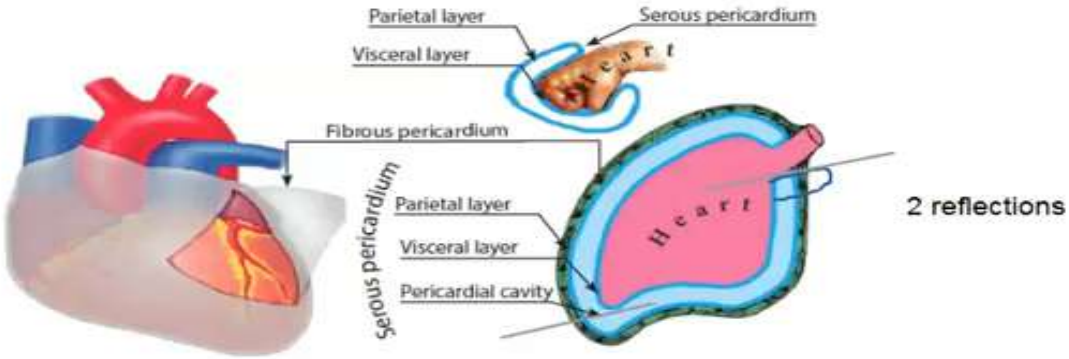
b) The other more posteriorly surrounding the veins (SVC, IVC, and pulmonary veins).

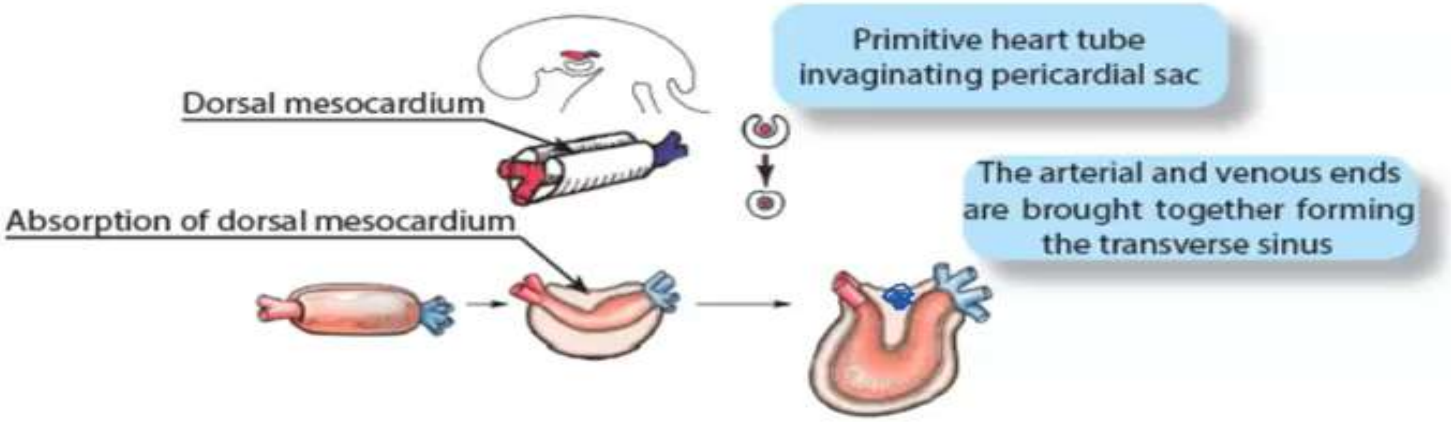


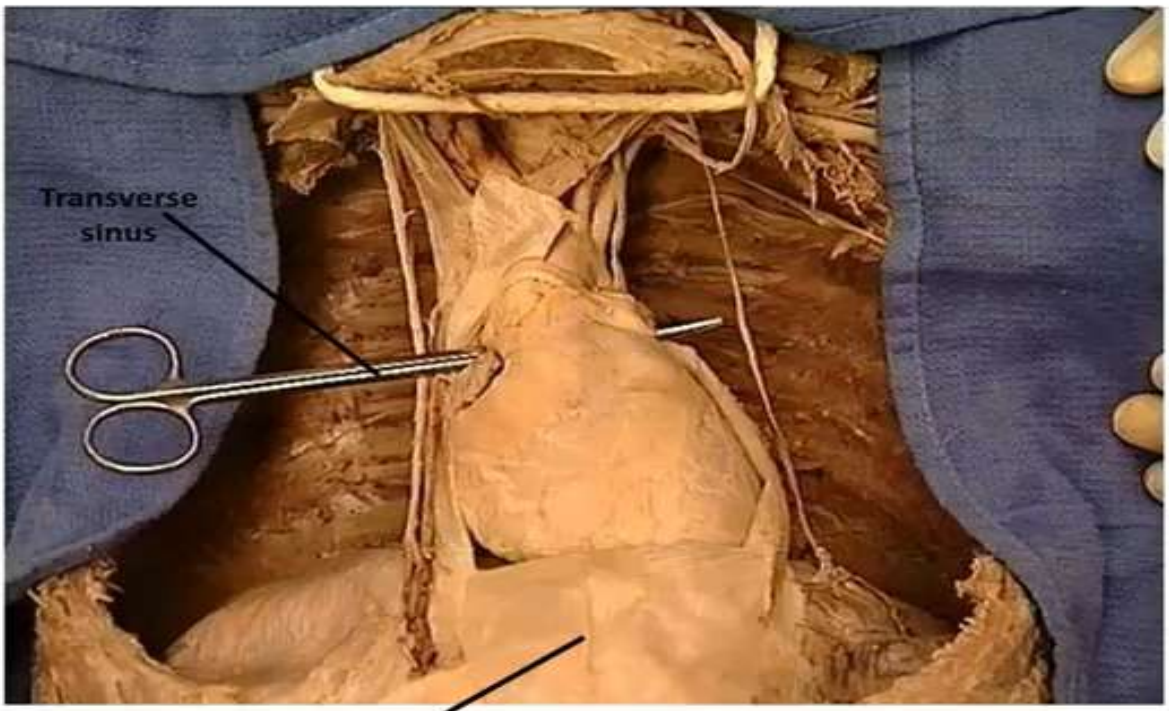
Locate the site, and identify the boundaries of each sinus?






Parietal layer
Serous pericardium
Visceral layer
Heart
Fibrous pericardium
Serous pericardium
Parietal layer
Visceral layer
Pericardial cavity
Heart
2 reflections


Primitive heart tube invaginating pericardial sac
Dorsal mesocardium
The arterial and venous ends are brought together forming the transverse sinus
Absorption of dorsal mesocardium


Transverse sinus
Fibrous pericardium open and reflected


Oblique
sinus

Blood and nerve supply

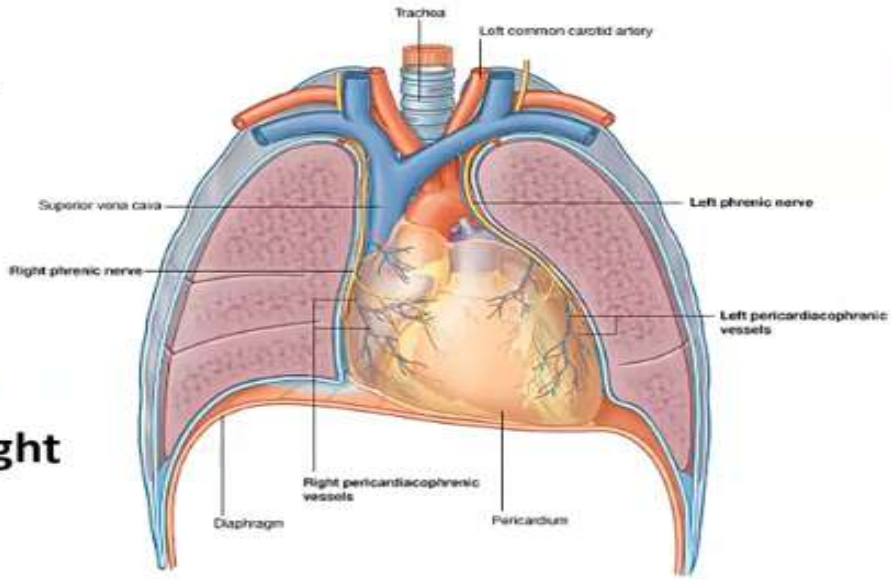
Arteries: musculophrenic, pericardiophrenic, and descending thoracic aorta.

Venous drainage: azygos system.

Nerves:

Fibrous + parietal layer of serous= somatic nerves (right and left phrenic nerves).

Visceral layer of serous = autonomic nerves = insensitive to pain.



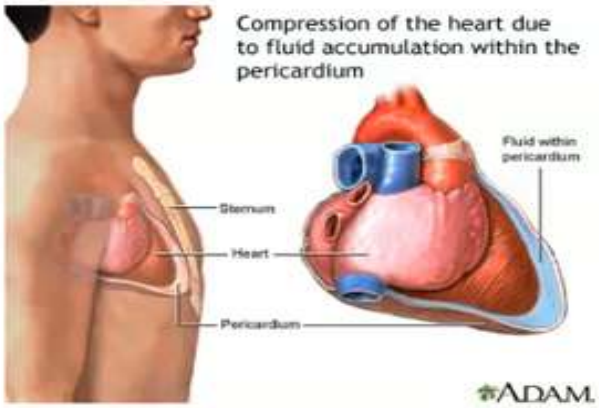
Ahmed Farid

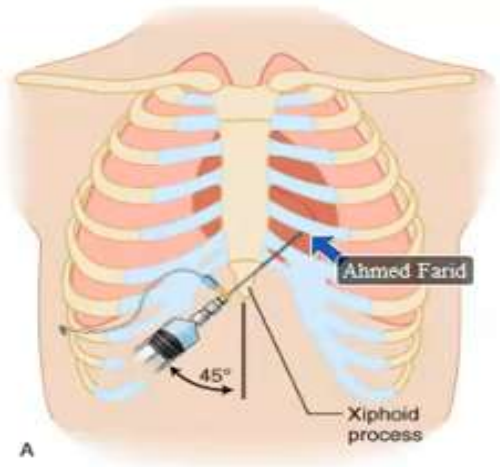
Applied anatomy:

- pericarditis: means inflammation of the serous pericardium, with roughening of the 2 layers of it, making friction rub with cardiac contraction.

- Pericardial fluid may accumulate in excess amount causing pericardial effusion which compresses the heart and interferes with atrial filling with venous return (cardiac tamponade).

- Pericardiocentesis: the procedure by which excessive pericardial fluid can be aspirated by needle through the pericardial sac. The needle is inserted through the left 5th or 6th intercostal space near the sternum (WHY?).





The pericardium

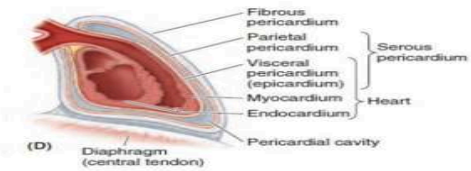
*Definition: it is a fibroserous sac surrounding the heart & the roots of the great vessels arising from or entering it.

* Site: it is situated in the middle mediastinum

*Structure: it consists of 2 sacs:

(a) an outer fibrous sac called **fibrous pericardium**

(b) an inner double-layered sac called **serous pericardium**



(A) Fibrous Pericardium

*Shape: conical sac made up of strong fibrous tissue. It has apex, base, ant. surface, post. surface & 2 lat. surfaces (Rt. & Lt.)

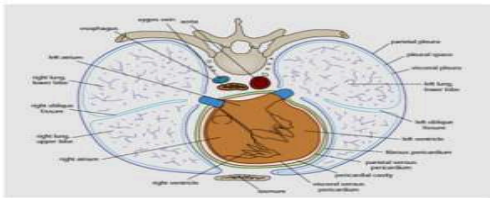
*Attachments:

- **the base (below):** is attached to the central tendon of the diaphragm.

-**The apex (above):** extends upwards to the level of the sternal angle where it blends with the adventitia of the big vessels (ascending aorta, pulmonary trunk & S.V.C) and with the pre-tracheal fascia

*Relations:

(1) **Anterior surface**:
-It is related to the sternum & costal cartilages but separated from them by ant. borders of both lungs & pleurae.
-it is connected to the back of the sternum by sup. & inf. sternopericardial ligaments



(2) **Rt. & Lt. Lateral surfaces**:
Each lat. surface is related to:
(a) mediastinal surface of lung & pleura.
(b) Phrenic n. & pericardiacophrenic vessels.

(3) **Posterior Surface**:
-It forms the ant. boundary of the post. mediastinum.
-It is pierced by 4 pulmonary veins.
-It is related to the principal bronchi & contents of post. mediastinum:
(a) oesophagus (b) descending aorta (c) azygos vein (d) thoracic duct.

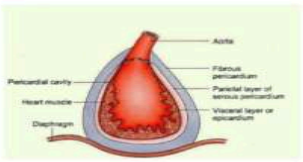
*Functions of Fibrous pericardium:

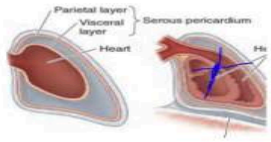
(1) Maintains a constant position for the heart.

(2) Being non-elastic, it prevents over distension of the heart.

(3) Keeps the mouths of the blood vessels open, not affected by cardiac or respiratory movements

(B) Serous Pericardium





***Definition**: it is a closed serous sac which is invaginated from above & behind by the heart & the roots of its big vessels

***Structure**: this invagination converts the serous sac into potential space having 2 layers:

(a)visceral (b)parietal

(a) Parietal layer: lines the fibrous pericardium & is reflected to the roots of the large vessels to become continuous with the visceral layer.

(B) Visceral layer: covers the heart & is called the **epicardium**.

(C) The pericardial cavity: is a potential space between the visceral & parietal layers of the serous pericardium. It contains a thin film of serous fluid secreted by the cells of serous pericardium.

*Functions of the Serous Pericardium:

(1) lubrication of the heart preventing friction during its movements.

(2) prevents adhesions between the heart & the surrounding organs.

***Blood Supply of the Pericardium**: the fibrous & parietal serous pericardium are supplied by branches from: (a) internal thoracic arteries b) musculophrenic arteries

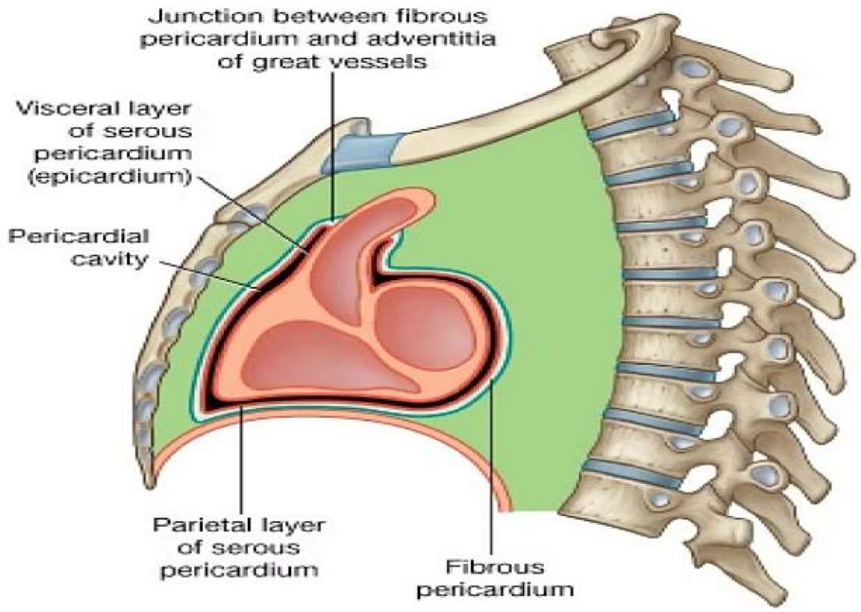
(c) pericardicophrenic aa. (d) descending thoracic aorta.

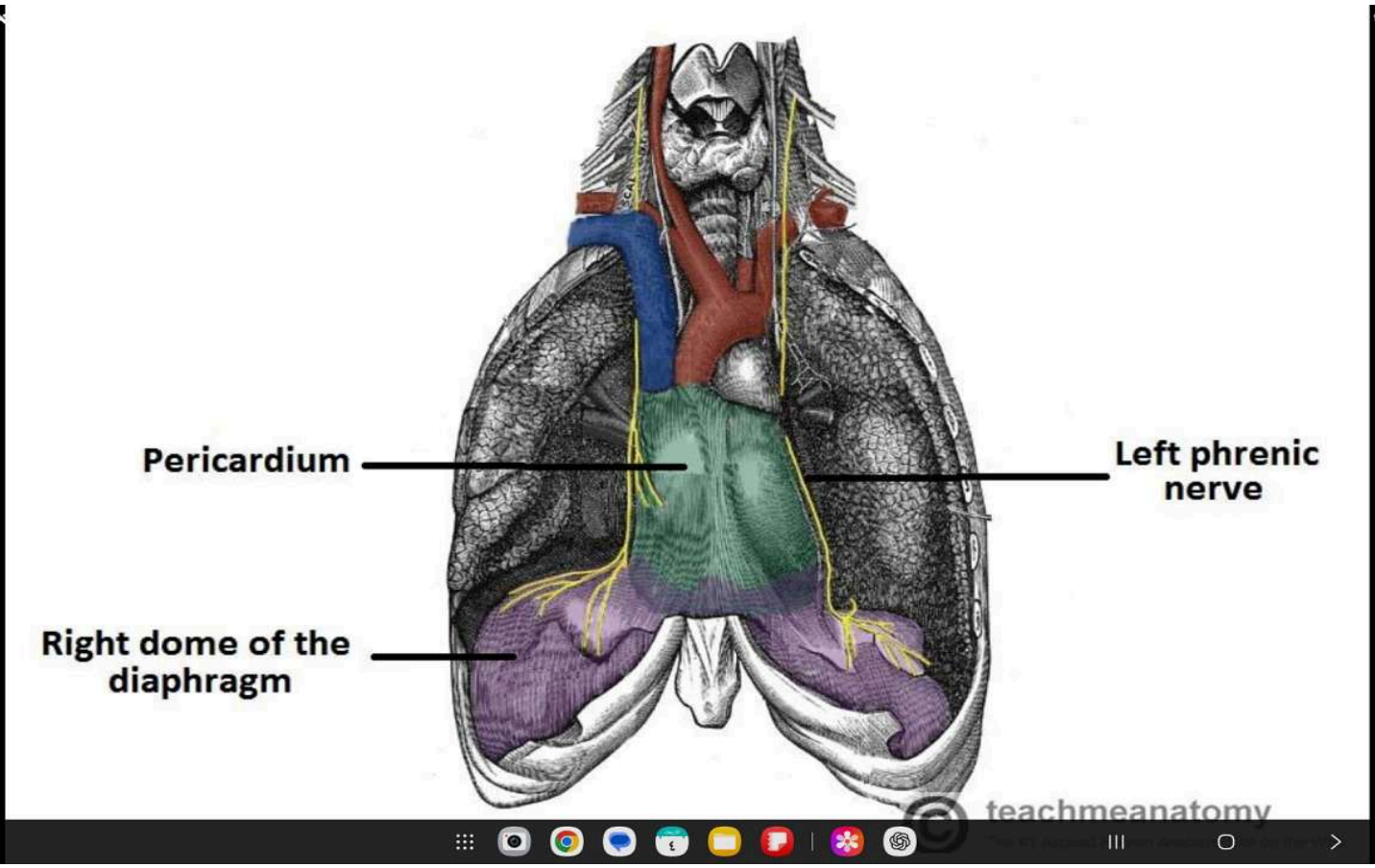
*Nerve Supply of the Pericardium:

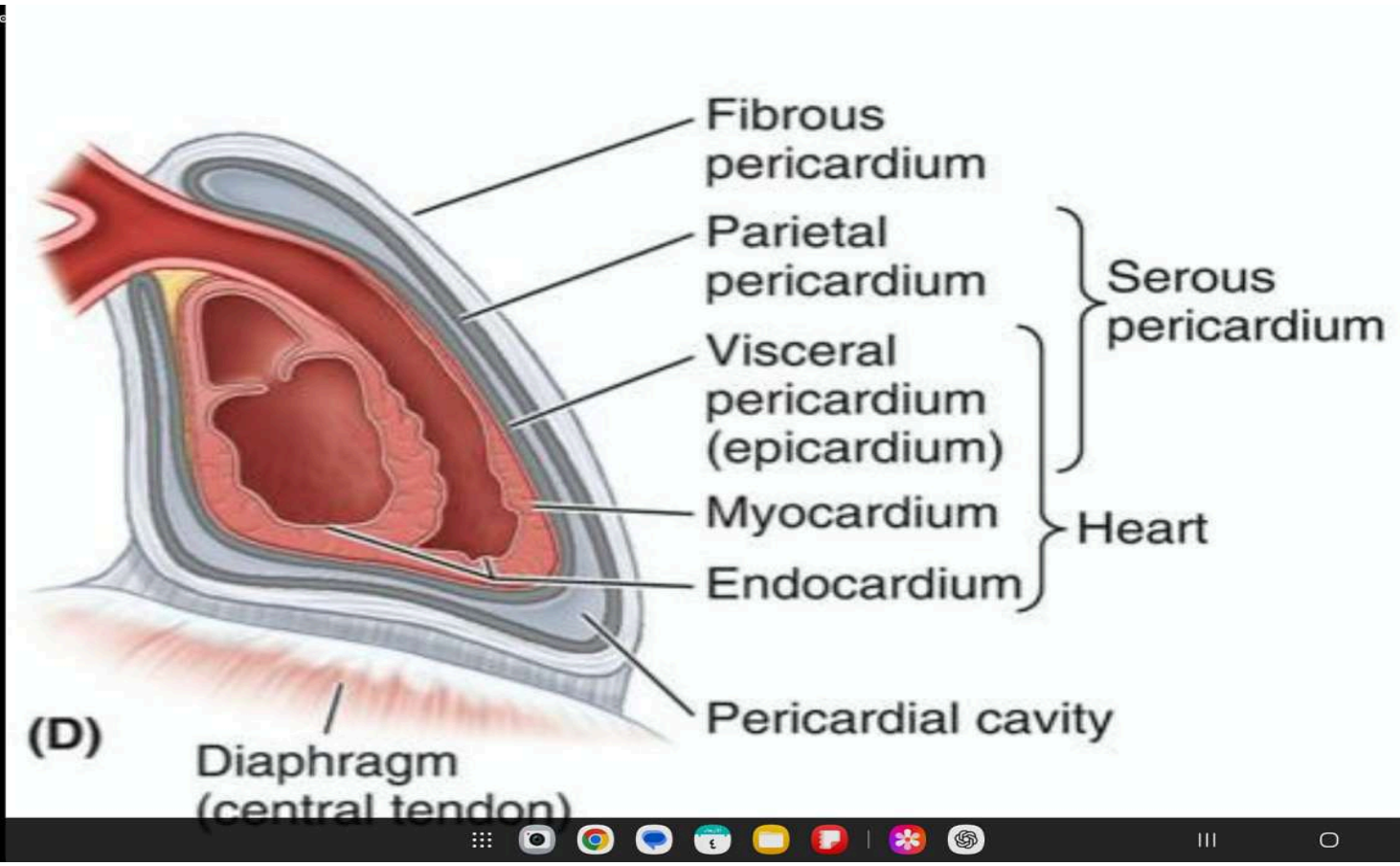
(a) Fibrous pericardium + parietal serous pericardium are supplied by the phrenic n.
They are sensitive to pain (as the pain of pericarditis).

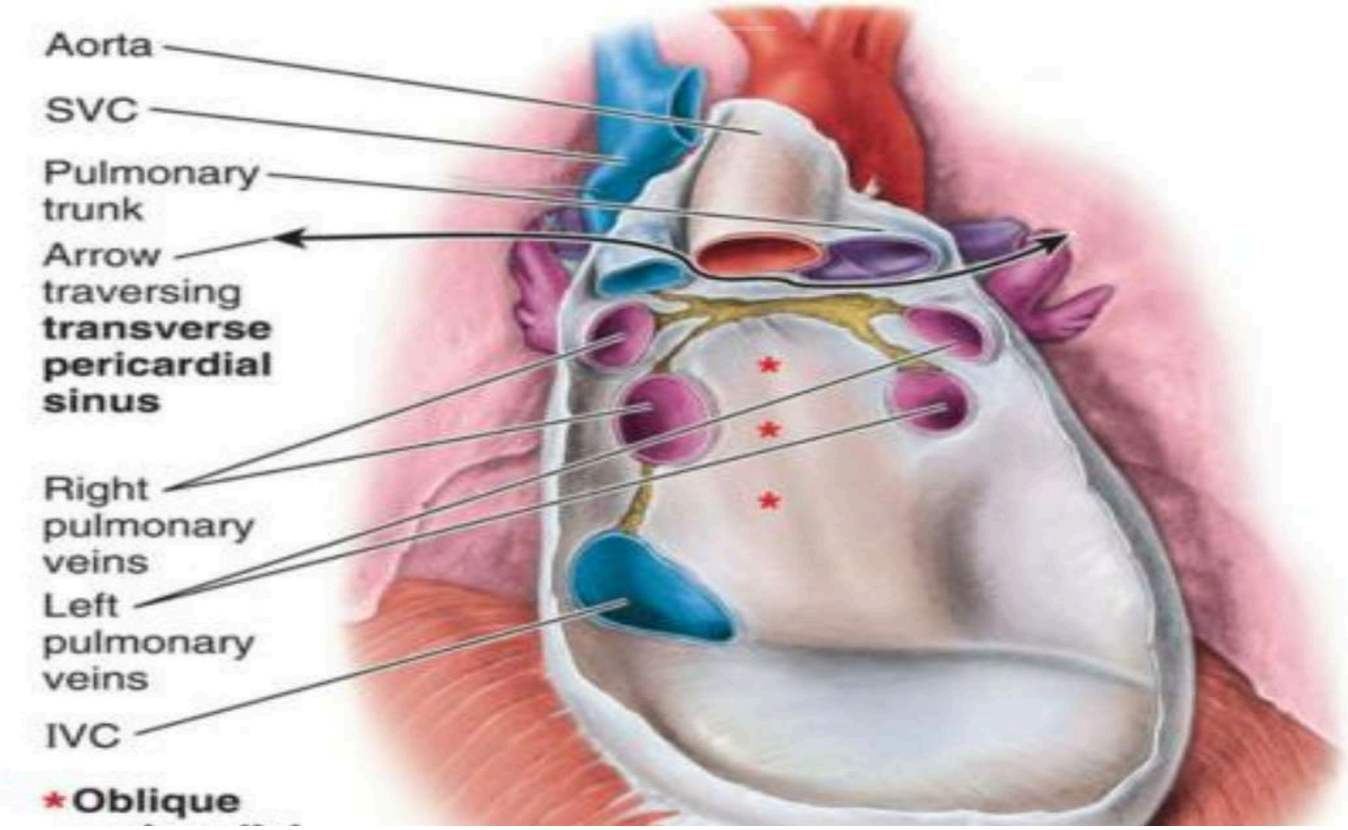
(b) Visceral layer of serous pericardium, like the heart, is supplied by autonomic nerves (sympathetic & parasympathetic). It is insensitive to pain but sensitive to ischemia

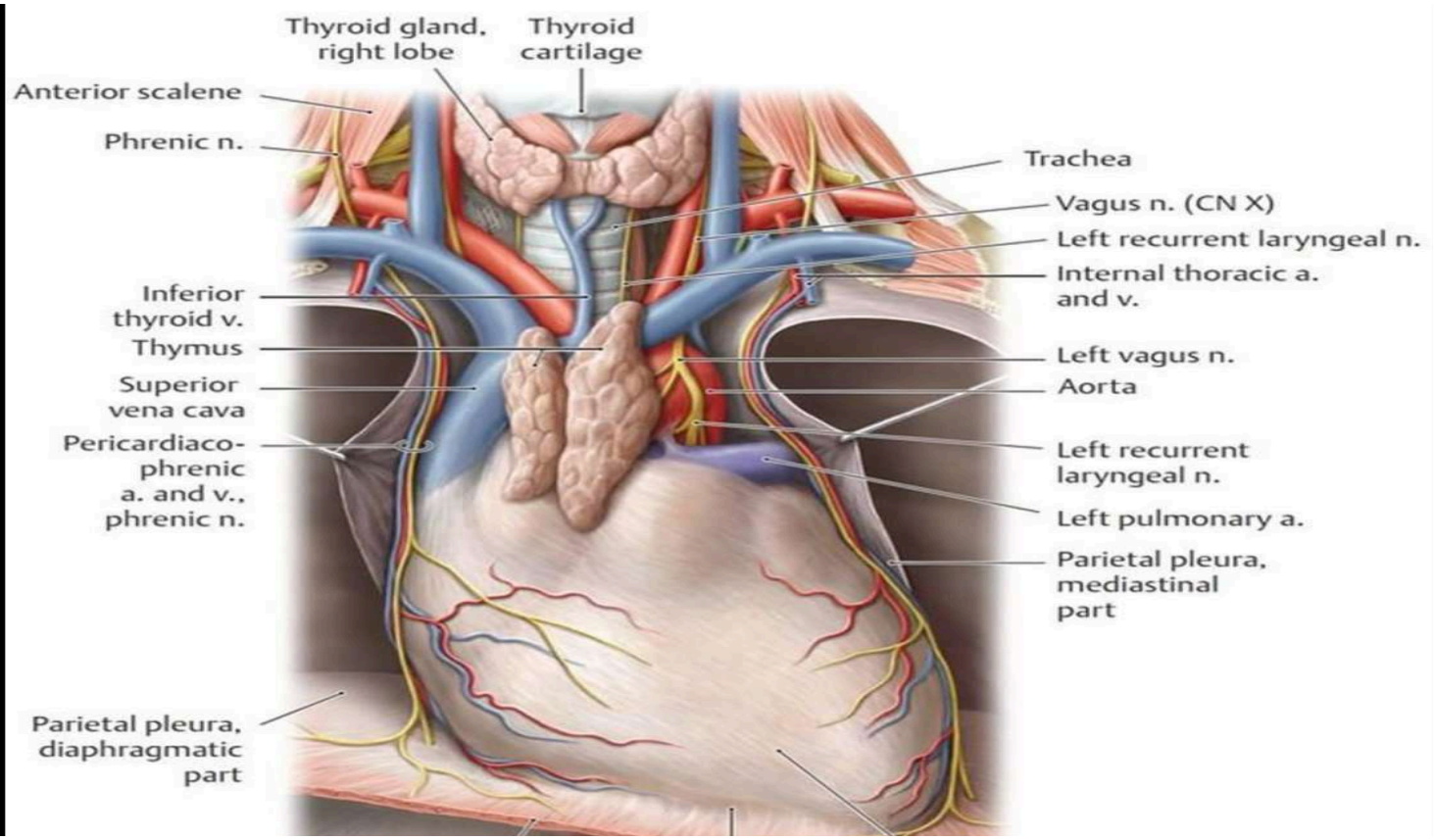
Sagittal section : Pericardium

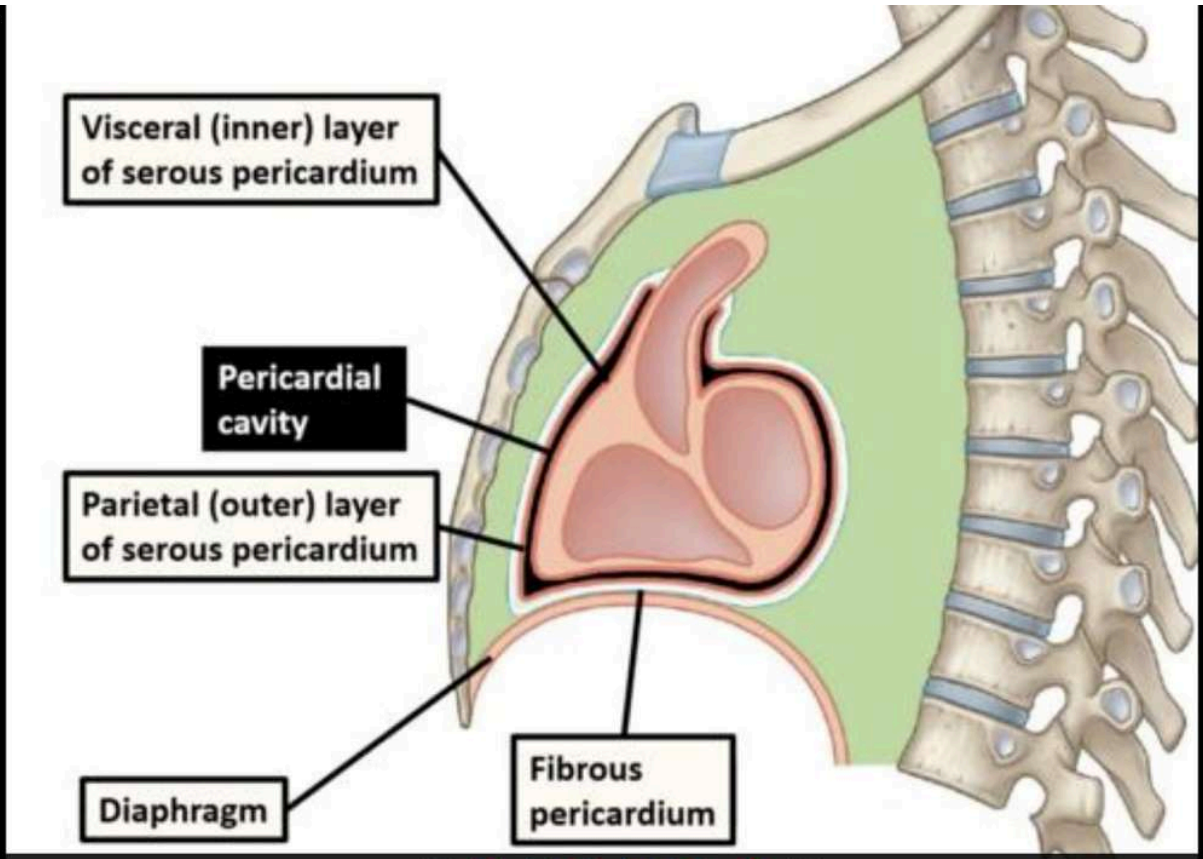


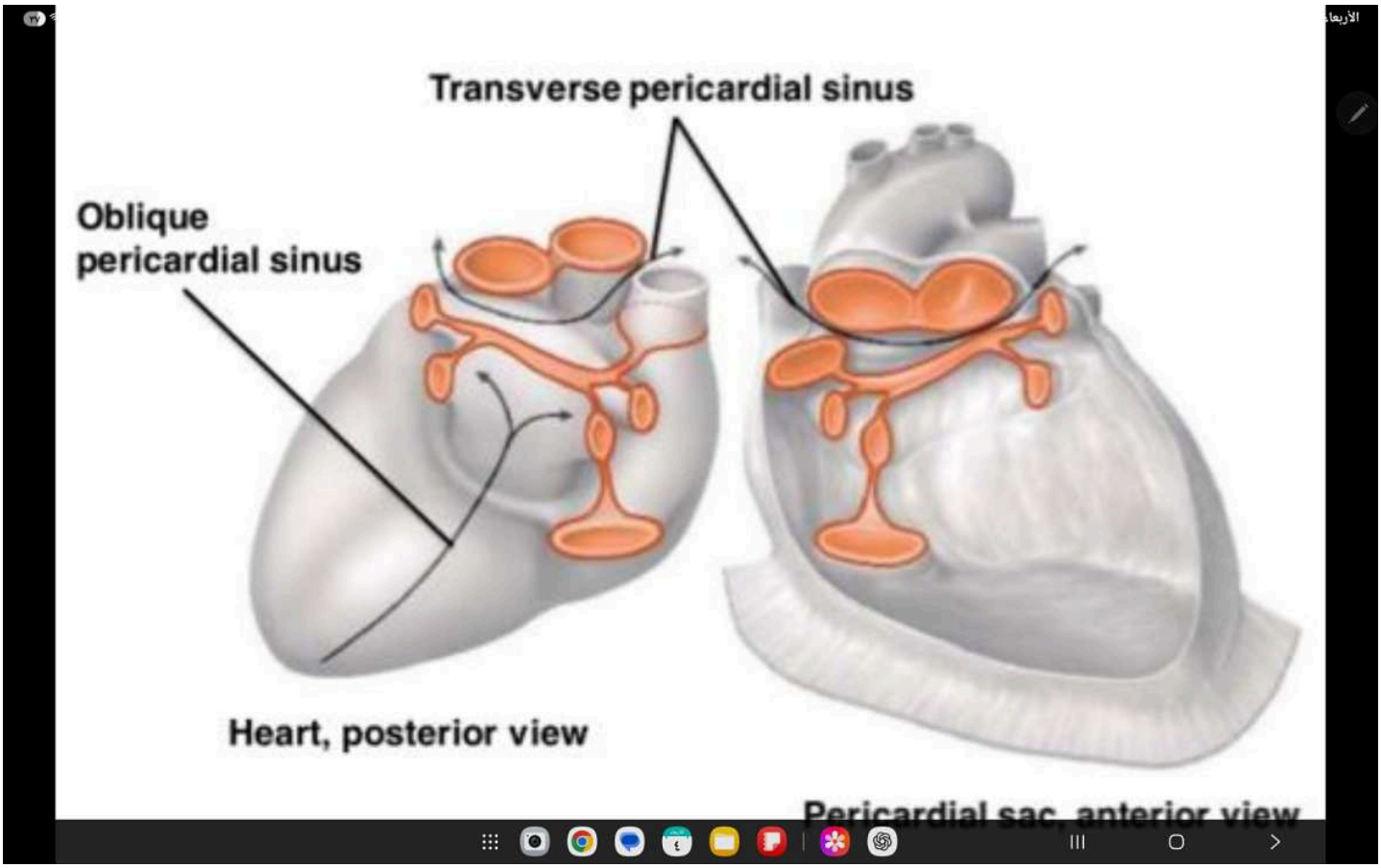

Pericardium
Left phrenic nerve
Right dome of the diaphragm
© teachmeanatomy


Fibrous pericardium
Parietal pericardium
Visceral pericardium (epicardium)
Myocardium
Endocardium
Serous pericardium
Heart
Pericardial cavity
(D)
Diaphragm (central tendon)


Aorta
SVC
Pulmonary trunk
Arrow traversing **transverse pericardial sinus**
Right pulmonary veins
Left pulmonary veins
IVC
* **Oblique**


Thyroid gland, right lobe
Thyroid cartilage
Anterior scalene
Phrenic n.
Trachea
Vagus n. (CN X)
Left recurrent laryngeal n.
Internal thoracic a. and v.
Inferior thyroid v.
Thymus
Left vagus n.
Superior vena cava
Aorta
Pericardiaco-phrenic a. and v., phrenic n.
Left recurrent laryngeal n.
Left pulmonary a.
Parietal pleura, mediastinal part
Parietal pleura, diaphragmatic part


Visceral (inner) layer of serous pericardium
Pericardial cavity
Parietal (outer) layer of serous pericardium
Diaphragm
Fibrous pericardium


Transverse pericardial sinus
Oblique pericardial sinus
Heart, posterior view
Pericardial sac, anterior view

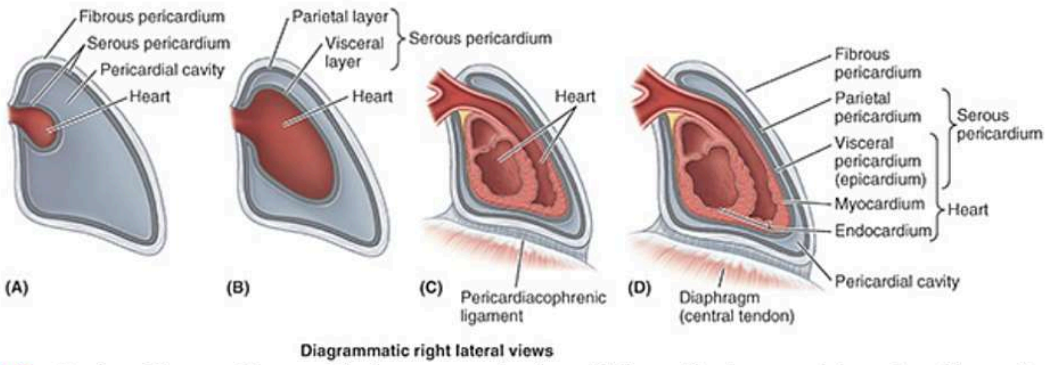


FIGURE 4.43. Pericardium and heart. The heart occupies the middle mediastinum and is enclosed by pericardium, composed of two parts. The tough, outer fibrous pericardium stabilizes the heart and helps prevent it from overdilating. Between the fibrous pericardium and the heart is a "collapsed" sac, the serous pericardium. **A and B.** Schematically, the embryonic heart invaginates the wall of the serous sac and enlarges. **C and D.** Ultimately, it practically obliterates the pericardial cavity, leaving only a potential space between the layers of serous pericardium. The pericardiacophrenic ligament (**D**) is the continuity of the fibrous pericardium with the central tendon of the diaphragm.

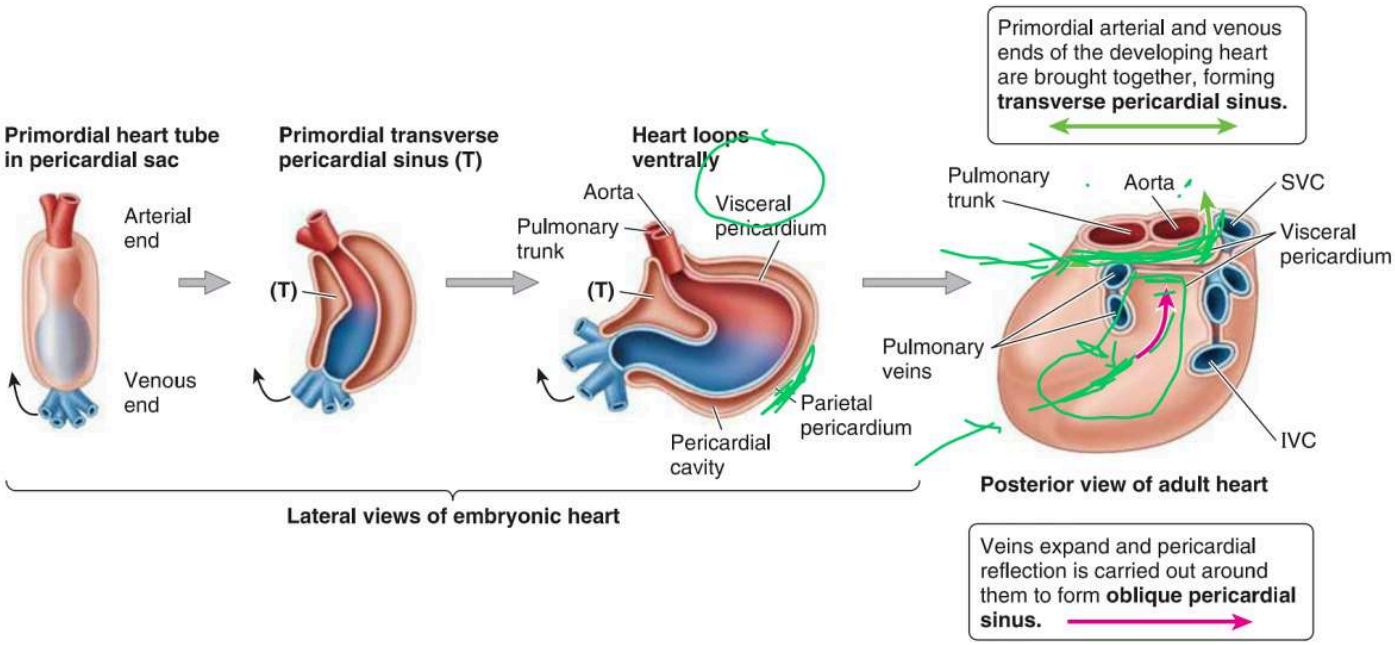


FIGURE 4.24. Development of heart and pericardium. The primordial, longitudinal heart tube invaginates the double-layered pericardial sac (somewhat like placing a hotdog in a bun). The primordial heart then "loops" ventrally, bringing the primordial arterial and venous ends of the heart together and creating the transverse pericardial sinus (*T*) between them. With growth of the embryo, the veins expand and spread apart inferiorly and laterally. The pericardium reflected around them forms the boundaries of the oblique pericardial sinus. *IVC*, inferior vena cava; *SVC*, superior vena cava.

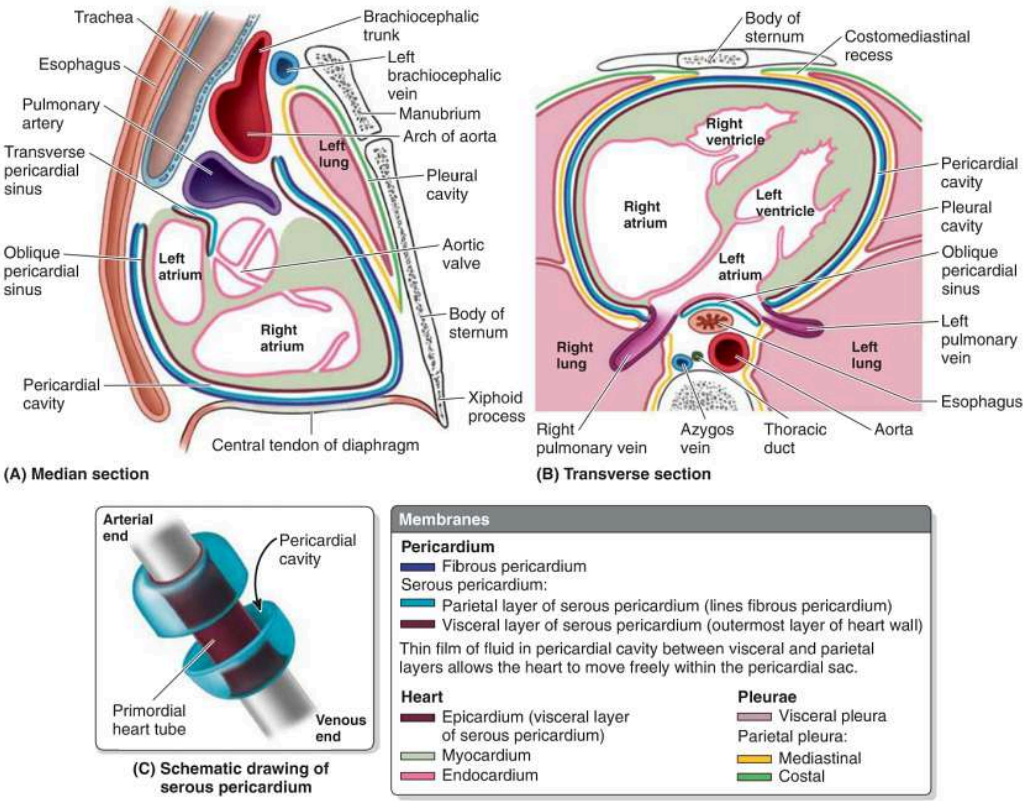
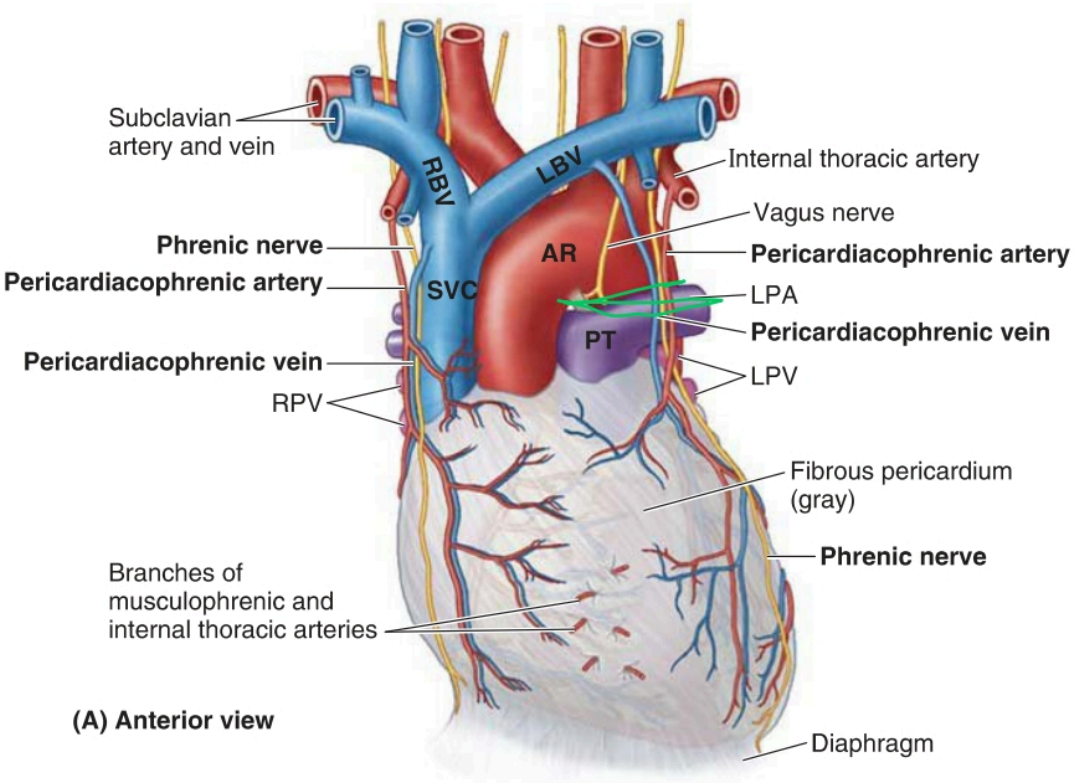


FIGURE 4.25. Layers of pericardium and pericardial cavity.


Subclavian artery and vein
RBV
LBV
Internal thoracic artery
Vagus nerve
Phrenic nerve
AR
Pericardiacophrenic artery
Pericardiacophrenic artery
SVC
LPA
Pericardiacophrenic vein
PT
Pericardiacophrenic vein
LPV
RPV
Fibrous pericardium (gray)
Phrenic nerve
Branches of musculophrenic and internal thoracic arteries
(A) Anterior view
Diaphragm

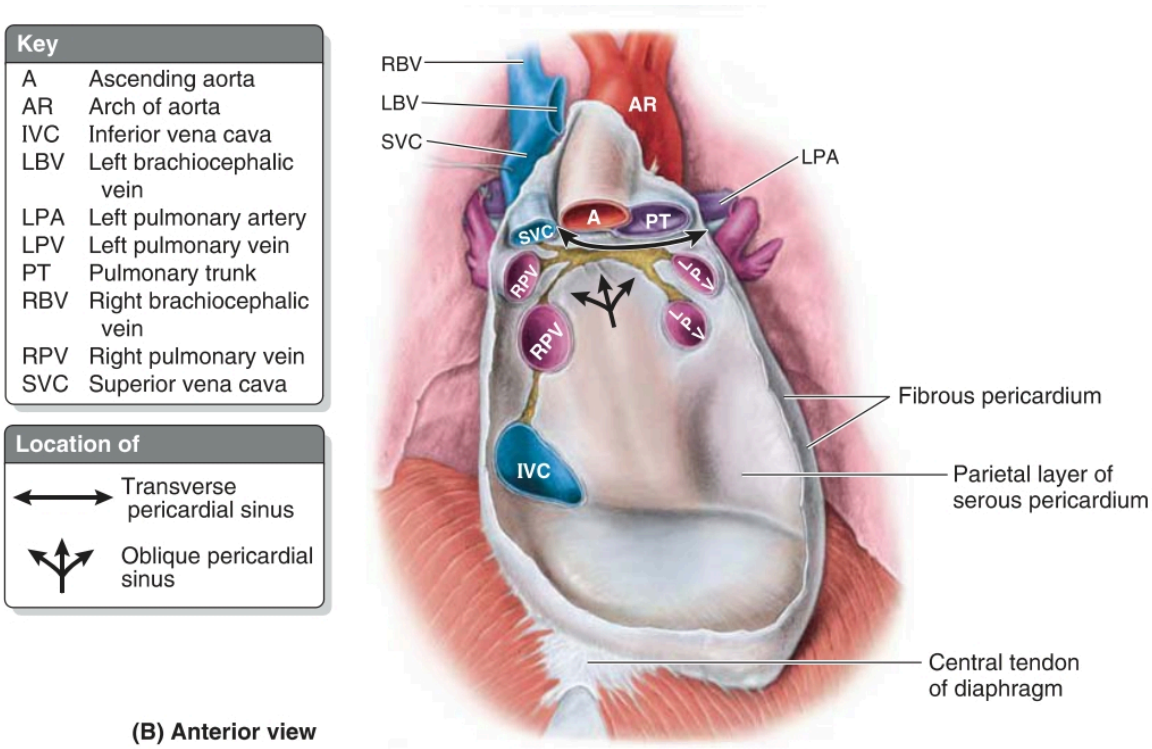


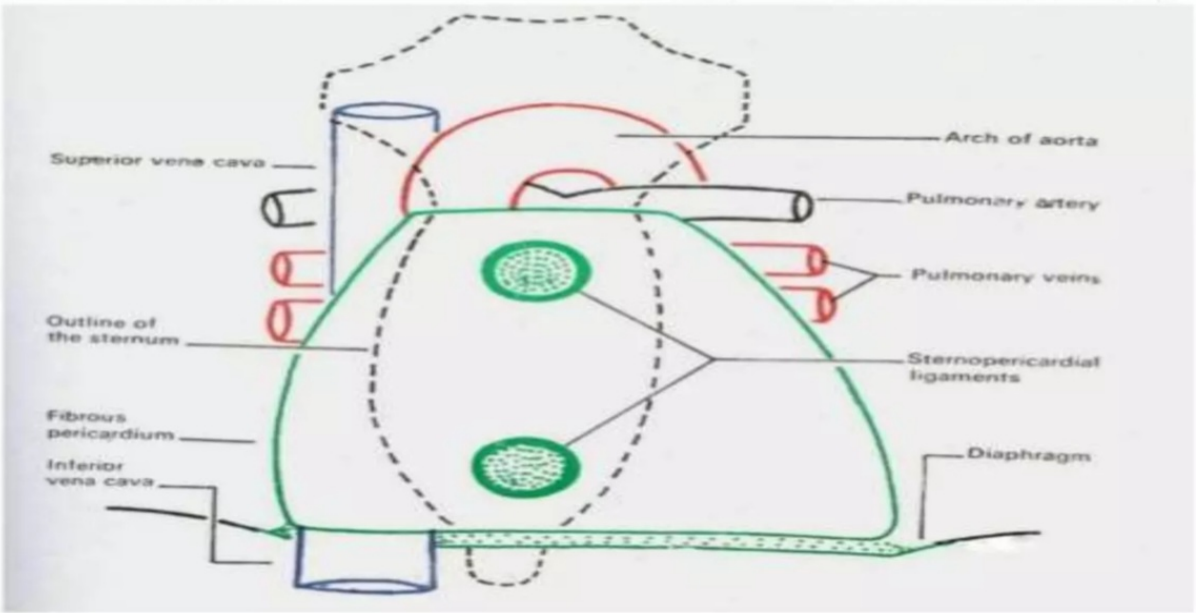
FIGURE 4.26. **Pericardium. A.** Arterial supply and venous drainage. **B.** Interior of pericardial sac, after removal of the heart, showing the location of transverse and oblique pericardial sinuses.

Fibrous pericardium

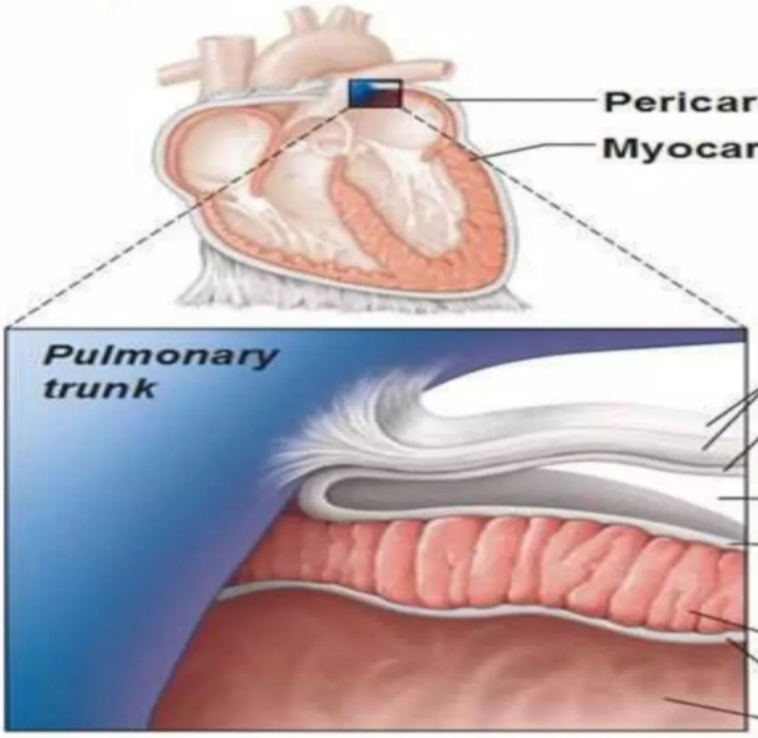
- Fibrous pericardium is a conical sac made up of fibrous tissue and is relatively non-distensible.
- Its rigid structure protects the heart against sudden overfilling and prevents over expansion of the heart.
- The parietal layer of serous pericardium is attached to its deep surface.
- The apex is blunt and fused with the root of the great vessels.
- The base is broad and blended with central tendon of diaphragm.
- Anteriorly it is connected to body of sternum by sternopericardial ligaments.

- Posteriorly attached with principal bronchi, esophagus and descending thoracic aorta
- On each side, it is related to the Mediastinal pleura, the Mediastinal surface of the lung.
- Aorta, SVC, Pulmonary arteries & Pulmonary veins receive fibrous prolongations from fibrous layer.
- As IVC enters the pericardium through the central tendon of diaphragm So receives no covering from fibrous layer.

Fibrous Pericardium - relations

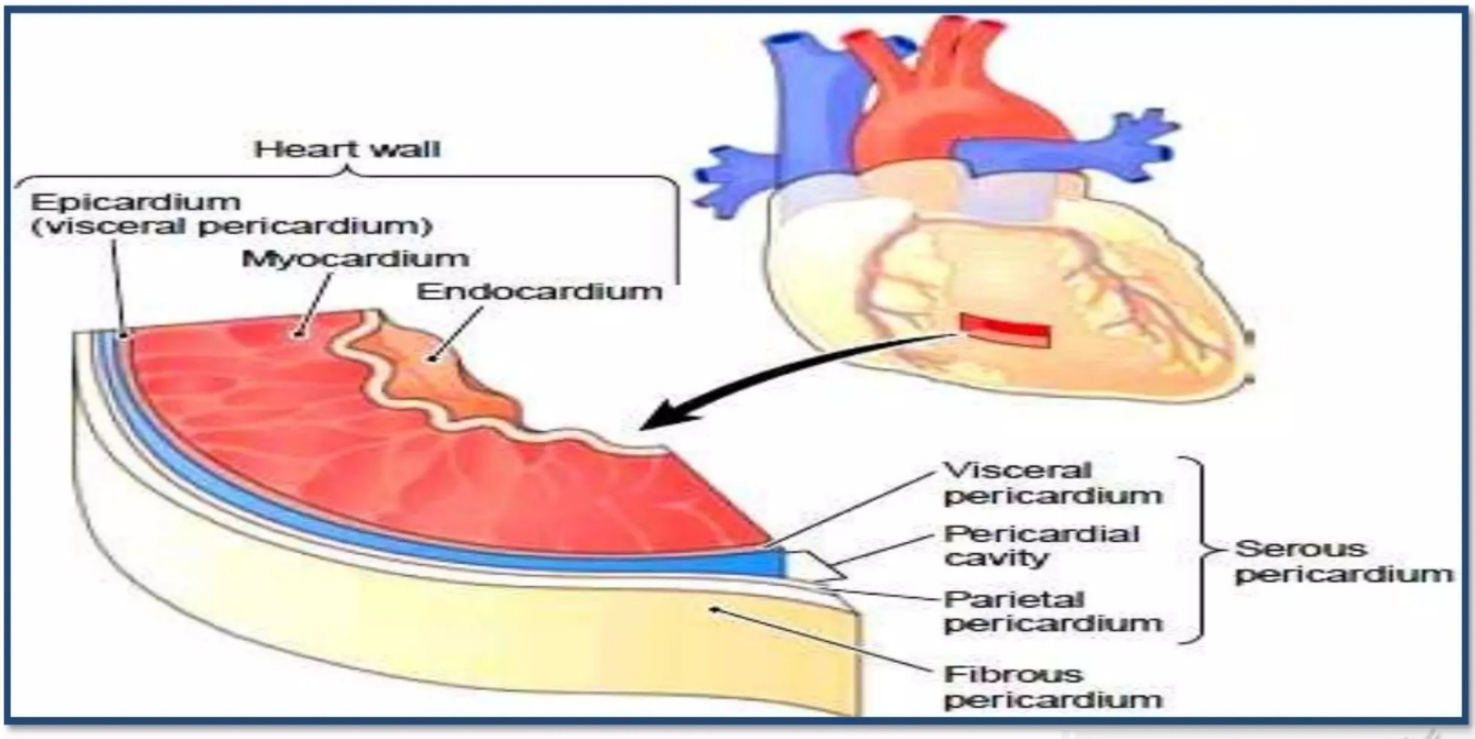


Layers of the Pericardium and of the Heart Wall



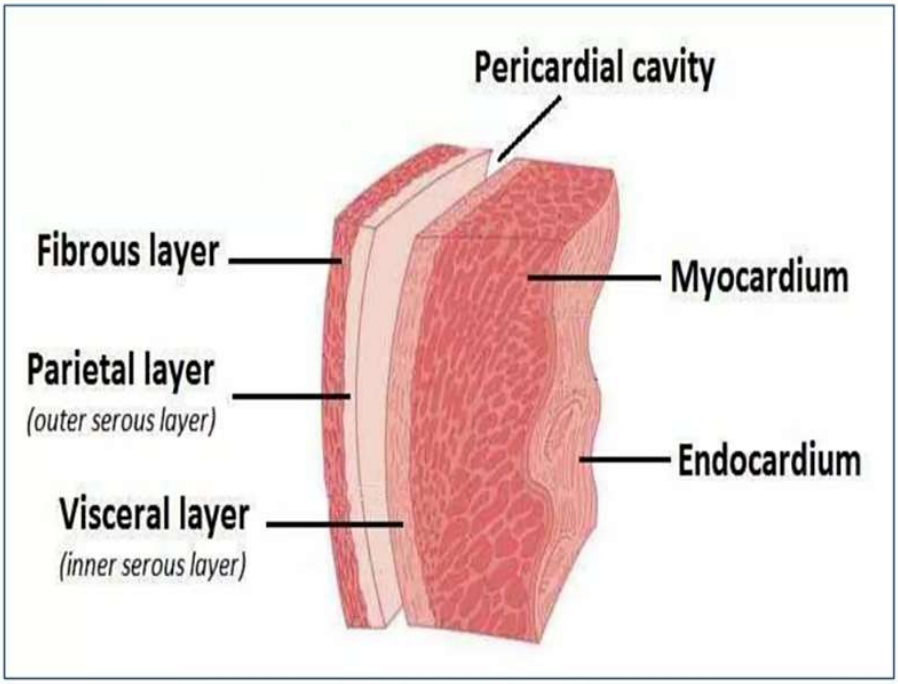
Serous pericardium

- Serous pericardium is thin, double-layered serous membrane lined by mesothelium. It is made up of parietal serous layer and visceral serous layer.
- The outer layer or parietal pericardium is fused with fibrous pericardium.
- The inner layer or the visceral pericardium, or epicardium is fused to the heart.

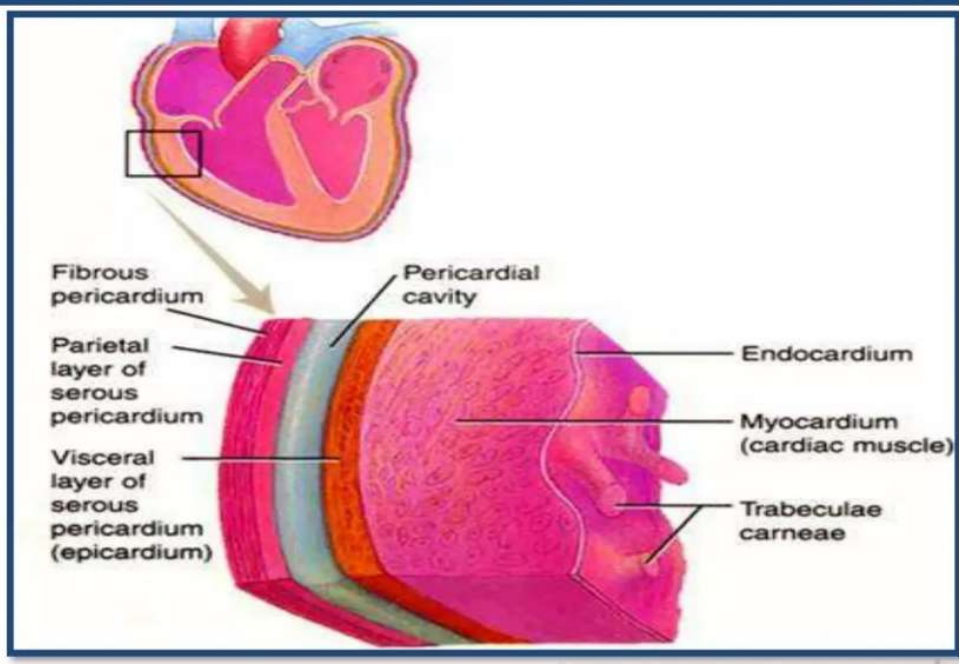

Heart wall
Epicardium
(visceral pericardium)
Myocardium
Endocardium
Visceral
pericardium
Pericardial
cavity
Parietal
pericardium
Fibrous
pericardium
Serous
pericardium

Pericardial cavity

- The pericardial cavity is a potential space between the parietal pericardium and the visceral pericardium.
- Pericardial cavity contains only a thin film of serous fluid known as pericardial fluid.

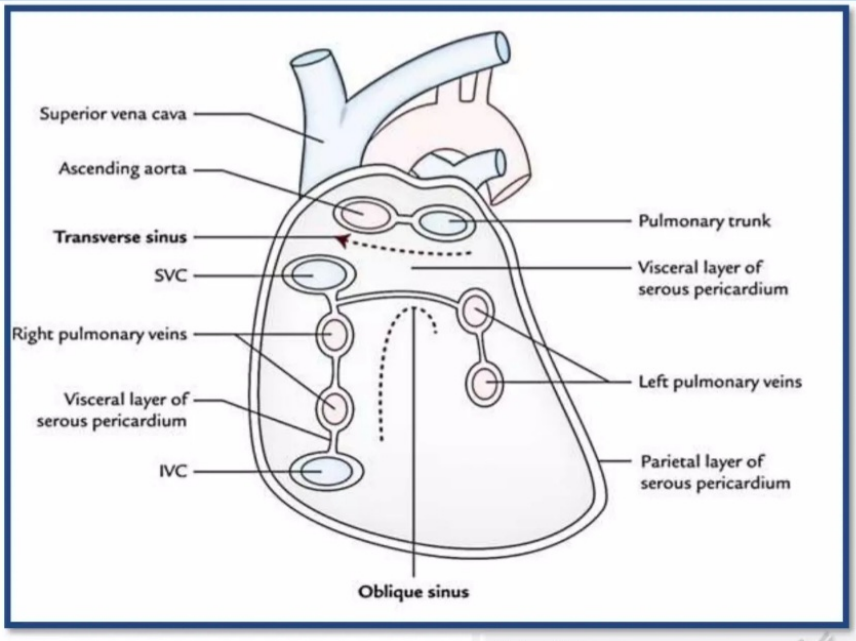


▪The pericardial fluid reduces friction within the pericardium by lubricating the epicardial surface allowing the membranes to glide over each other with each heart beat.



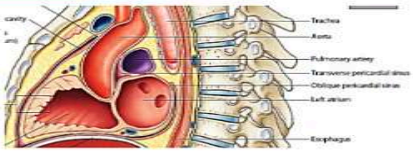
Pericardial sinuses

- The Sinuses of Pericardium lie between the parietal and visceral layers of serous pericardium and are 2 in number: Transverse sinus & oblique sinus.
- They’re created because of the reflection of visceral layer of serous pericardium around great vessels of the heart.



Pericardial sinuses

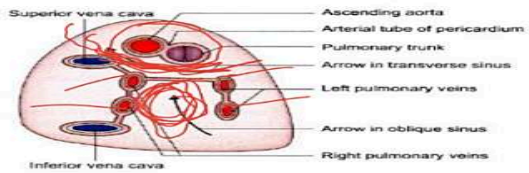
Sinuses of the Serous Pericardium

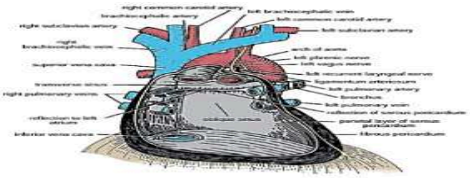


***Definition:** they are recesses inside the pericardial cavity (lined by serous pericardium). They include the transverse sinus & the oblique sinus.

***Introduction**: the epicardium at the roots of the great vessels is arranged to form 2 tubes

(a) ant. tube surrounding the arterial end of the heart (aorta & pulm. trunk) (b) post. tube enclosing the venous end of the heart (venae cavae & pulm. veins). The passage between the 2 tubes is known as the **transverse sinus** while the recess behind the venous end is known as the **oblique sinus** of pericardium.



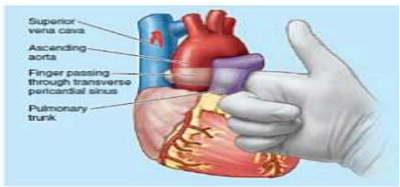


(A) Transverse Sinus of Pericardium:

- it is a transverse passage between the arterial & venous ends of the heart tube & connecting the Rt. & Lt. parts of the pericardial cavity

***Boundaries**: **-infront**: ascending aorta & pulmonary trunk
-behind & above: the Rt. pulmonary artery.
-behind & below: the S.V.C. & 2 atria (especially the Lt.)

- **to demonstrate the transverse sinus**: put a finger infront of the lower part of the S.V.C. & push it to the left behind the ascending aorta & pulmonary trunk. The finger traverses the sinus & appears at its left end between the pulmonary trunk & the Lt. auricle.



(B) Oblique Sinus of the Pericardium:

- it is a blind recess of the pericardial cavity that lies behind the base of the heart.

***Boundaries:**

(a) Anteriorly: the post. wall of Lt. atrium (base of the heart).

(b) **Posteriorly**: the fibrous pericardium (lined by parietal layer of serous pericardium)

(c) **Above**: it is closed by the reflection of serous pericardium on the back of Lt. atrium

(d) **On the Rt. side**: reflection of serous pericardium on to the 2 Rt. pulmonary veins & the 2 venae cavae.

(e) **On the Lt. side**: reflection of the serous pericardium on to the 2 Lt. pulmonary veins

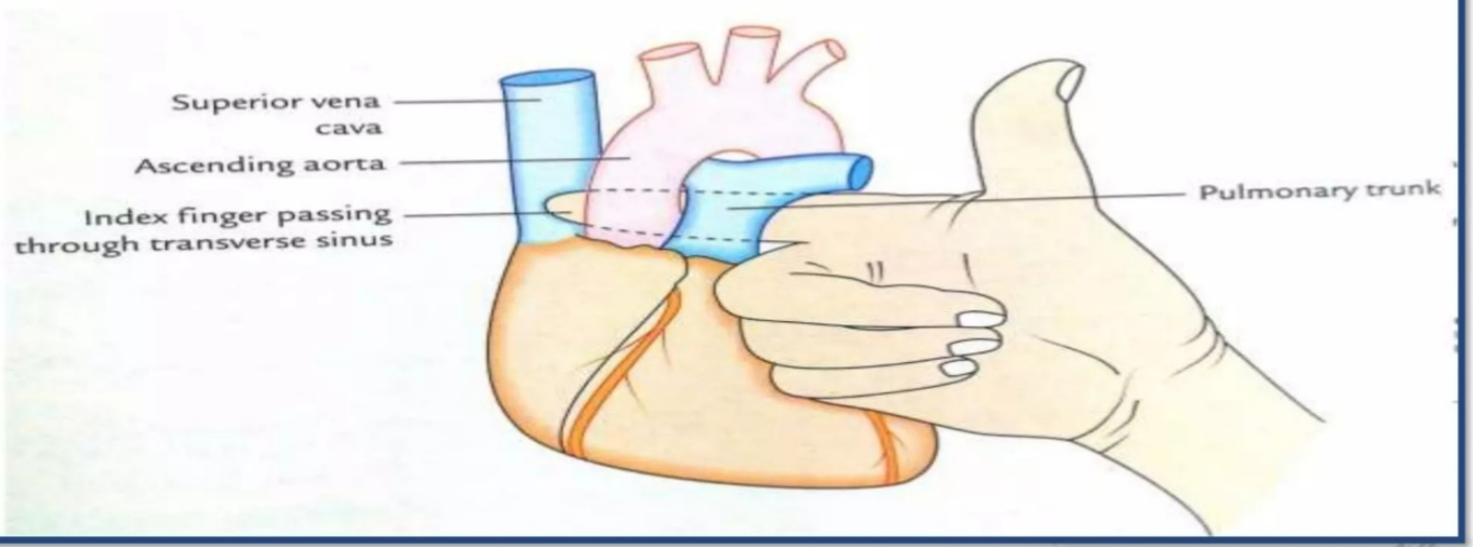
-**Below, the sinus is open** to the general pericardial cavity & its entrance is bounded by the I.V.C. (below & to the Rt.) & the lower Lt. pulmonary v. (above & to the Lt.).

-**Significance of the oblique sinus:** it acts as a bursa behind Lt. atrium permitting its pulsations.

Transverse pericardial sinuses

- Transverse sinus is the horizontal gap between the arterial ends of the heart tubes in front and venous end behind .
- transverse pericardial sinus separates the heart's arterial outflow (aorta, pulmonary trunk) from its venous inflow (superior vena cava, pulmonary veins).
- It is located: Posterior to the ascending aorta and pulmonary trunk .Anterior to the superior vena cava. Superior to the left atrium.
- During cardiac surgery, following the pericardial sac is opened anteriorly, a finger is gone through the transverse sinus of pericardium.
- A temporary ligature is gone through the transverse sinus around the aorta and pulmonary trunk.

Transverse Sinus

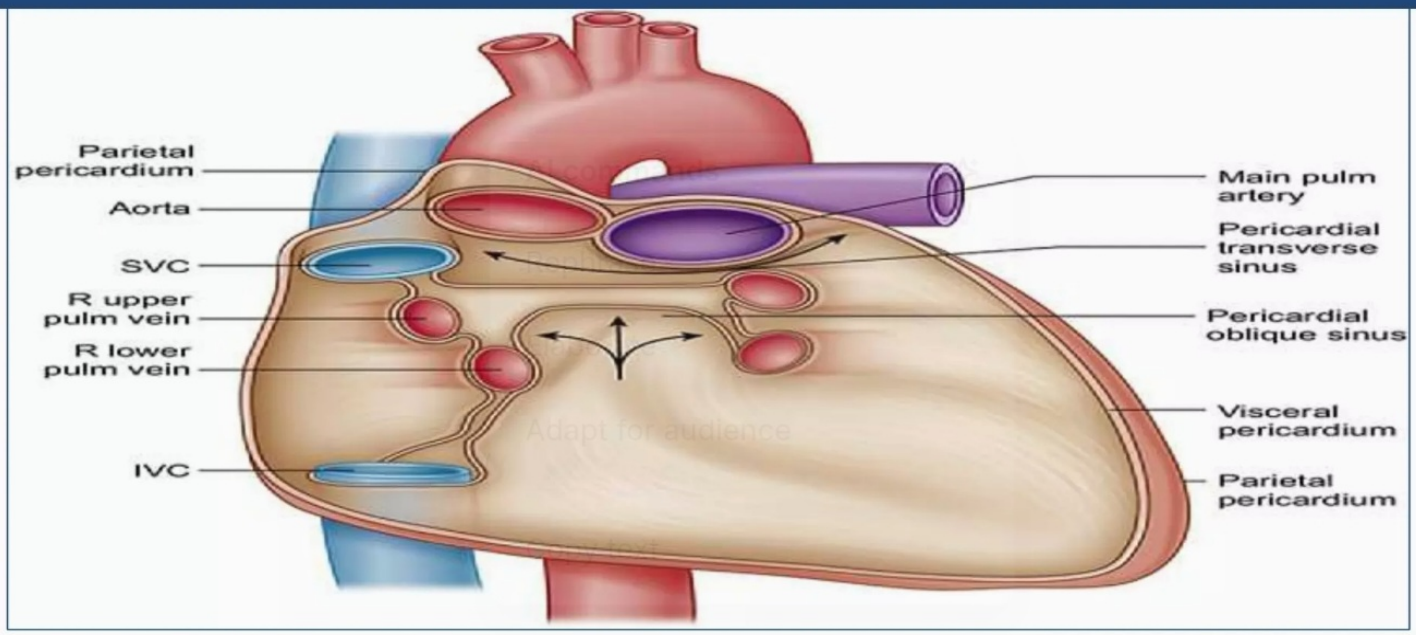


Transverse sinus

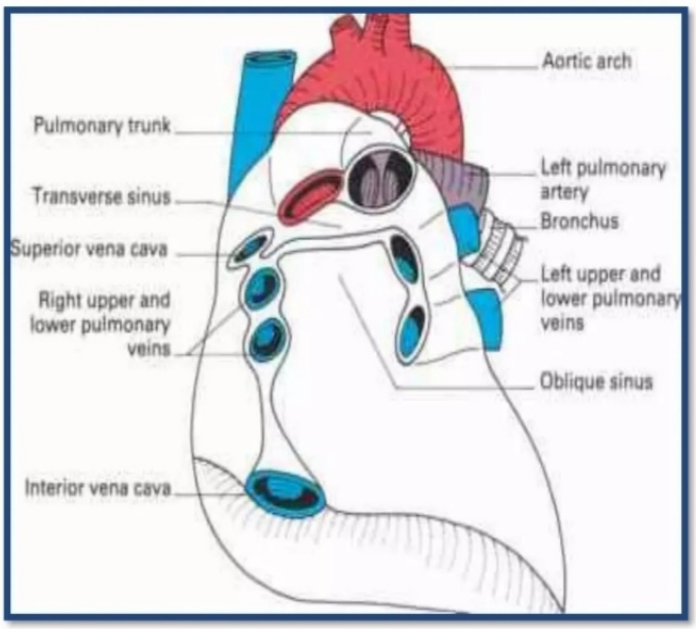
Oblique pericardial sinuses

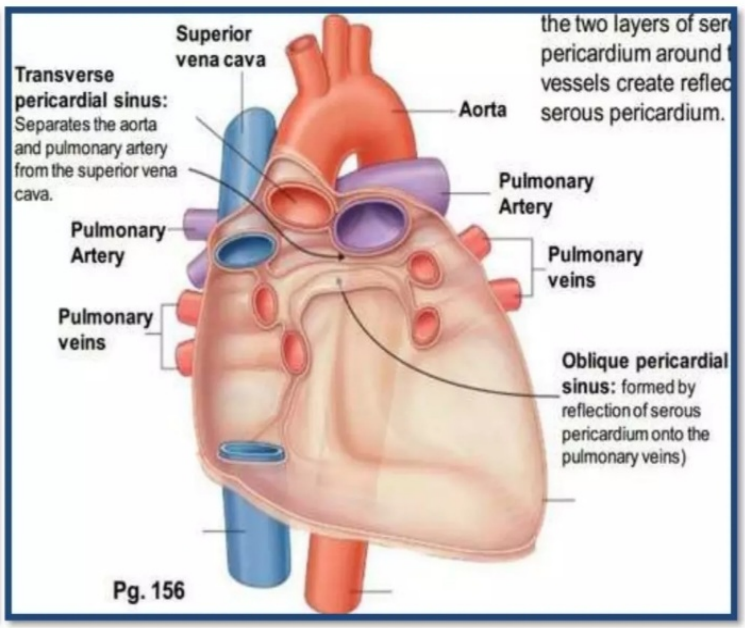
- The oblique sinus of the pericardial cavity is a blind ending passage posterior to the heart formed by the reflections of the visceral and parietal pericardium onto the vessels traversing the space.
- With the heart in its anatomical position and the dorsal pericardium intact, it forms an inverted U-shaped recess that can be probed to reveal the structures forming its walls:

Image source : Google



Pericardial sinuses



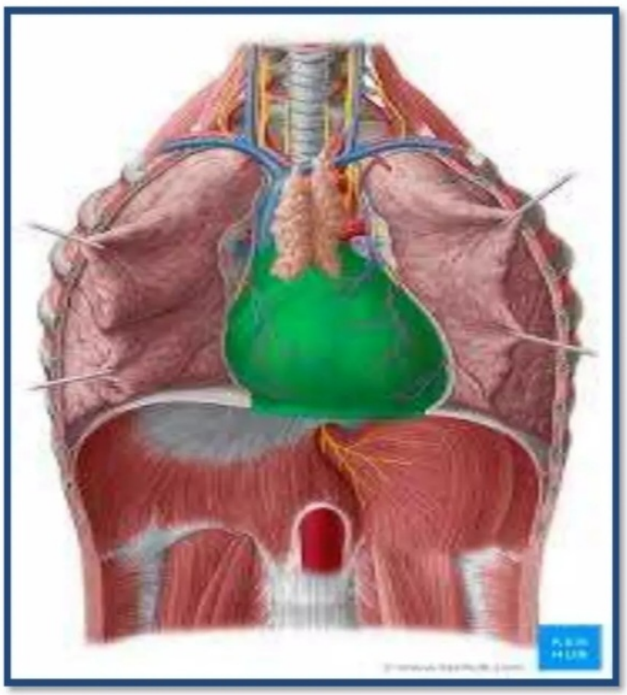


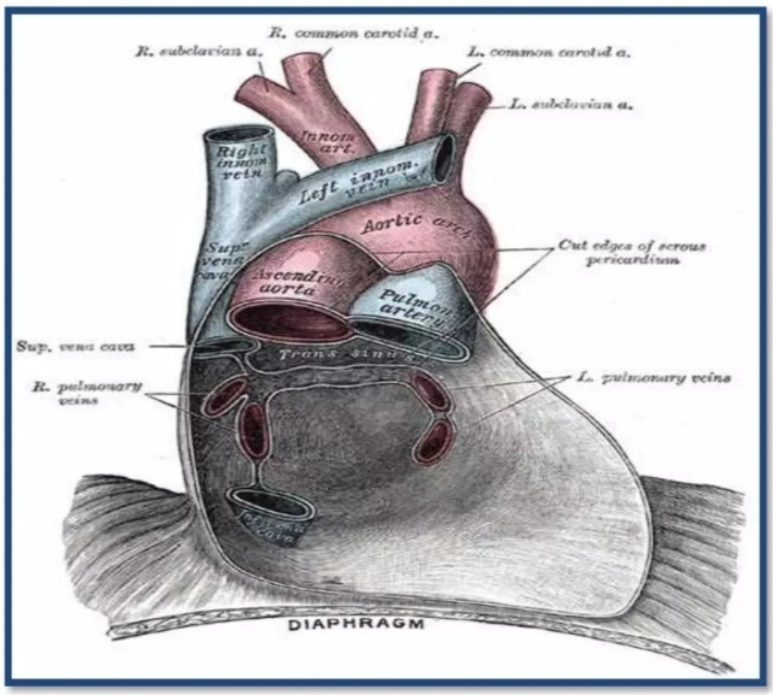
Pericardial sinuses

Content of the pericardium

- Heart with cardiac vessels and nerves
- Ascending aorta.
- Pulmonary trunk.
- Lower half of the superior vena cava.
- Terminal part of the inferior vena cava
- The terminal parts of the pulmonary veins

Content of the pericardium





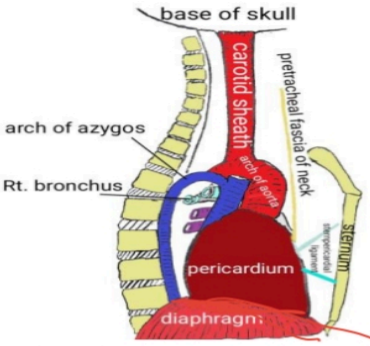
Functions of the pericardium

- Fixes the heart in the mediastinum and limits its motion.
- Prevents overfilling of the heart.
- Lubrication: A thin film of fluid between the two layers of the serous pericardium reduces the friction generated by the heart as it moves within the thoracic cavity
- Protection from infection. The fibrous pericardium serves as a physical barrier between the muscular body of the heart and adjacent organs prone to infection, such as the lungs.

Fixation of the pericardium & heart in position

"Cardiac Sling Mechanism"

The heart is kept in position inside the thorax by the following mechanisms:



(A) Fixation of the pericardium to the adjacent structures:

(1) To the sternum by the sterno-pericardial ligaments.

(2) To the base of the skull by the carotid sheath.

(3) To the deep fascia of neck by the pretracheal fascia.

(4) To the diaphragm below.

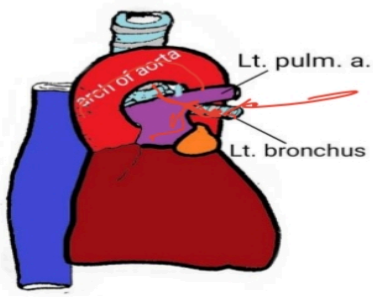
(B) Hooking of the great vessels of the heart around thoracic structures:

(1) The hooking of the arch of aorta & the Lt. pulmonary a. around the Lt. bronchus.
(2) The hooking of the arch of azygos v. around the Rt. bronchus before opening into the S.V.C.
(3) The arching of the subclavian vessels across The first rib of each side at the root of neck.

(C) The pleurae & lungs support the pericardium & the heart from the sides.

(D) The diaphragm supports the heart inferiorly.

***NB:** The cardiac sling mechanism induces its fixation effect in the upper post. part (base of the heart) leaving the lower ant. (ventricular) part free to contract accompanying inspiratory movements.*



Functions

1.**Stabilization of the heart** within the thoracic cavity by virtue of its ligamentous attachments -- limiting the heart's motion.

2. **Protection of the heart** from mechanical trauma and infection from adjoining structures.

3. The pericardial fluid functions as a **lubricant** and decreases friction of cardiac surface during systole and diastole.

4. **Prevention of excessive dilation** of heart especially during sudden rise in intra-cardiac volume (e.g. acute aortic or mitral regurgitation).

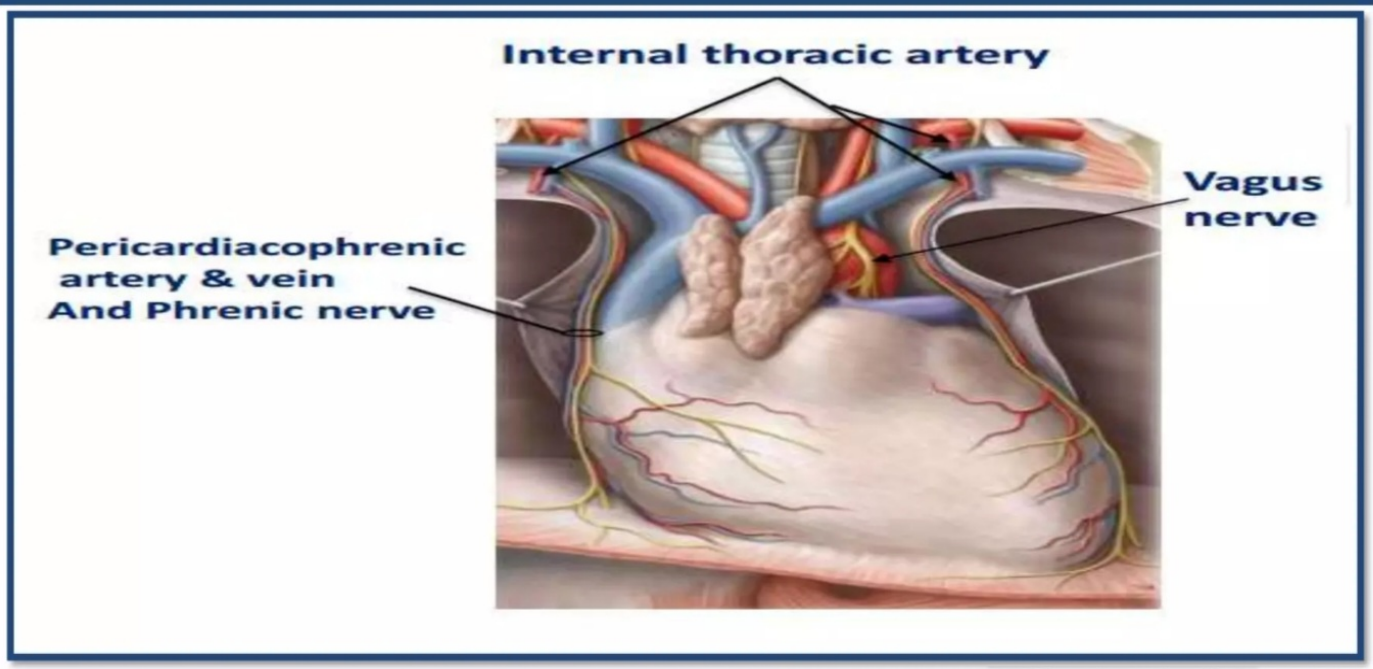
Blood supply of the pericardium

- Arterial supply:

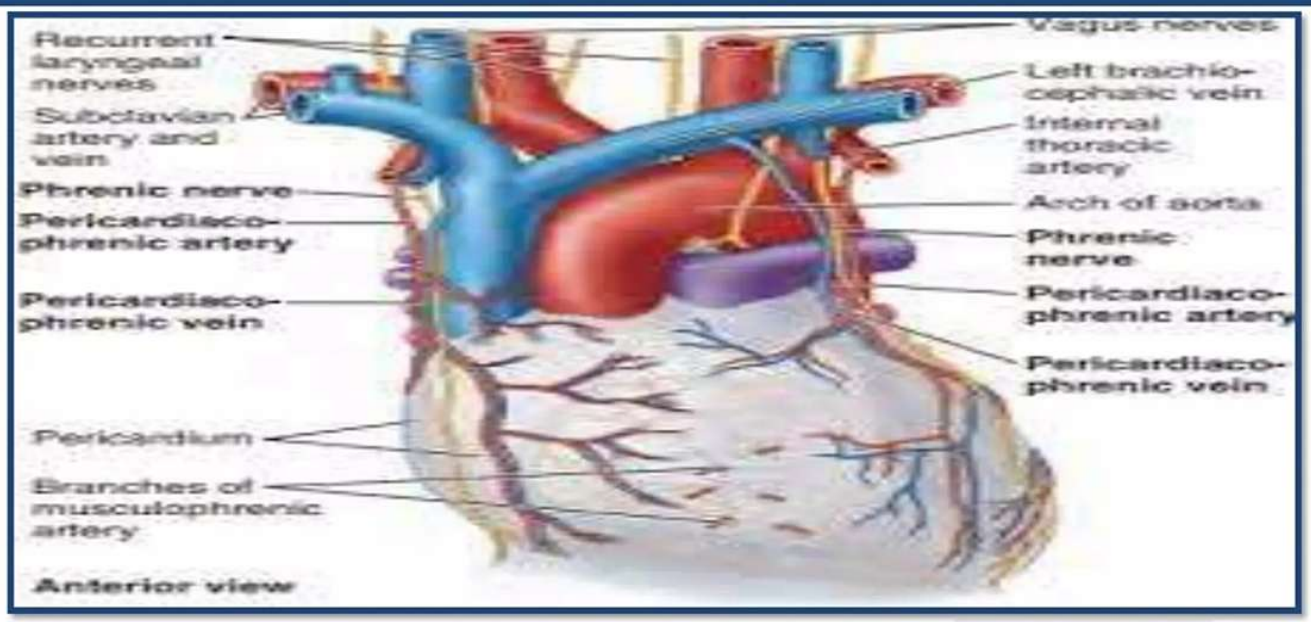
1. Pericardiophrenic artery
2. Musculophrenic artery .
3. Additional branches Bronchial ,Oesophageal , Superior phrenic arteries .



- Pericardiophrenic artery is main artery supplies the pericardium and diaphragm except the visceral pericardium.
- Pericardiophrenic artery & musculophrenic artery are branches of internal thoracic artery.
- Note: Coronary artery supplies only the visceral layer of serous pericardium(Epicardium).



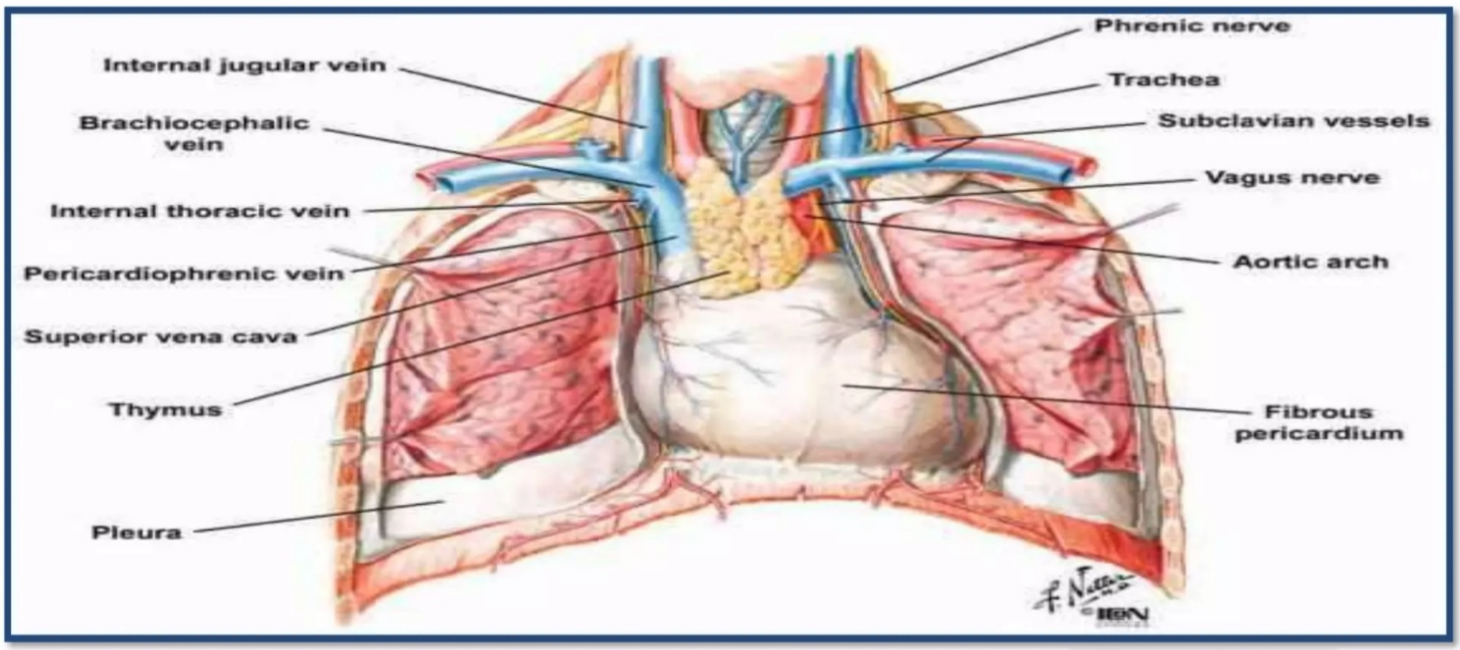
Blood supply of pericardium



Blood supply of pericardium

Venous drainage of the pericardium

- Deoxygenated blood is drained into the azygos venous system and the Pericardiophrenic veins.
- Azygos venous system and Pericardiophrenic veins are tributaries of the brachiocephalic vein.

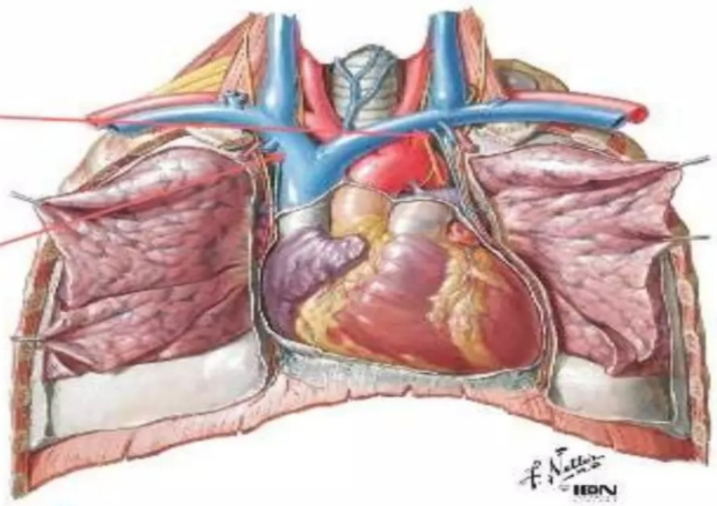


Venous drainage of pericardium

remedē® System Can Provide Unilateral Stimulation From Two Locations

2 stimulation locations

- Left Pericardiophrenic Vein
- Right Brachiocephalic Vein



39

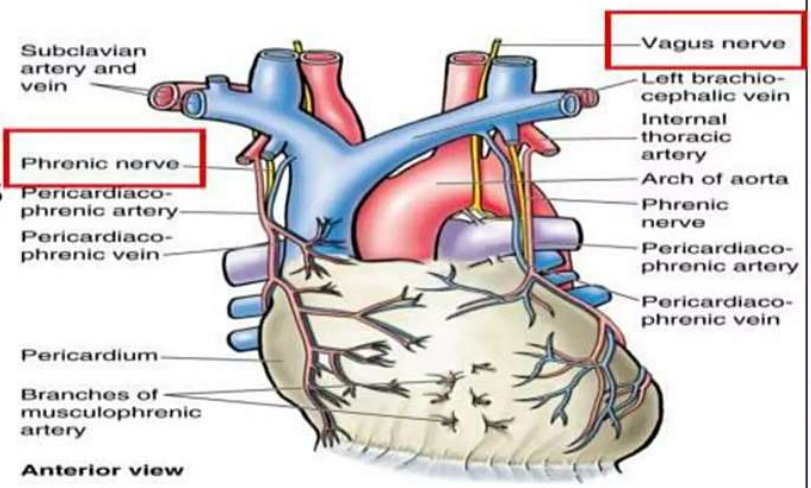
Venous drainage of pericardium

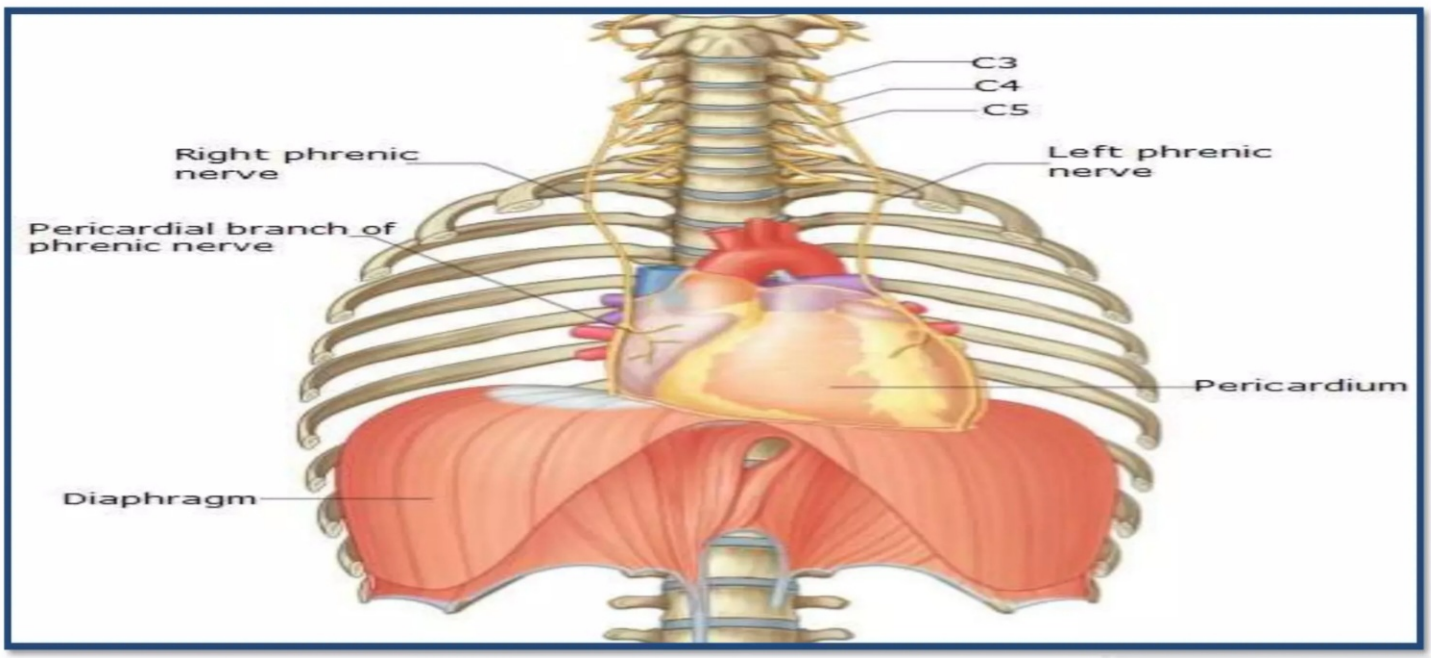
Nerve supply of the pericardium

- A. Phrenic nerve(C3-C5) provides somatic afferent innervation. It innervates fibrous and parietal layer of serous pericardium. It is Sensitive to pain. Phrenic nerve is a common source of referred pain. Thus shoulder pain is experienced as a result of pericarditis.
- B. The sympathetic trunk provides postganglionic vasomotor fibres. The Visceral layer of Serous pericardium is innervated by branches of sympathetic trunk
- C. vagus nerve (ANS): Visceral layer (Epicardium) is pain insensitive.

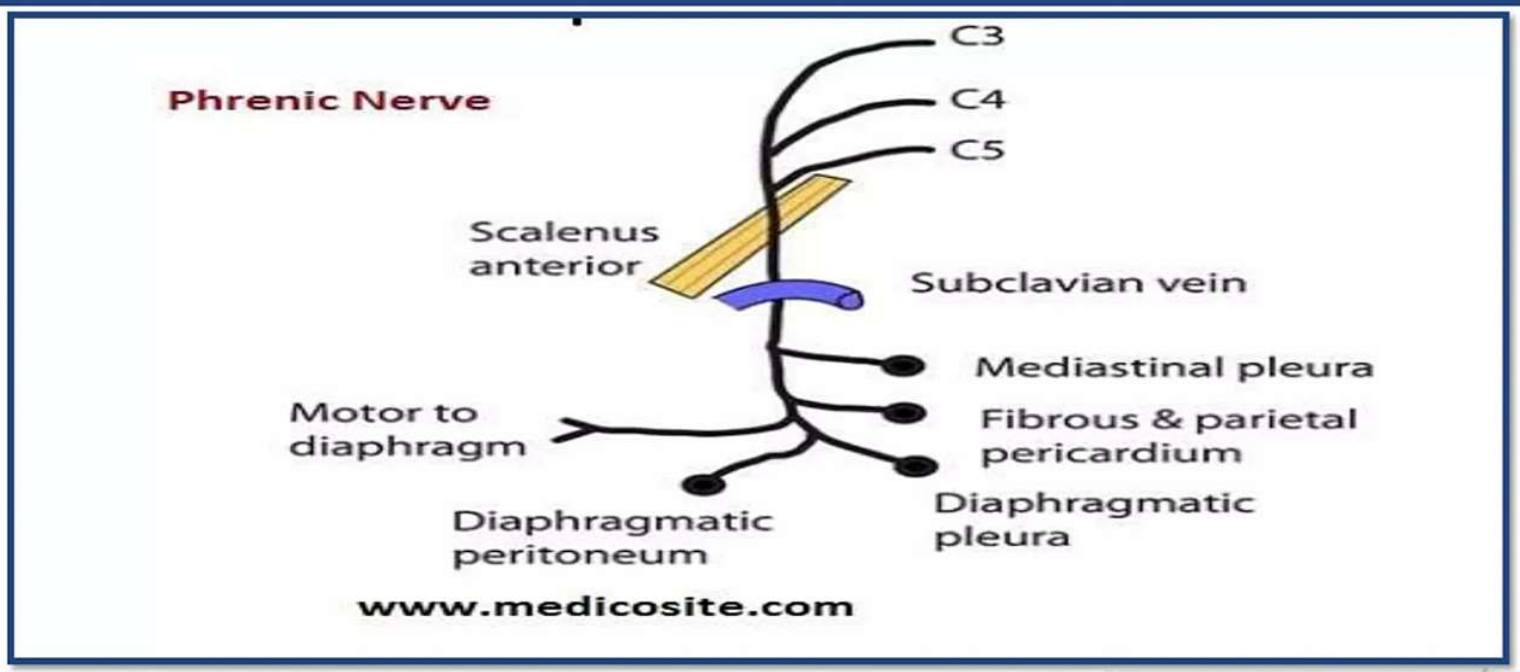
Nerve supply of the pericardium

1. **Phrenic nerves** (C3 - C5), primary source of sensory fibers; pain sensations conveyed by these nerves are commonly referred to the skin (C3 - C5 dermatomes) of the ipsilateral supraclavicular region (top of the shoulder of the same side).
2. Vagus nerves, function uncertain.
3. Sympathetic trunks, vasomotor





Phrenic nerve



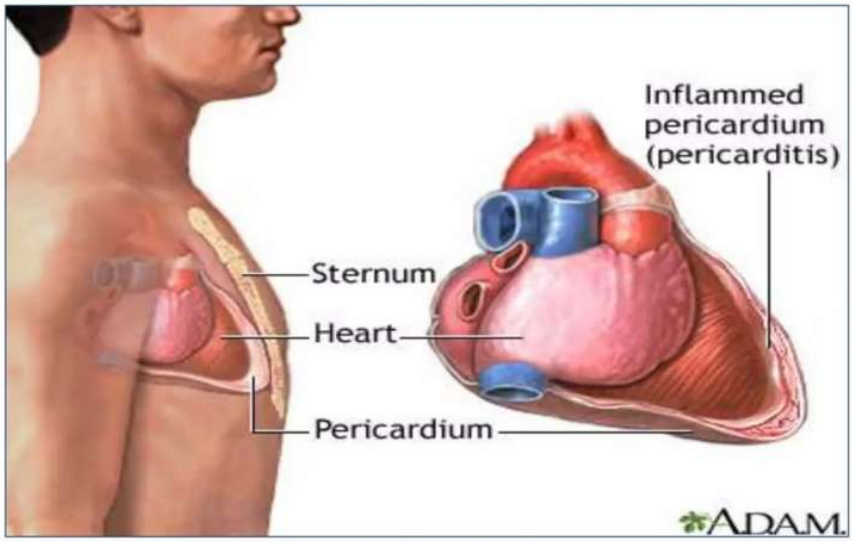
Phrenic nerve

Clinical relevance of the pericardium

- Pericarditis
- Pericardial Effusion
- Cardiac Tamponade
- Pericardiocentesis

Pericarditis

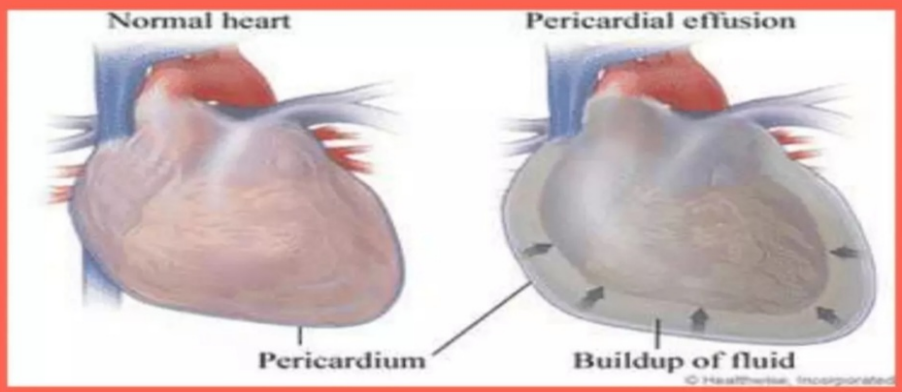
- Pericarditis is an Inflammation of the lining around the heart(the pericardium).
- Pericardium causes chest pain and accumulation of fluid around the heart .



Pericarditis

Pericardial effusion

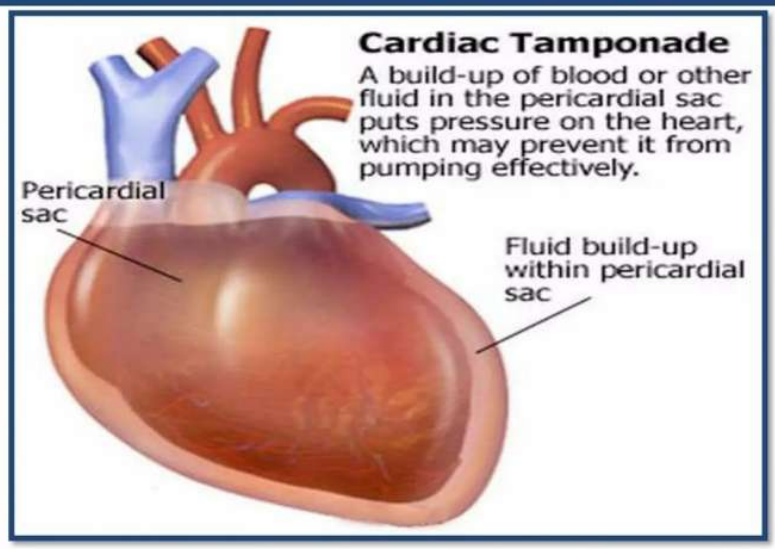
- Pericardial effusion(fluid around the heart) is an abnormal accumulation of the fluid , which is between the heart and pericardial cavity .



Pericardial effusion

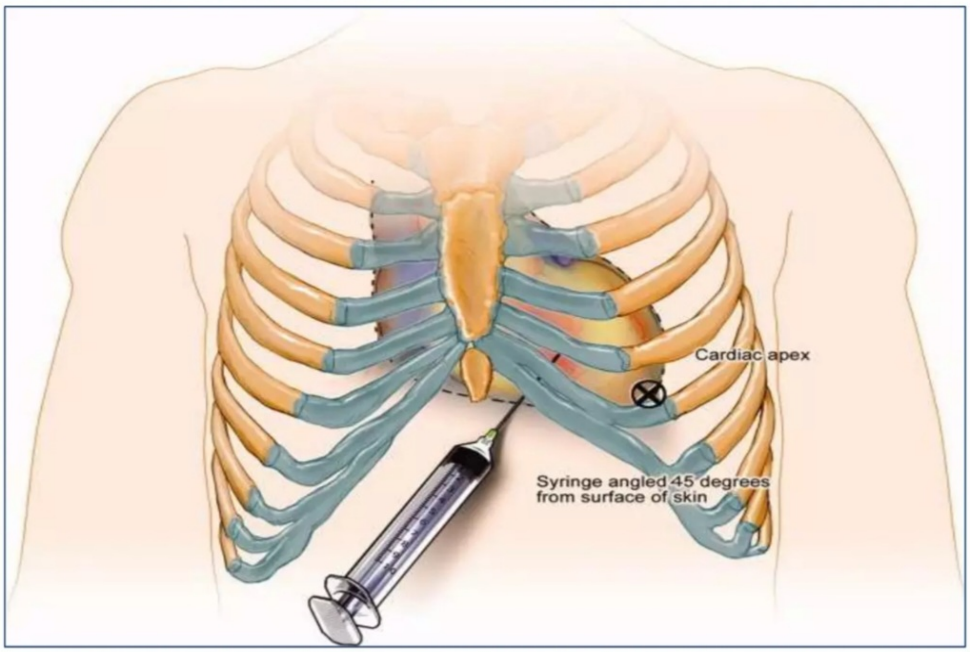
Cardiac tamponade

- Cardiac tamponade is a clinical syndrome caused by the accumulation of fluid in the pericardial space.
- It results in reduced ventricular filling .
- It is a medical emergency

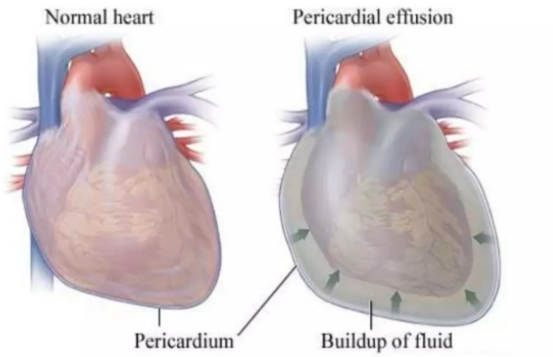


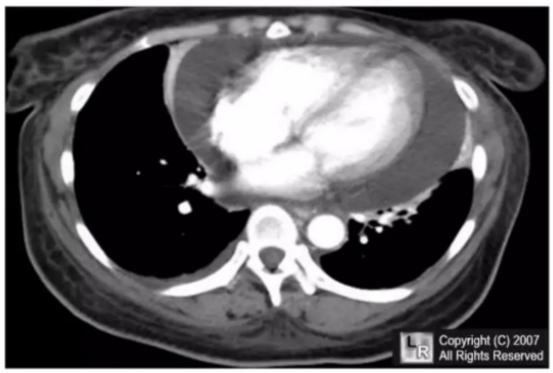
Pericardiocentesis

- Pericardiocentesis, also called a pericardial tap.
- It is a procedure in which a needle and catheter remove fluid from the pericardium, the sac around heart.



Pericardiocentesis


Normal heart
Pericardial effusion
Pericardium
Buildup of fluid


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Grooves

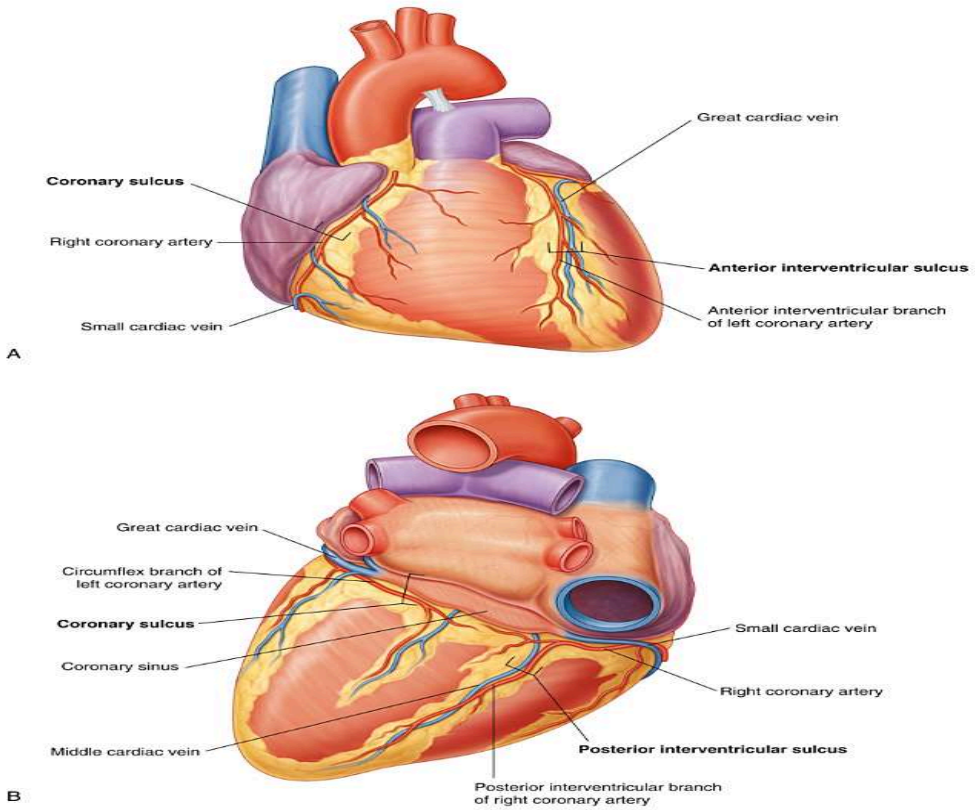
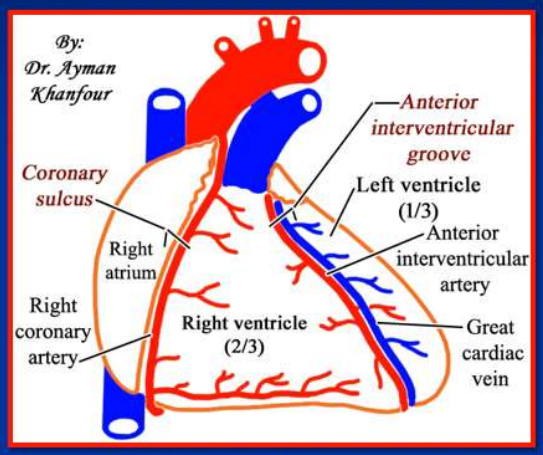
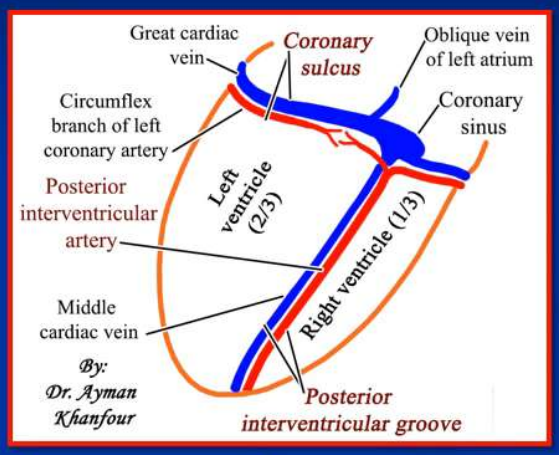


Fig. 3.70 Sulci of the Heart. **(A)** Anterior surface of the heart. **(B)** Diaphragmatic surface and base of the heart.

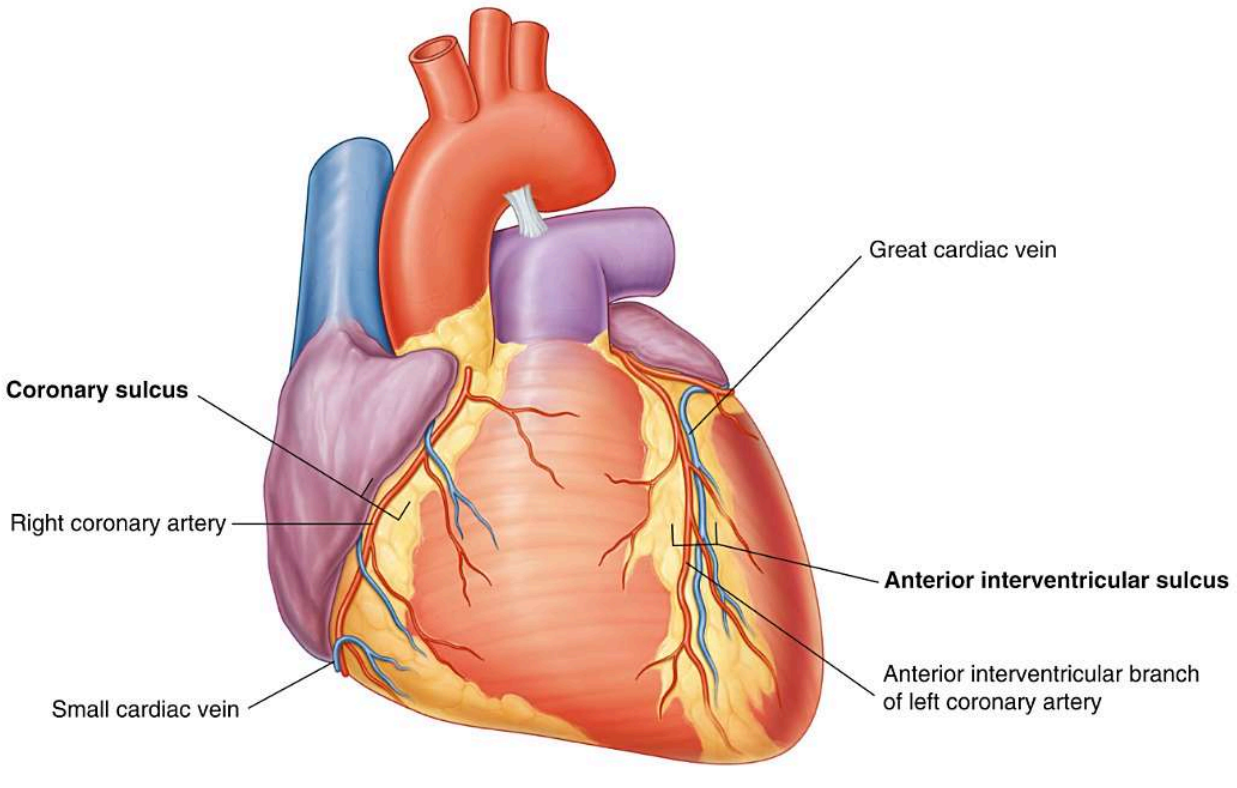
Grooves

I. Atrio-ventricular Groove (coronary sulcus)



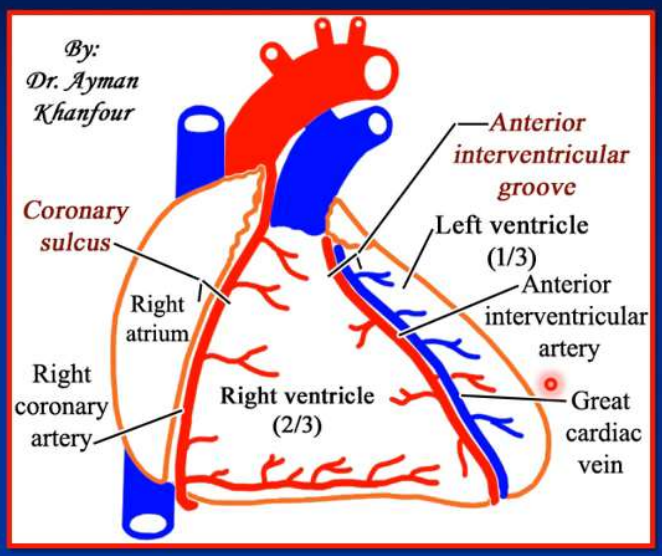


- **Site:** Between the *two atria* and the *two ventricles* (circular groove).
- **Contents:**
  a. The main trunks of the right and left coronary arteries.
  b. Circumflex branch of the left coronary artery.
  c. Coronary sinus.


Great cardiac vein
Coronary sulcus
Right coronary artery
Anterior interventricular sulcus
Anterior interventricular branch
of left coronary artery
Small cardiac vein

A

II. Anterior Inter-ventricular Groove



- **Site: Between the *two ventricles* on the sternocostal surface.**
- **Contents:**
  - **a. Anterior interventricular artery.**
  - **b. Great cardiac vein.**

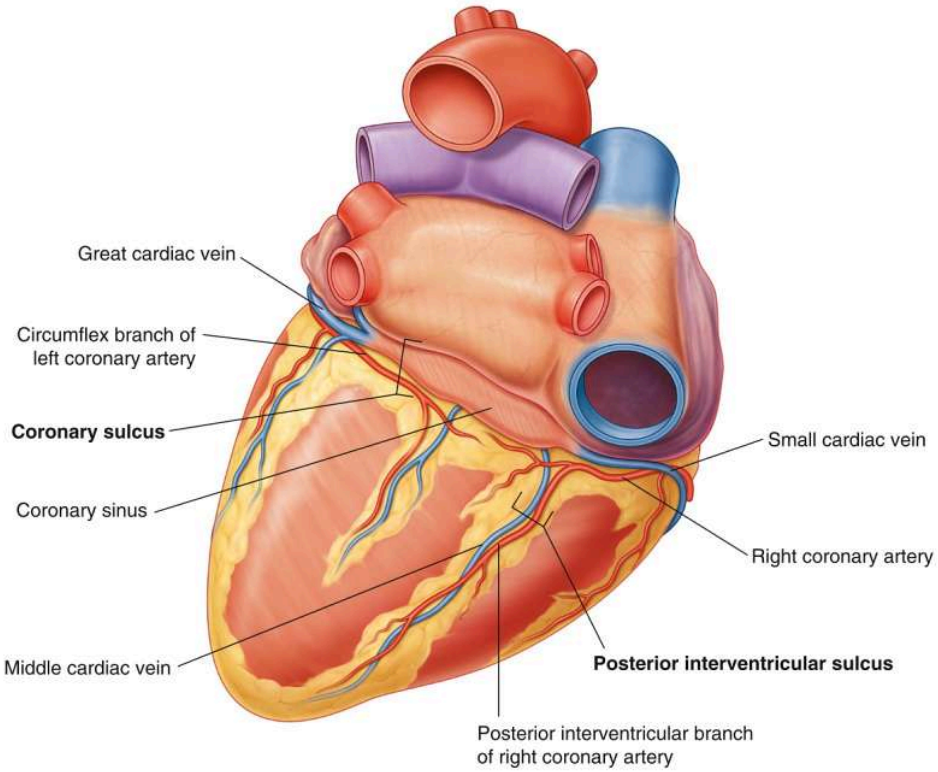
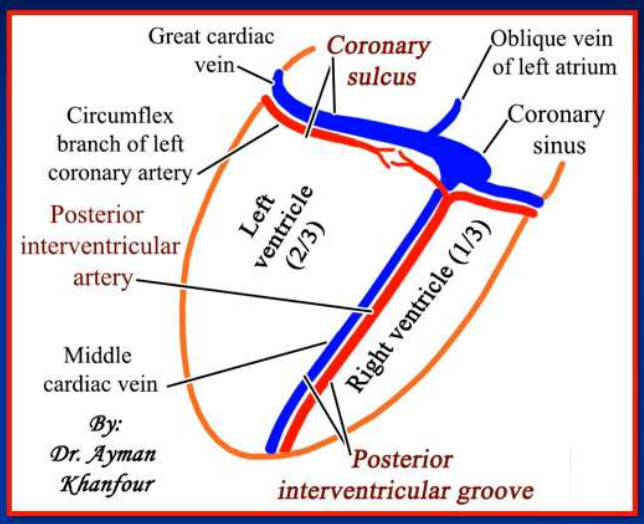


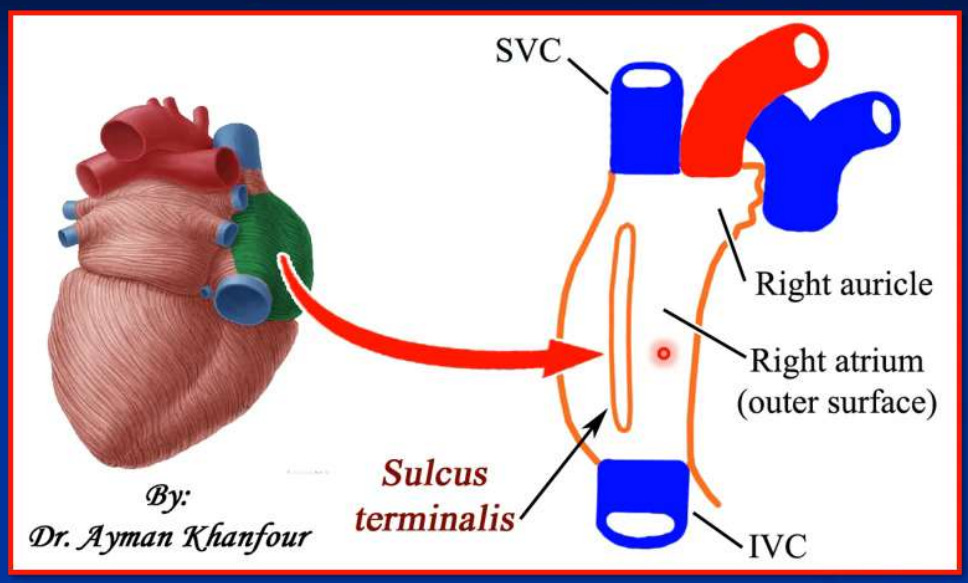
Fig. 3.70 Sulci of the Heart. (**A**) Anterior surface of the heart. (**B**) Diaphragmatic surface and base of the heart.

III. Posterior Inter-ventricular Groove

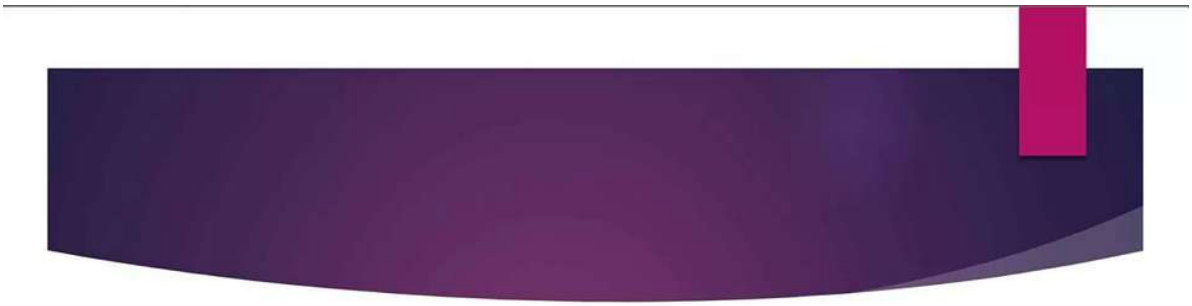


- **Site: Between the *two ventricles* on the inferior surface.**
- **Contents:**
  - **a. Posterior interventricular artery.**
  - **b. Middle cardiac vein.**

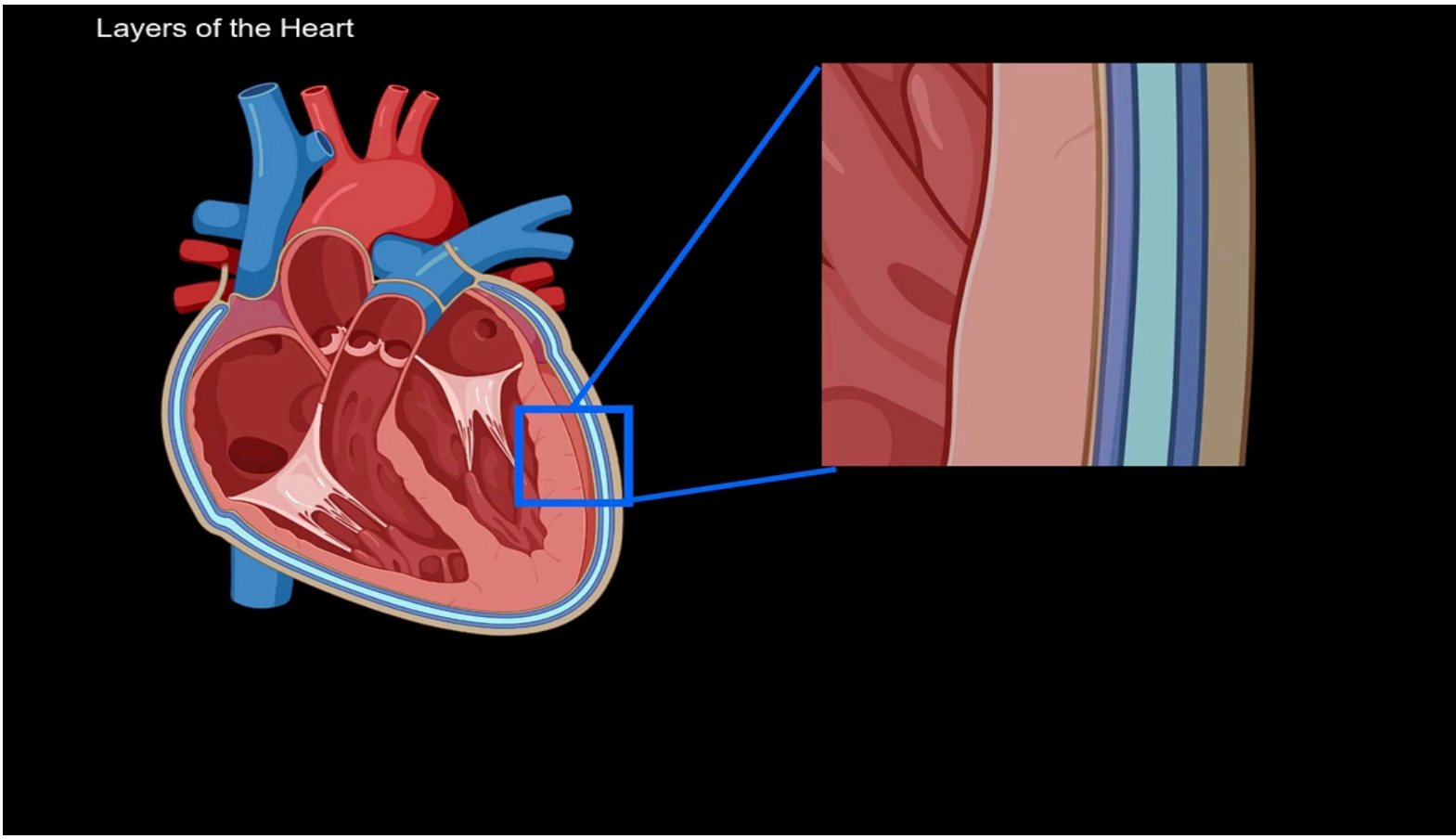
IV. Sulcus terminalis

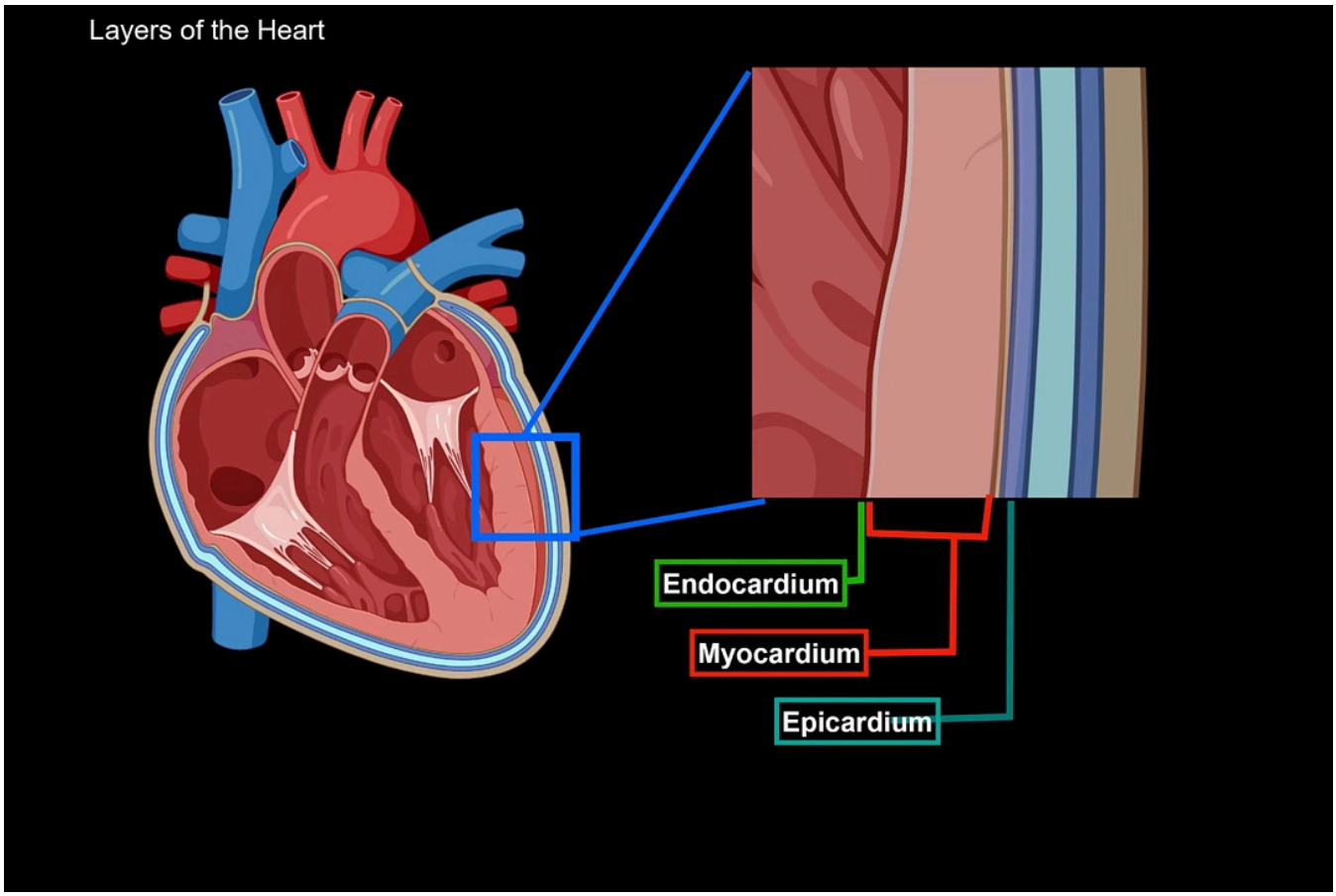


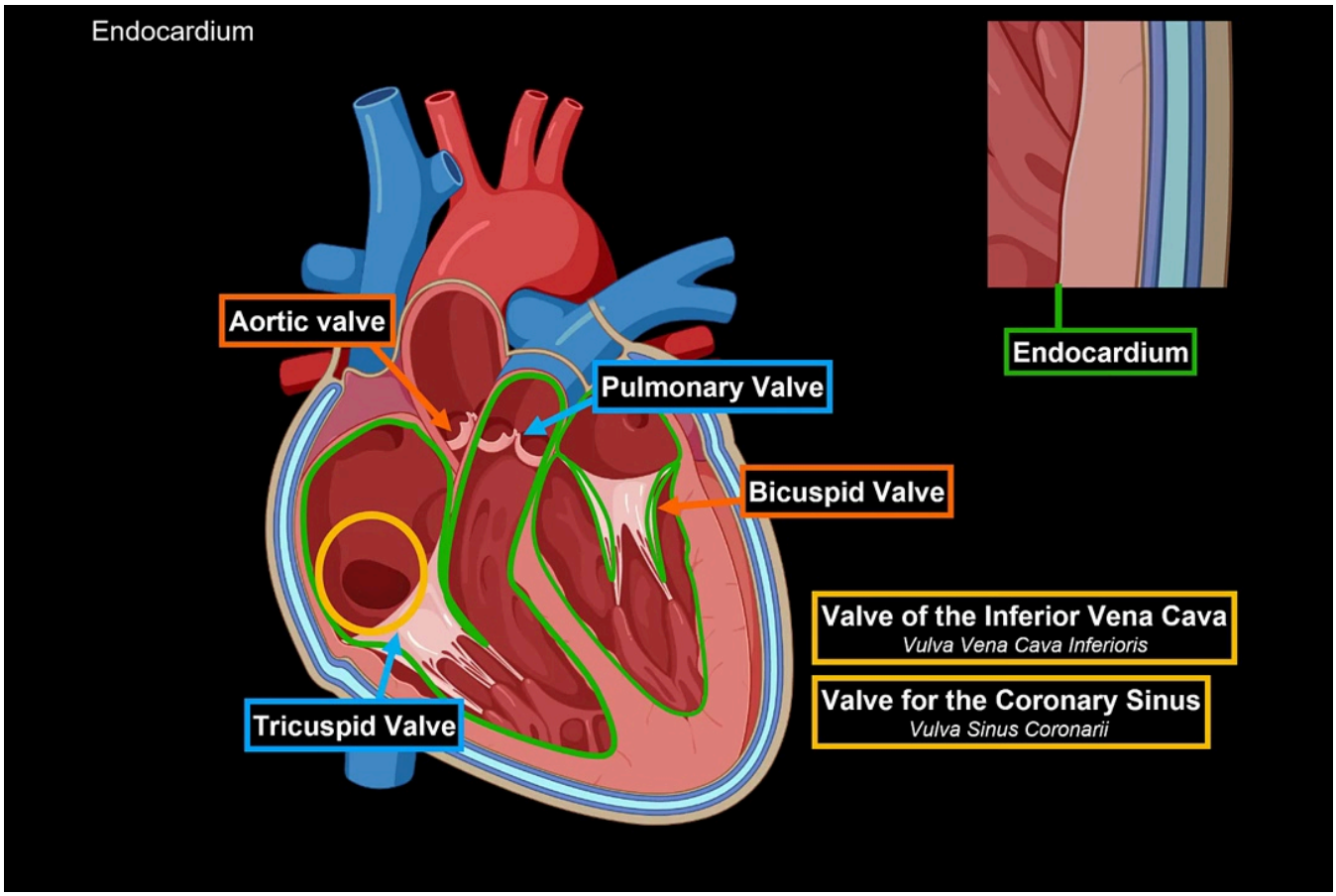
- It is present on the outer surface of the right atrium between the SVC and IVC
- It is represented on the inner surface by the crista terminalis.

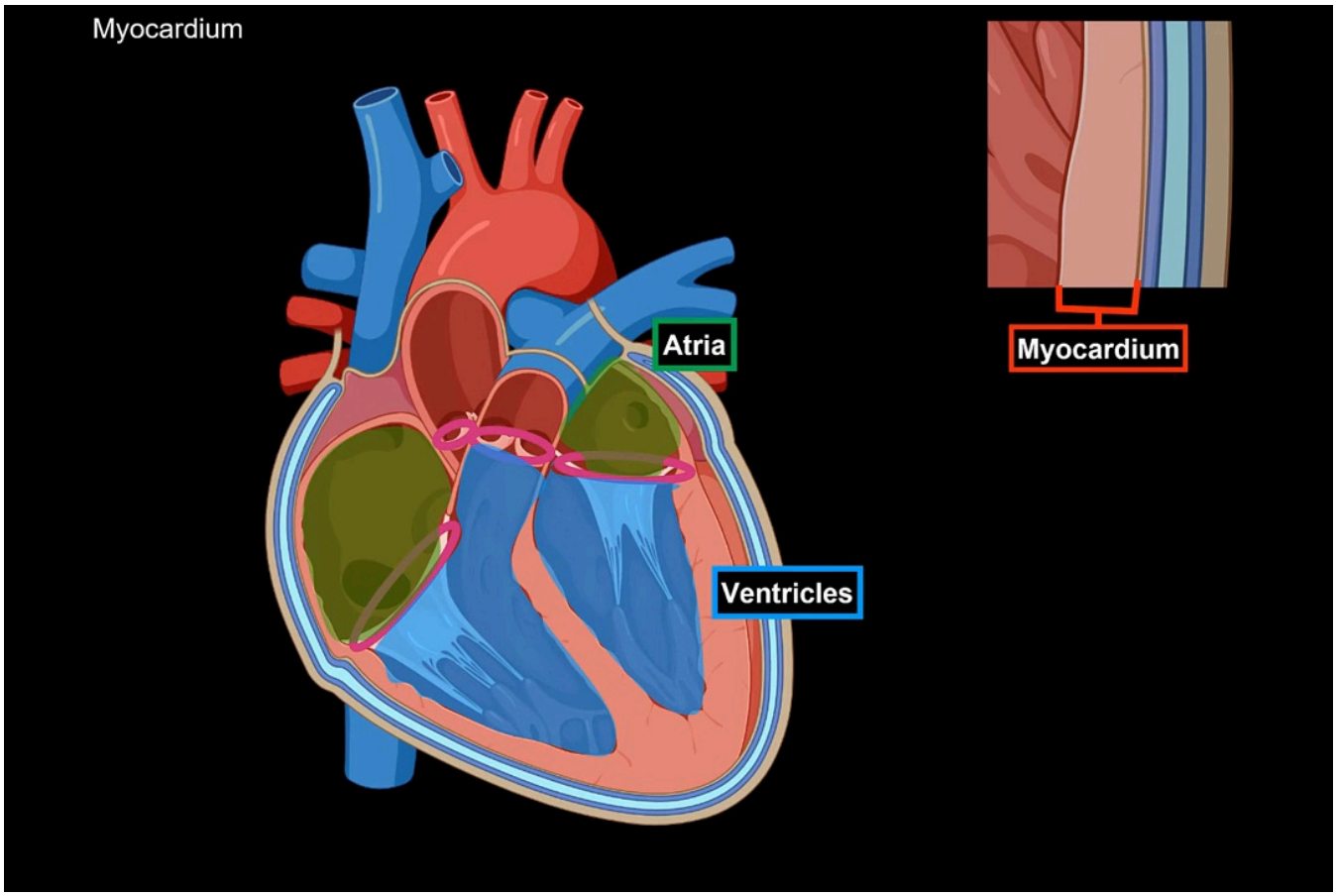


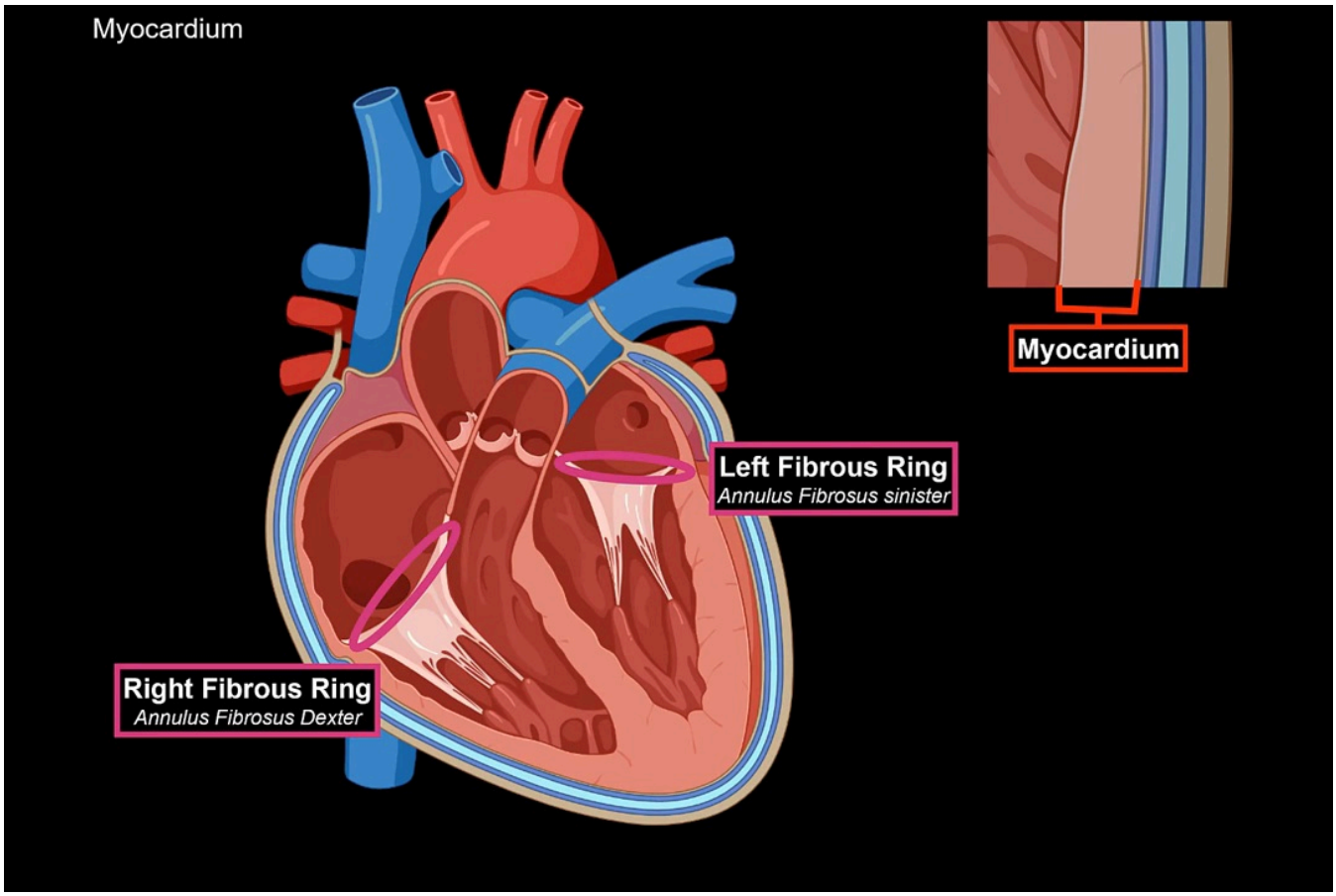
THANKS

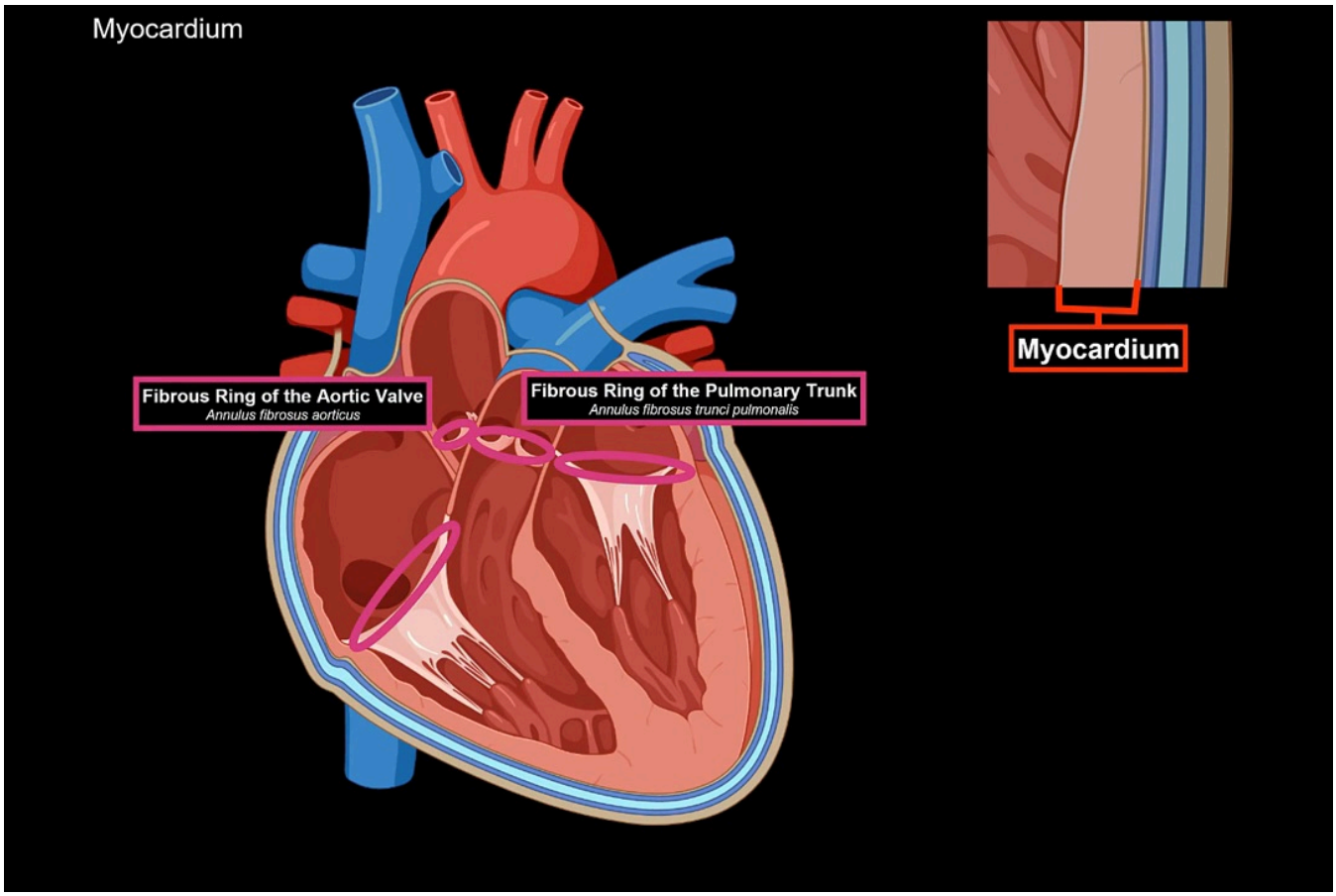

Layers of the Heart

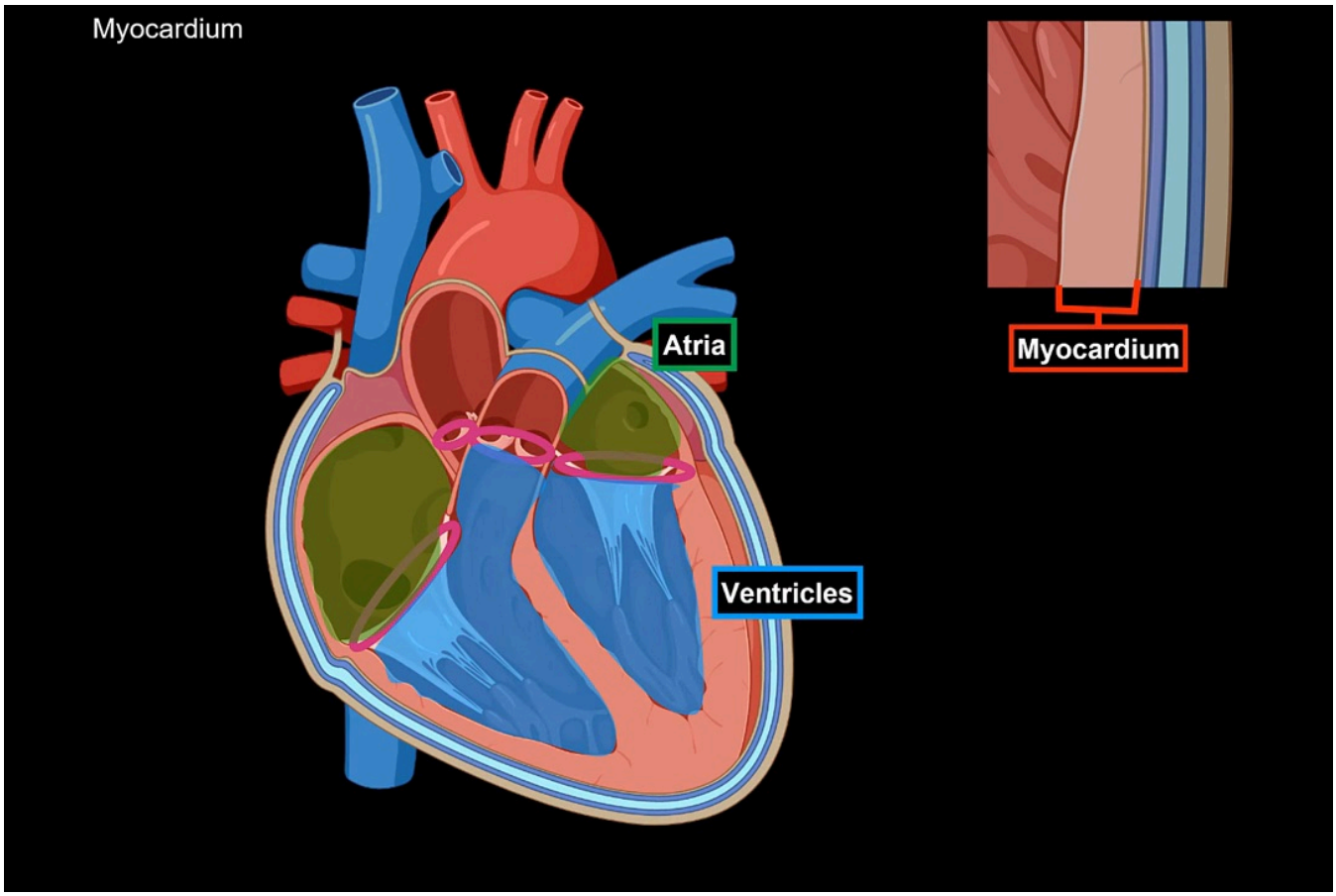

Layers of the Heart
Endocardium
Myocardium
Epicardium

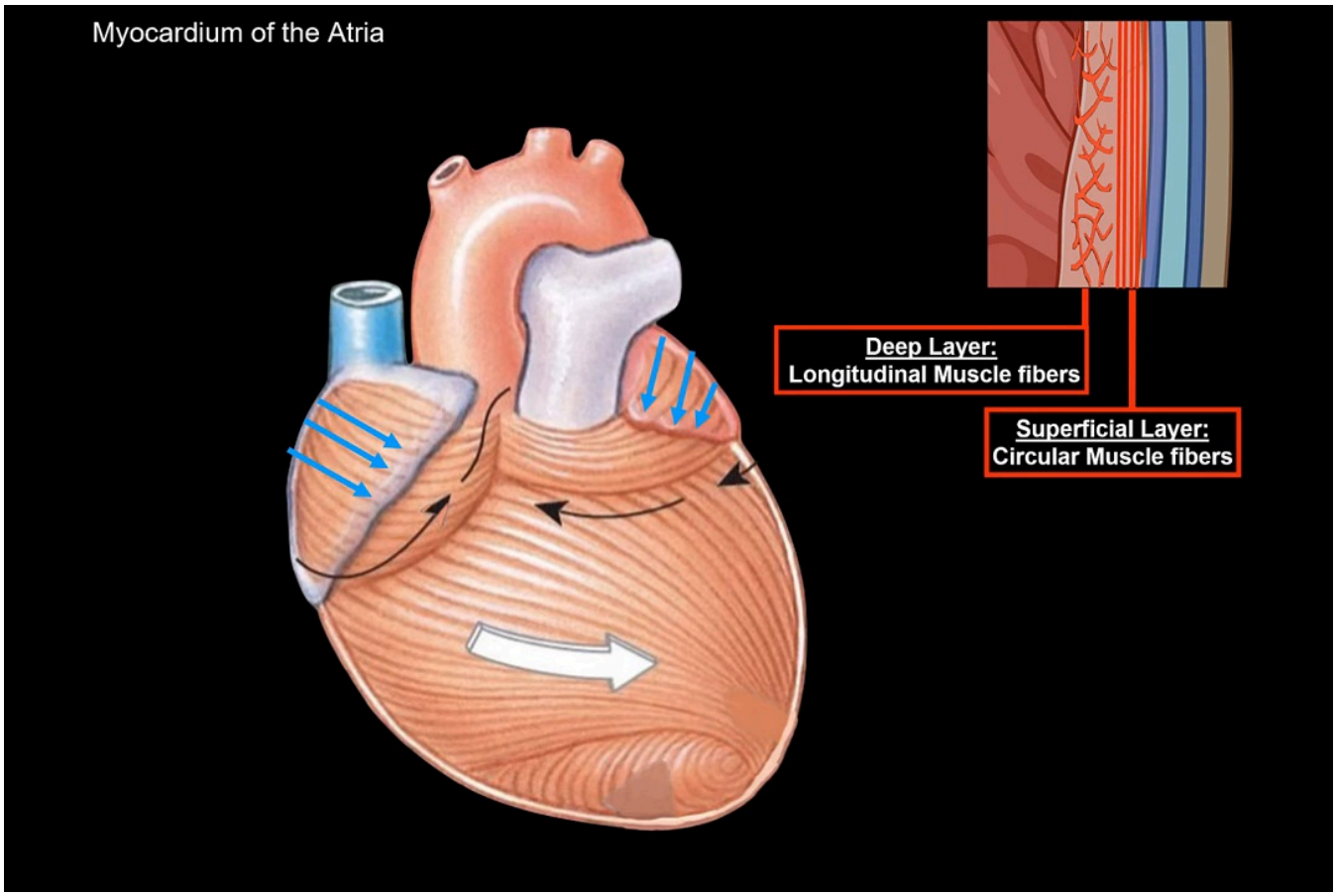

Endocardium
Aortic valve
Pulmonary Valve
Bicuspid Valve
Tricuspid Valve
Valve of the Inferior Vena Cava
Vulva Vena Cava Inferioris
Valve for the Coronary Sinus
Vulva Sinus Coronarii
Endocardium

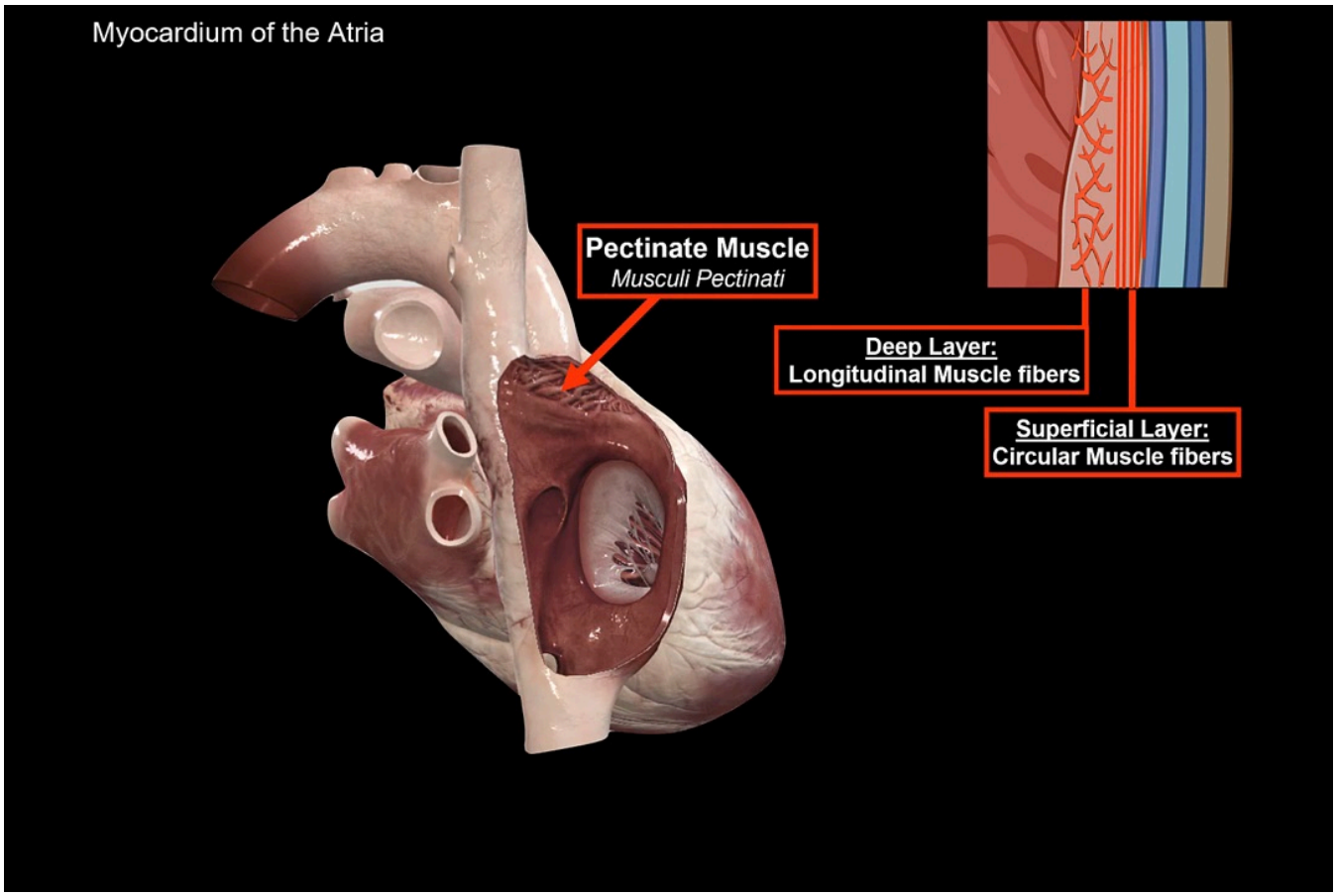

Myocardium
Atria
Ventricles
Myocardium

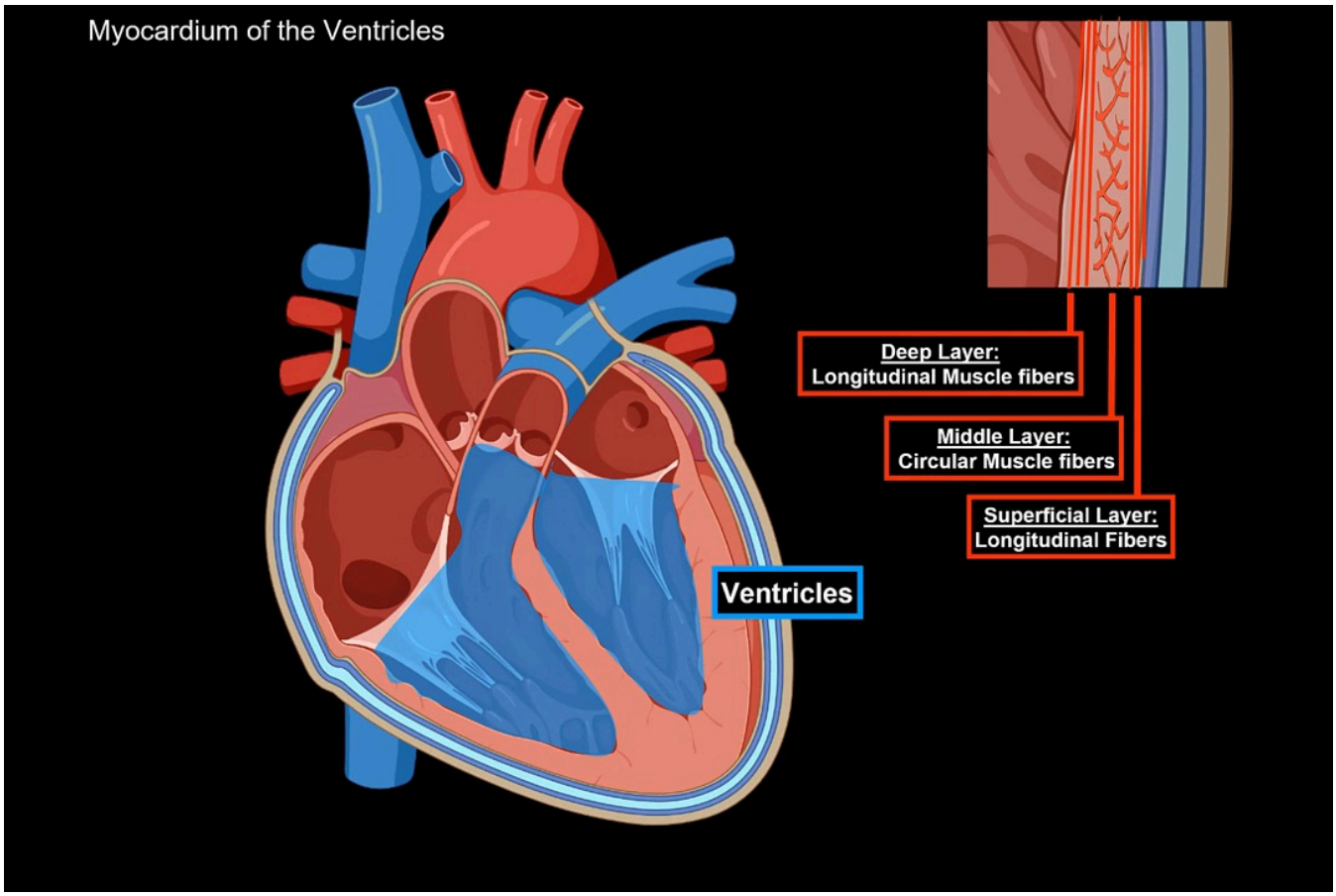

Myocardium
Myocardium
Left Fibrous Ring
Annulus Fibrosus sinister
Right Fibrous Ring
Annulus Fibrosus Dexter


Myocardium
Fibrous Ring of the Aortic Valve
Annulus fibrosus aorticus
Fibrous Ring of the Pulmonary Trunk
Annulus fibrosus trunci pulmonalis
Myocardium

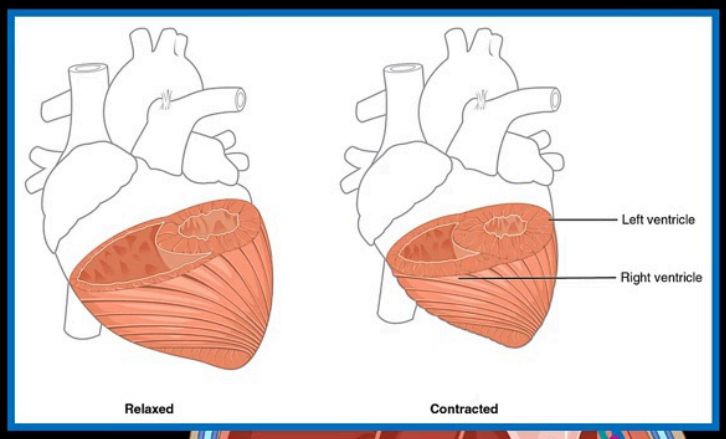

Myocardium
Atria
Ventricles
Myocardium

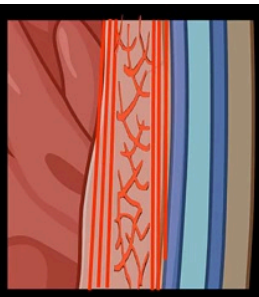

Myocardium of the Atria
Deep Layer:
Longitudinal Muscle fibers
Superficial Layer:
Circular Muscle fibers


Myocardium of the Atria
Pectinate Muscle
Musculi Pectinati
Deep Layer:
Longitudinal Muscle fibers
Superficial Layer:
Circular Muscle fibers


Myocardium of the Ventricles
Ventricles
Deep Layer:
Longitudinal Muscle fibers
Middle Layer:
Circular Muscle fibers
Superficial Layer:
Longitudinal Fibers

Myocardium of the Ventricles





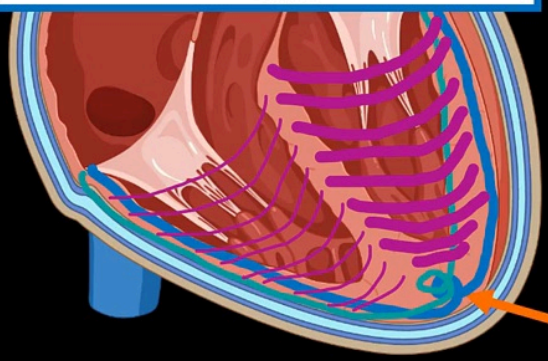
Longitudinal Muscle fibers

Superfical —> Apex —> Deep

Deep —> Apex —> Superficial

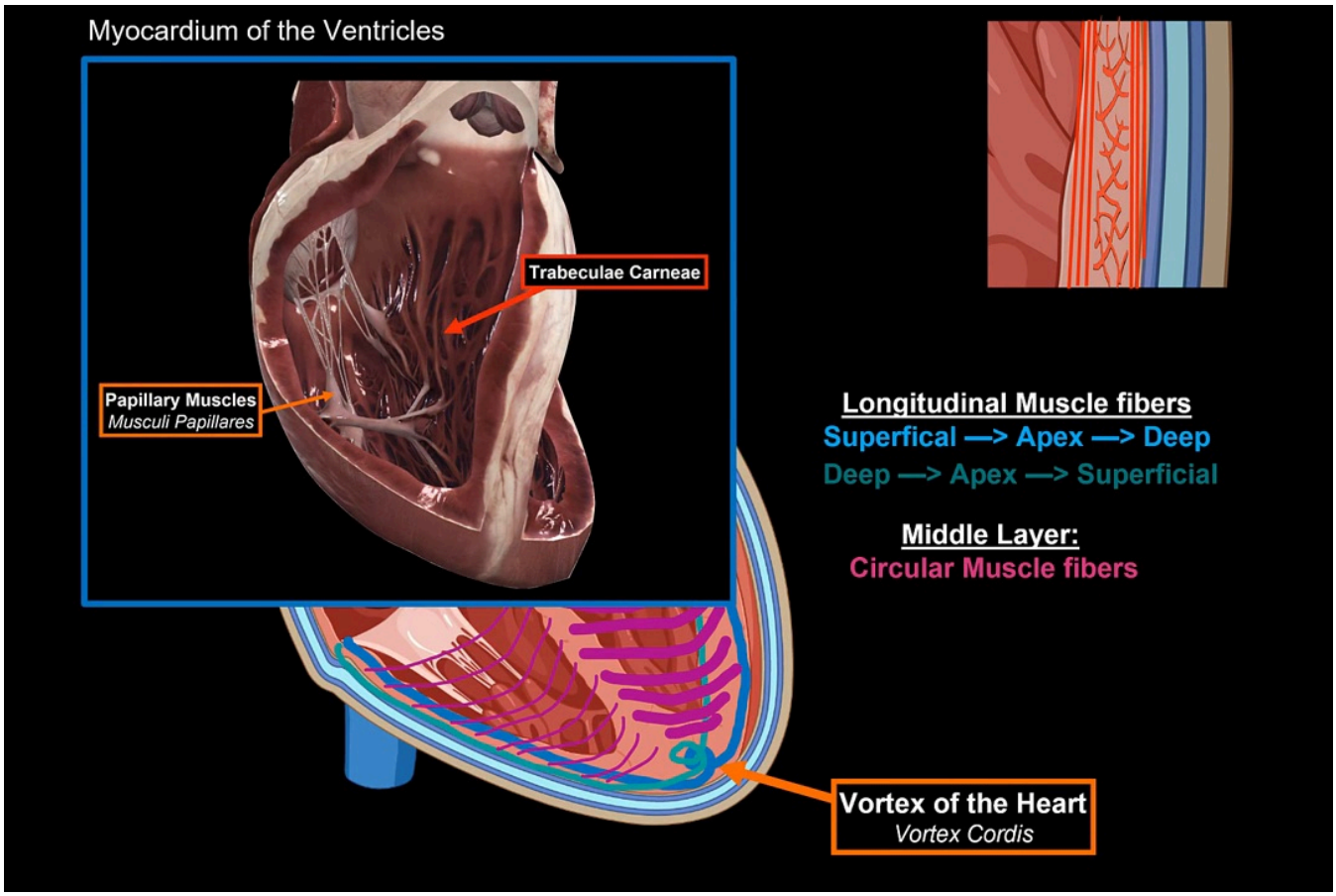
Middle Layer:

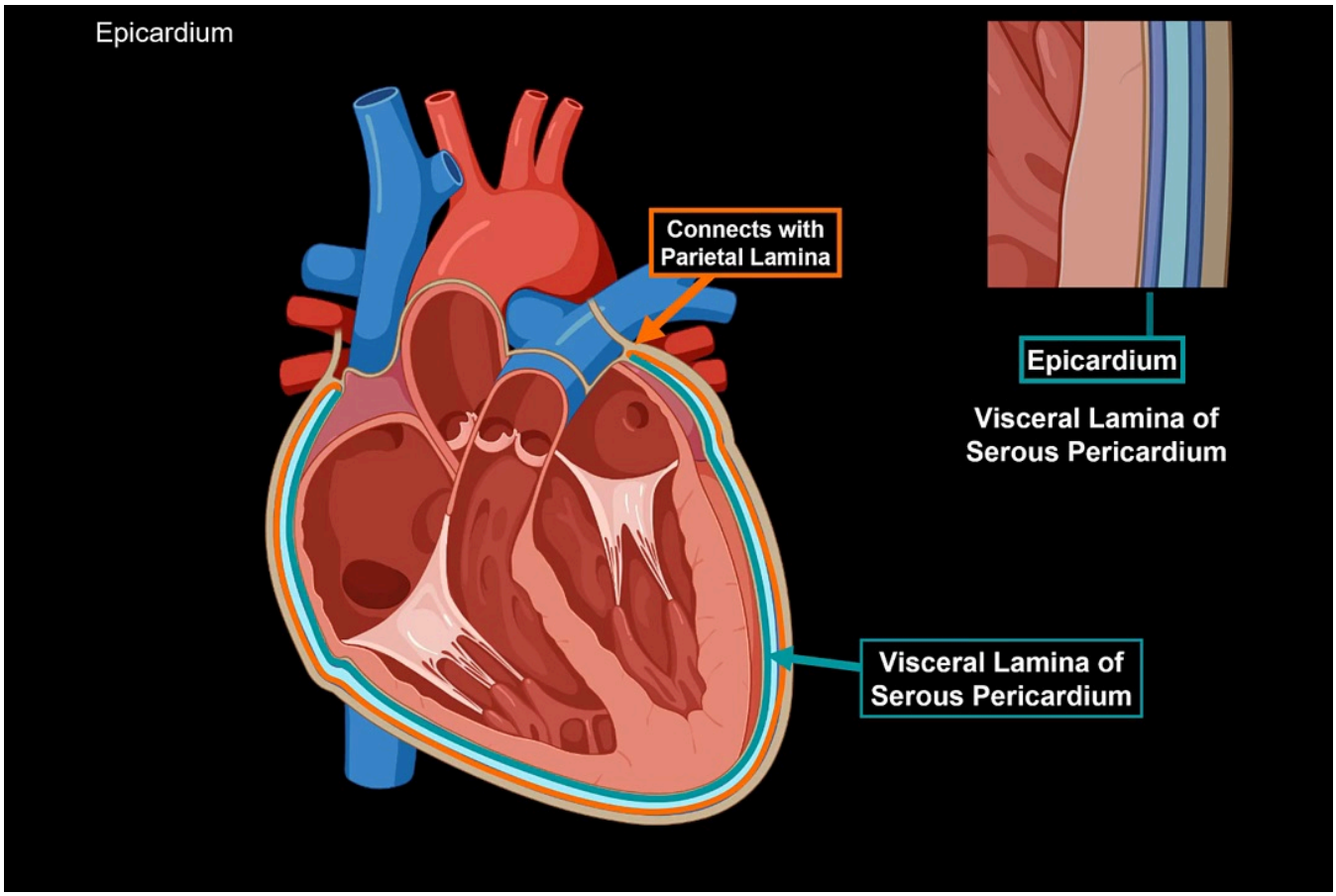
Circular Muscle fibers

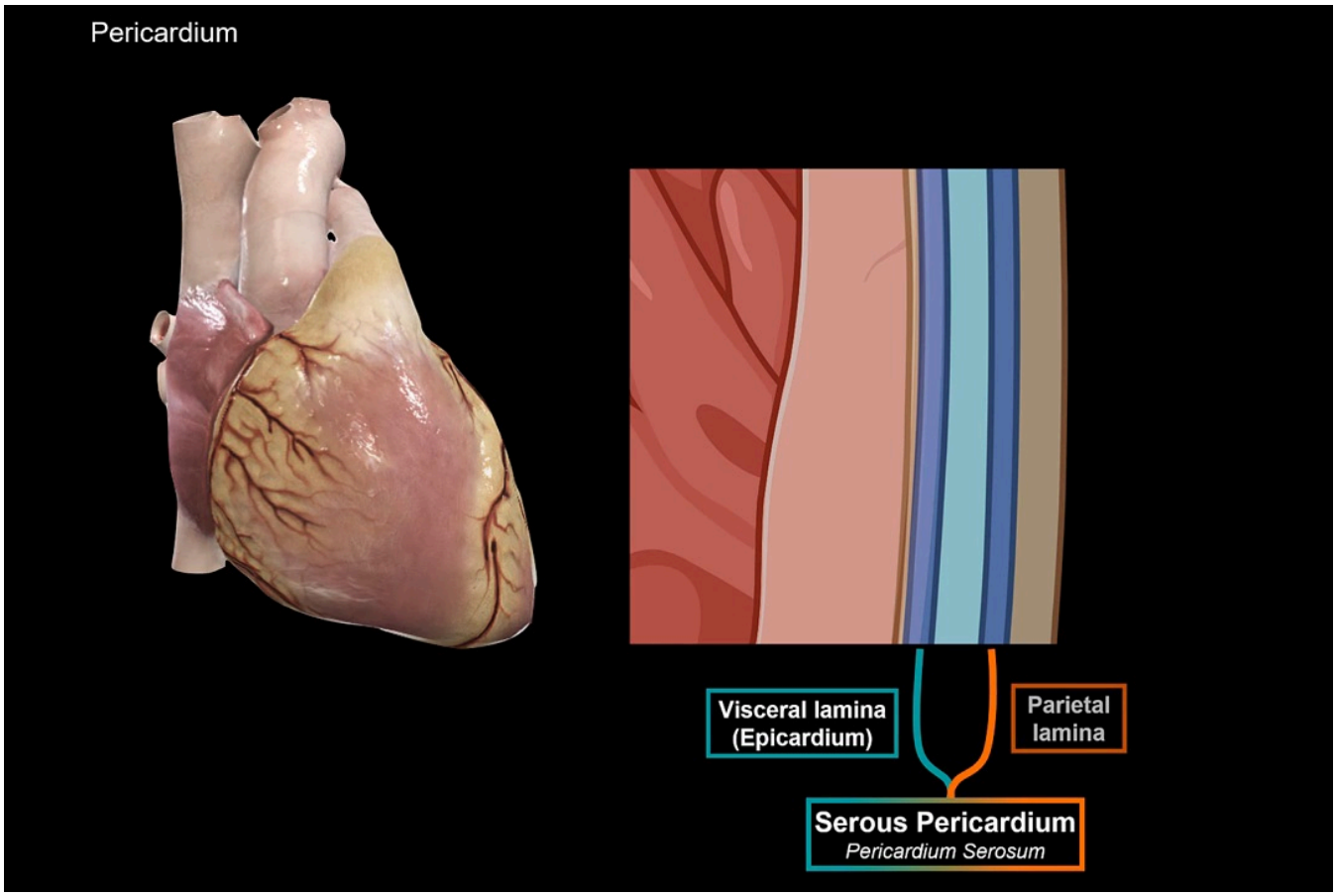


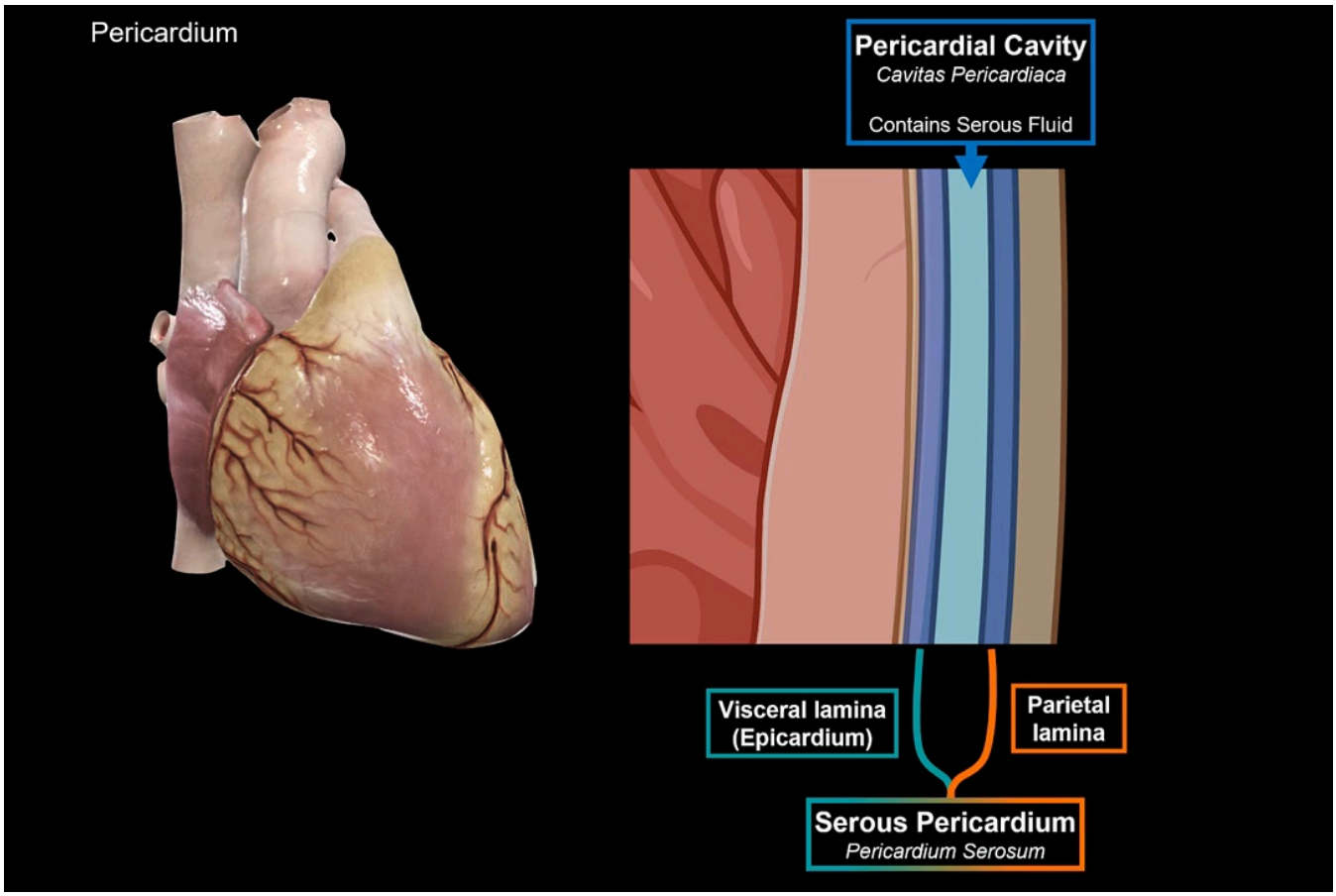
Vortex of the Heart

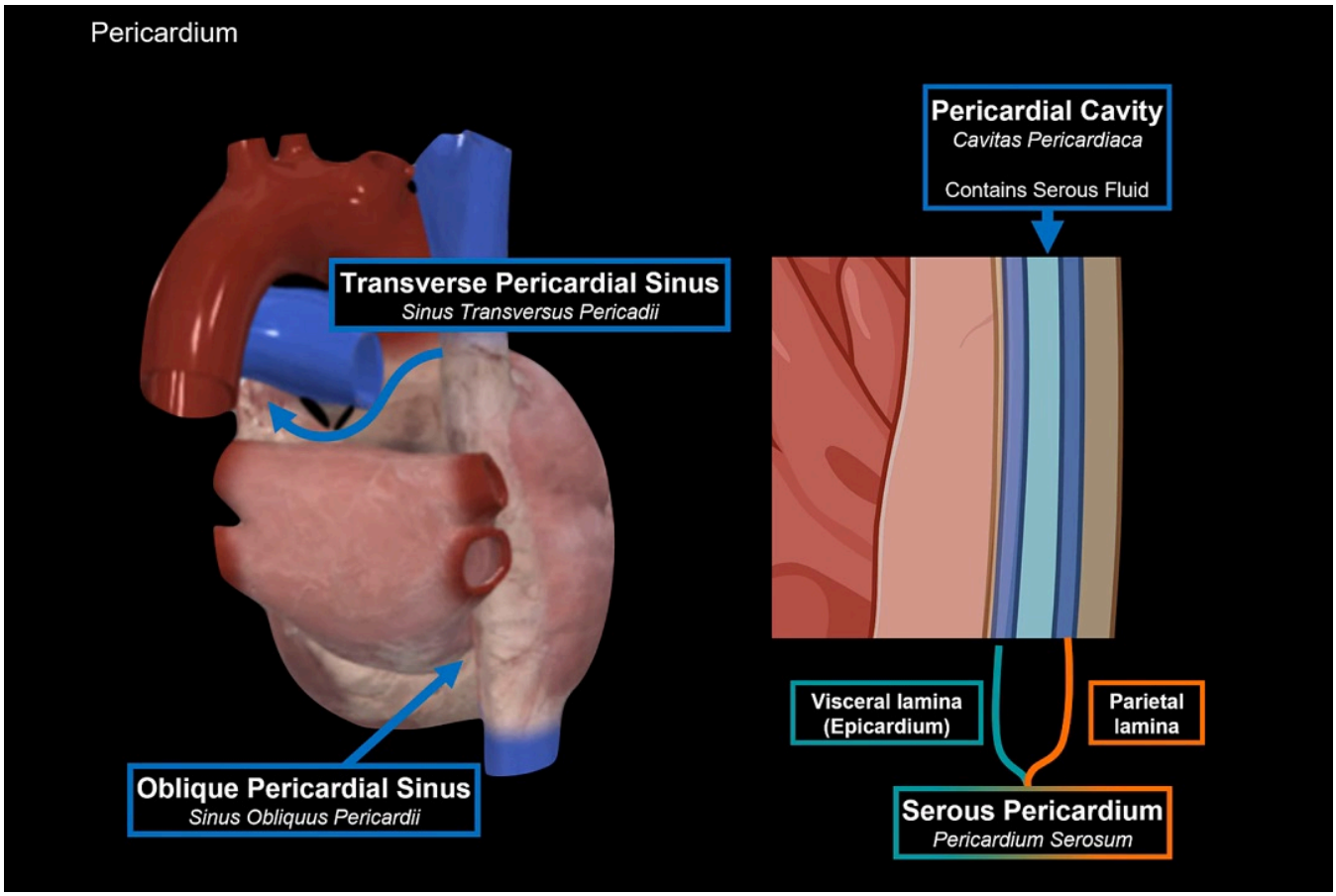
Vortex Cordis


Myocardium of the Ventricles
Trabeculae Carneae
Papillary Muscles
Musculi Papillares
Longitudinal Muscle fibers
Superfical —> Apex —> Deep
Deep —> Apex —> Superficial
Middle Layer:
Circular Muscle fibers
Vortex of the Heart
Vortex Cordis

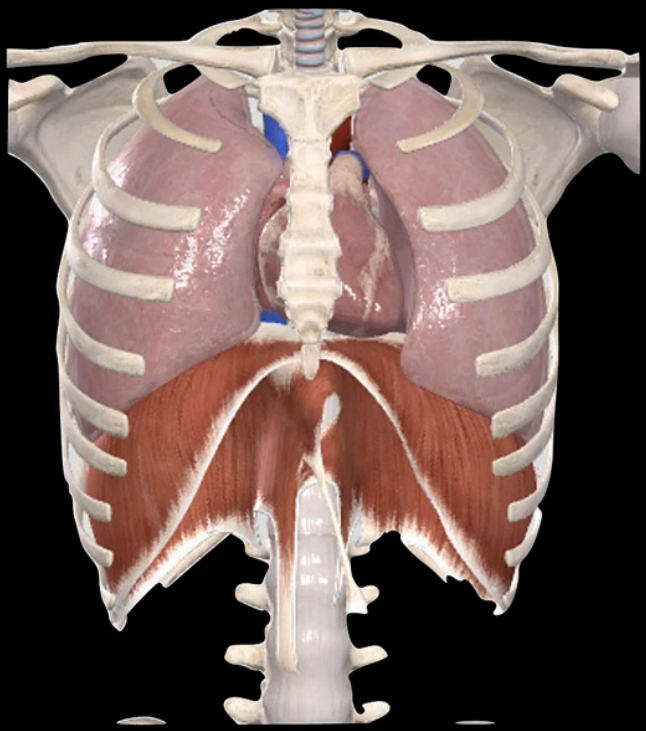

Epicardium
Connects with Parietal Lamina
Epicardium
Visceral Lamina of Serous Pericardium
Visceral Lamina of Serous Pericardium


Pericardium
Visceral lamina
(Epicardium)
Parietal
lamina
Serous Pericardium
Pericardium Serosum


Pericardium
Pericardial Cavity
Cavitas Pericardiaca
Contains Serous Fluid
Visceral lamina
(Epicardium)
Parietal
lamina
Serous Pericardium
Pericardium Serosum

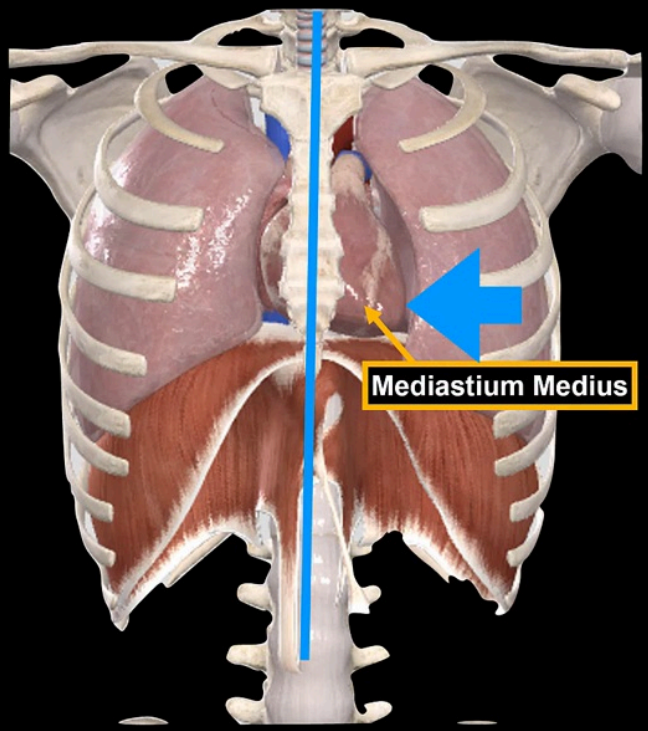

Pericardium
Pericardial Cavity
Cavitas Pericardiaca
Contains Serous Fluid
Transverse Pericardial Sinus
Sinus Transversus Pericadii
Oblique Pericardial Sinus
Sinus Obliquus Pericardii
Visceral lamina
(Epicardium)
Parietal
lamina
Serous Pericardium
Pericardium Serosum

Topography of the Heart

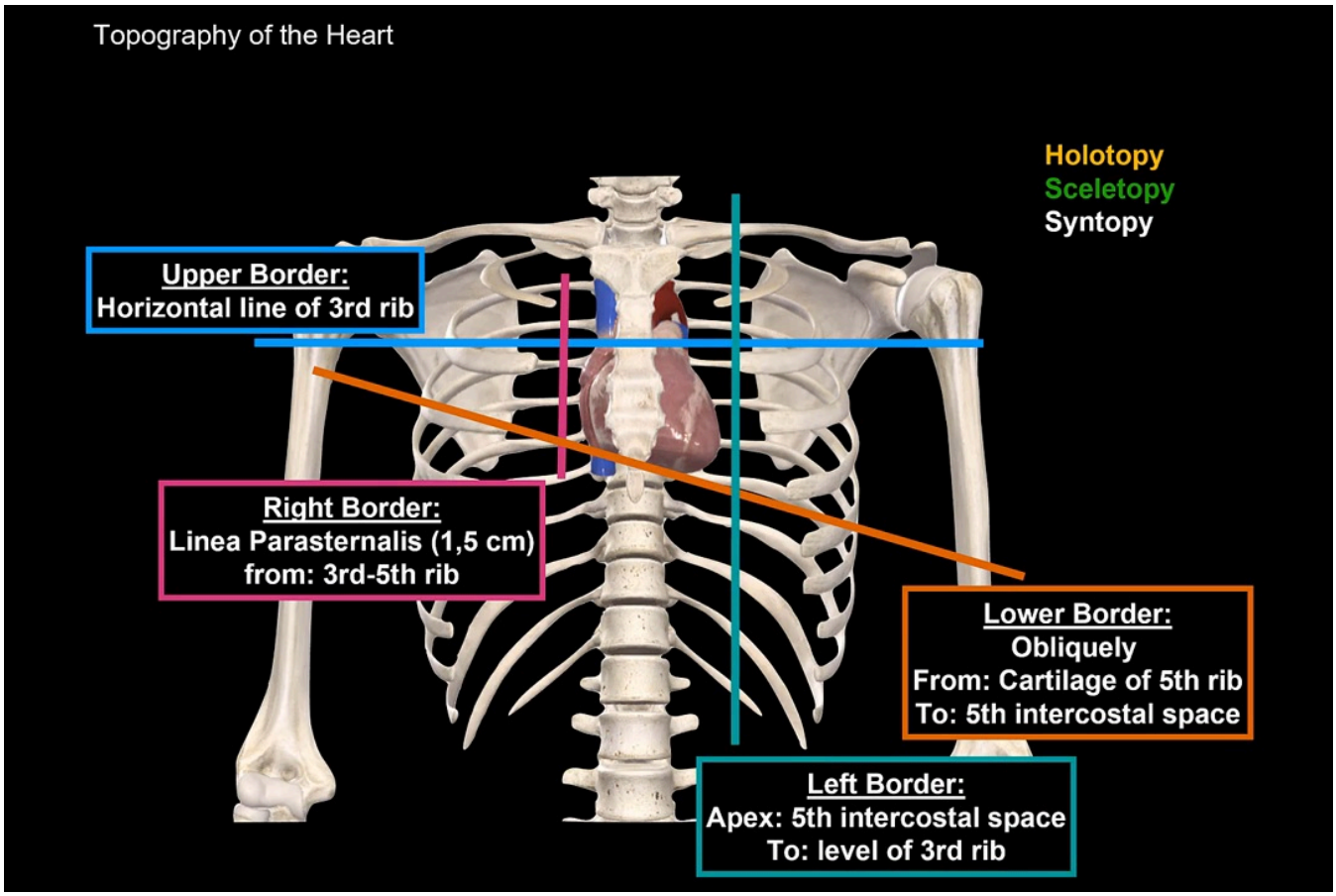


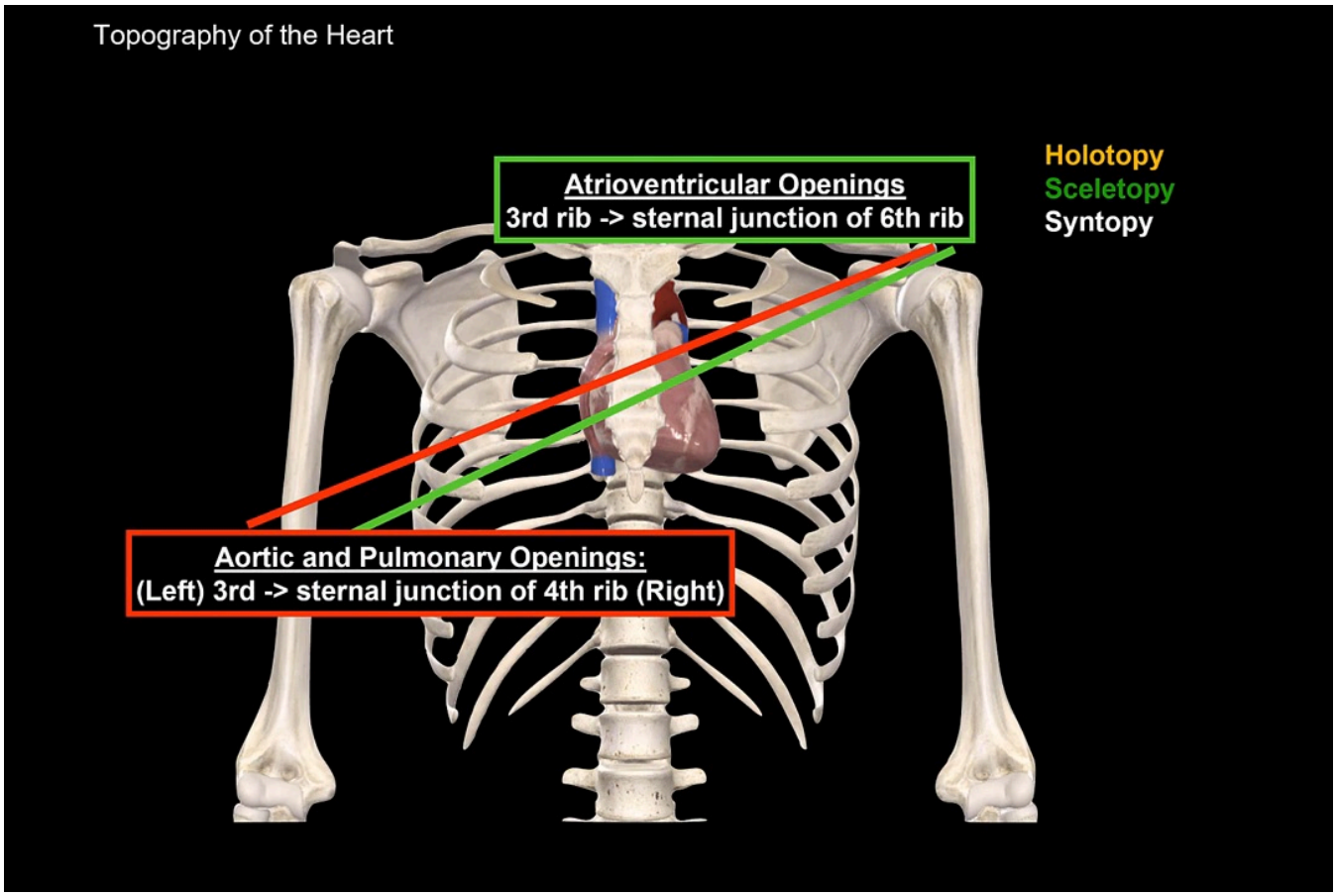
Holotopy
Sceletopy
Syntopy

Topography of the Heart

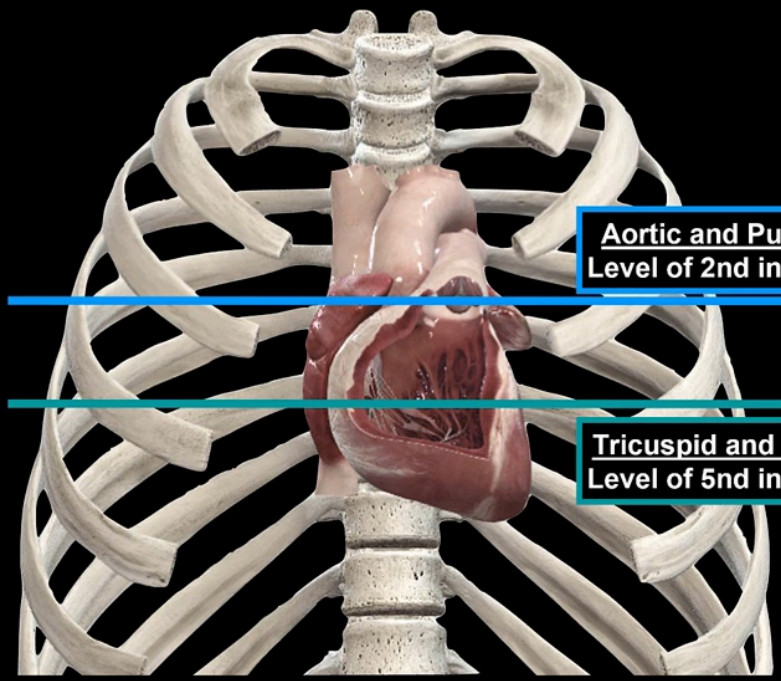


Holotopy
Sceletopy
Syntopy


Topography of the Heart
Holotopy
Sceletopy
Syntopy
Upper Border:
Horizontal line of 3rd rib
Right Border:
Linea Parasternalis (1,5 cm)
from: 3rd-5th rib
Lower Border:
Obliquely
From: Cartilage of 5th rib
To: 5th intercostal space
Left Border:
Apex: 5th intercostal space
To: level of 3rd rib


Topography of the Heart
Atrioventricular Openings
3rd rib -> sternal junction of 6th rib
Aortic and Pulmonary Openings:
(Left) 3rd -> sternal junction of 4th rib (Right)
Holotopy
Sceletopy
Syntopy

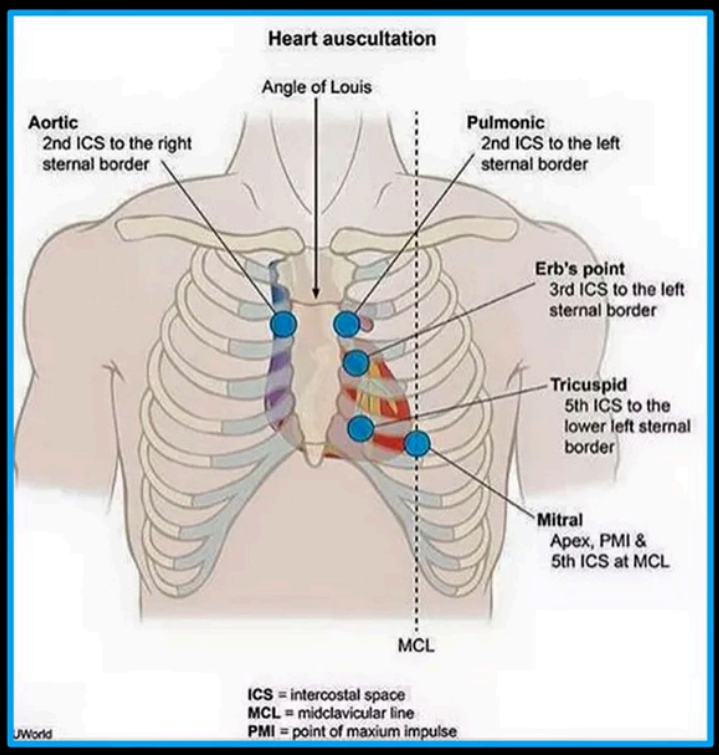
Topography of the Heart



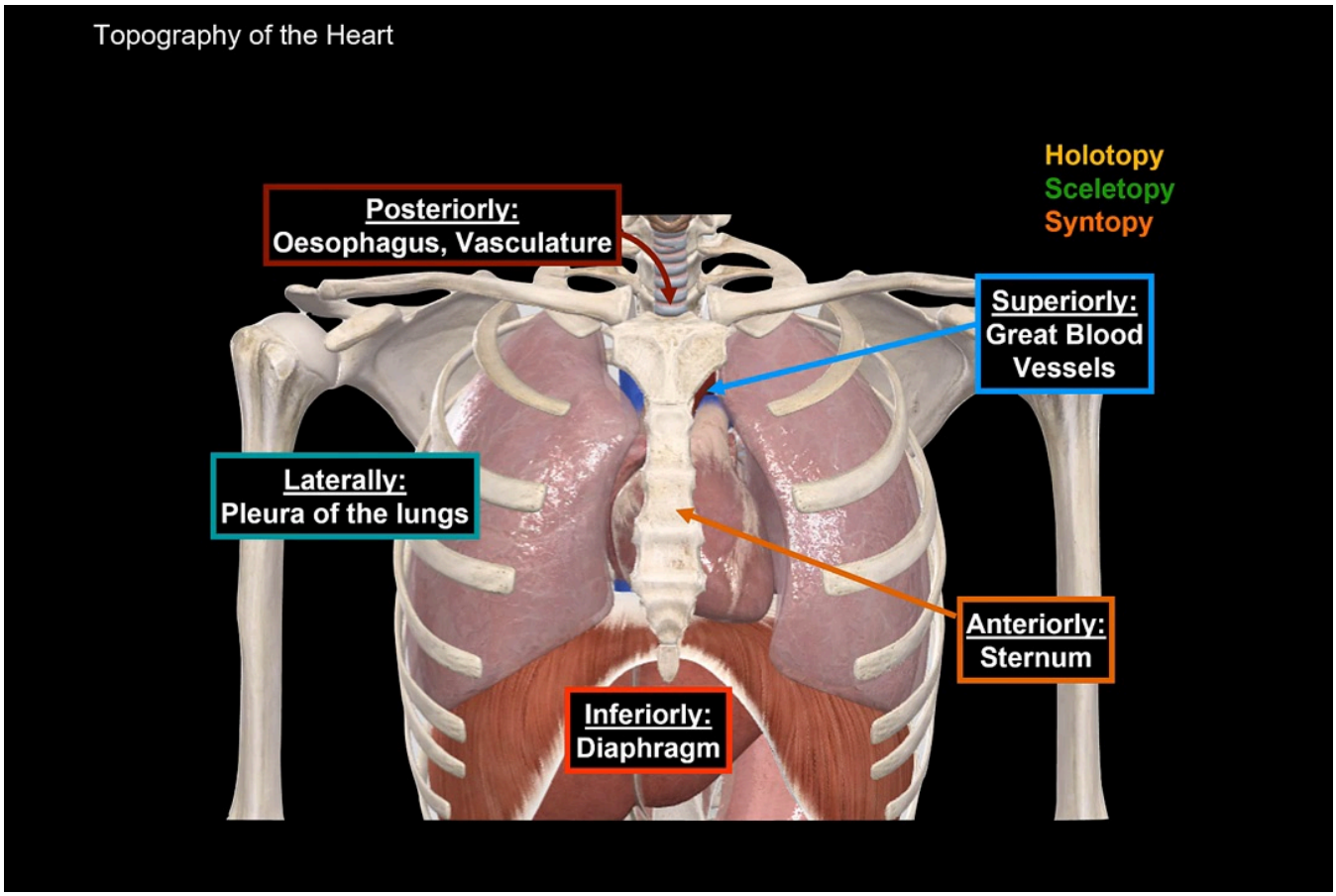
Holotopy
Sceletopy
Syntopy

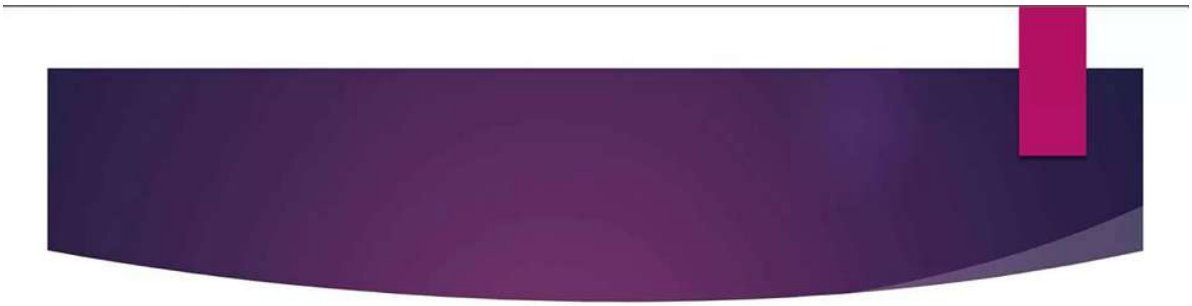
Aortic and Pulmonary Valve
Level of 2nd intercostal space

Tricuspid and Bicuspid Valve
Level of 5nd intercostal space


Heart auscultation
Angle of Louis
Aortic
2nd ICS to the right sternal border
Pulmonic
2nd ICS to the left sternal border
Erb's point
3rd ICS to the left sternal border
Tricuspid
5th ICS to the lower left sternal border
Mitral
Apex, PMI & 5th ICS at MCL
MCL
ICS = intercostal space
MCL = midclavicular line
PMI = point of maxium impulse
UWorld

Topography of the Heart





THANKS